gain energy from one or more planetary fly-by's around planets along the way (also known as Gravity Assists, abbreviated GA's; see Sec. 9.6.3): the energy is available for free, and the flight time might be reduced considerably. Ideally, on-board propellant would have to be taken along only for insertion in the orbit around the target planet and maintenance thereof. The first thing to do is to propose one or more scenarios of celestial bodies that one wants to pass along consecutively. This can be based on a plot of the positions of the planets during the interval of interest: the positions of the departure planet at this moment (e.g., "leave Earth in 2015") and the target planet at arrival (e.g., "be at Saturn within 5 years") will typically suggest a logical order of possibly useful GA planets.

It will be clear that a succession of Hohmann transfers would be difficult to use for such a purpose, if only for the strict requirements on geometry of the positions of the planets involved. Instead, the Lambert technique as presented in the book website in the web article is typically applied in these cases. Consider for instance a mission to Saturn, for which one wants to perform a GA at Jupiter. In such a case, the mission is a succession of 2 legs (Earth-Jupiter and Jupiter-Saturn). If the designer assumes epochs for launch, Jupiter GA, and Saturn arrival, it is possible to compute the parameters of the two connecting Kepler orbits, determine the excess velocity vector V_∞ before and after the Jupiter GA, and determine the ΔV's required at the Earth and at Saturn. Ideally, the required excess velocities before and after the Jupiter GA would be equal in magnitude (remember: they should be equal in case of a perfect hyperbola around the GA planet), and the gravity field of the GA planet should be strong enough to achieve the required bending angle (see Eq. 9-83).

However, the satellite heliocentric velocities directly before and directly after the GA result from solving two almost independent Lambert problems (they only share the epoch of the GA, and hence the position of the GA planet), so it is very unlikely that the magnitudes of the excess velocities are identical (irrespective of whether they are expressed in a heliocentric or a planetocentric frame). If these excess velocities indeed differ in magnitude, one needs to apply a ΔV when passing through the pericenter of the hyperbola around the GA planet: we have a so-called powered GA. Another possibility would be the situation when the gravity field of the GA planet is not strong enough (or the conditions prohibit the spacecraft from coming close enough to the surface of that planet—think of Jupiter's radiation belts) to bend the excess velocity from its original direction to the one required for the 2nd leg. In such a case, the mission also needs a ΔV, which is typically applied when the vehicle is about to leave the *SoI* of the GA planet; again we speak of a powered GA. The results (the summation of the 4 possible maneuvers: Earth escape, 1st maneuver during GA, 2nd maneuver during GA, Saturn capture) can be used in an optimization, in which the propellant usage can be minimized. As an example, a mission from the Earth to Neptune might be flown using a Venus-Venus-Earth-Jupiter-Saturn (VVEJS) sequence (required ΔV in parenthesis): launch from the Earth on May 25, 2012 (1.5972 km/s), GA around Venus (0.0010 km/s), Venus GA again (0.0030 km/s), GA around the Earth (0.0094 km/s), GA around Jupiter (0.4172 km/s), GA around Saturn (0.0081 km/s), and arrival at Neptune on December 18, 2029 (3.7831 km/s for orbit insertion at 4000 km altitude); the total ΔV amounts to 5.8190 km/s, whereas the total flight takes 17.6 years. For comparison: a direct Hohmann transfer would require a ΔV of 15.7 km/s and a transfer time of 30.6 years; this ΔV is far too large to be delivered by current launchers and upper stages alone.

In order to save propellant mass, one might also consider another option: fly a so-called *Aero Gravity Assist (AGA)*. In such a case, the vehicle performs a GA around a planet with an atmosphere, and uses the aerodynamic forces to achieve a larger total turn angle than would have been possible if flying a hyperbola only (lift, pointed downward unlike what we want with airplanes, forces the vehicle to follow a curved trajectory at a given altitude; with respect to the planetocentric frame, the vehicle will also loose some velocity because of drag). This exotic topic will not be treated here, however.

10.7.3 Round-Trip Missions

In many cases, one is not only interested in flying a mission to a distant object (a planet, an asteroid), but also in returning the vehicle to the Earth. Examples are sample retrieval missions like Stardust [NASA, 2010b], and Hayabusa [ISAS, 2010], to mention a few. It is obvious, right from the beginning, that this will complicate the mission considerably. First, it will require a larger amount of propellant: now it is also necessary to depart from the target planet, and a braking maneuver might be needed to end up in a parking orbit (for instance, for a rendezvous with the International Space Station) around the Earth—although the latter might be avoided if a direct entry into the Earth's atmosphere is employed, and a subsequent landing is made somewhere on the Earth's surface. Second, the total trip time can increase considerably: one can fly a one-way mission to Mars in 0.71 years using a Hohmann transfer (Table 10-33), but a back-and-forth mission using 2 Hohmann transfers would require a total mission time of 2.66 years (0.71 years for each leg plus another 1.24 years waiting at Mars for a favorable alignment of the two planets).

Timing

As for an analytical treatment of this round trip problem, see Wertz [2003], Wertz [2004], and Sarzi-Amade and Wertz [2009]. Here, we will treat the simple case in which (1) the departure and target planets move in circular, coplanar orbits around the Sun, (2) there is a single leg outbound and a single leg back, and (3) the transfers (outbound and return) are done by means of simple Hohmann trajectories. Other cases are extensively discussed in the references just mentioned.

Table 10-32◀, Fig. 10-33◀, Eq. 10-32◀

The geometry of the problem is sketched in Fig. 10-34, for the case in which we perform a trip to an outer planet and back. In this example, an Earth-Mars-Earth trip is considered, but of course the relevant parameters can be referred to any combination of planets.

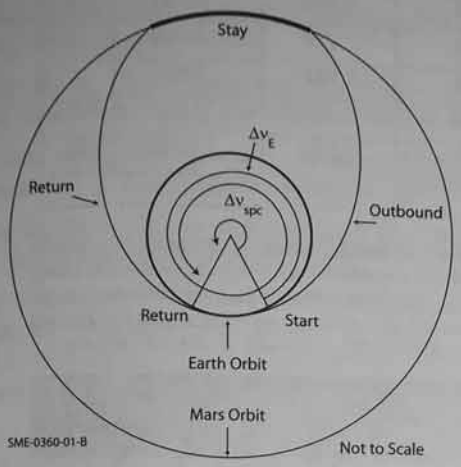


Fig. 10-34. Geometry of Round Trip Flight [Wertz, 2003]. Note: the transfers depicted here are non-Hohmann.

During the time of the mission, the Earth is assumed to cover an angle Δv_E, whereas the spacecraft covers an angle Δv_{spc}. In order to rendezvous at the Earth again at the end of the mission, the relation between the two angles must be the following:

$$\Delta v_E = \Delta v_{spc} + 2\pi W \tag{10-33}$$

where W is an integer number (which can be positive or negative). Without derivation (see Wertz, [2003]), the fastest round trip to Mars following a dual-Hohmann transfer is achieved when $W = 1$ (the case with $W = 0$ would require departure from Mars before even having arrived there). For the fastest possible trip to Venus, instead, we would have $W = -1$. In addition, the total transfer time T can be expressed as the summation of 2 Hohmann transfer times T_H plus a stay time at the target planet T_S:

$$T = 2T_H + T_S \tag{10-34}$$

If the angular velocities of the Earth and Mars around the Sun are expressed as ω_E and ω_M, respectively, one can express the total angle Δv_E covered by the Earth as:

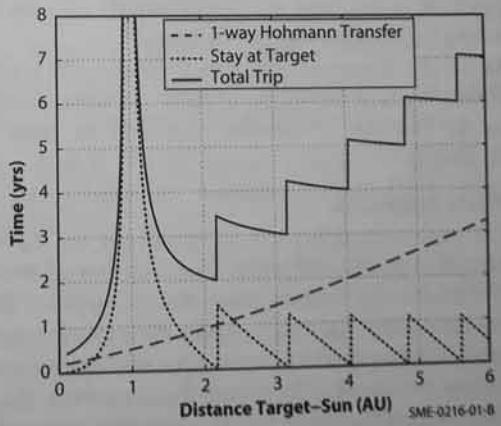
$$\begin{aligned} \Delta v_E &= \omega_E T = \Delta v_{spc} + 2\pi W \\ &= \pi + \omega_M T_S + \pi + 2\pi W \end{aligned} \tag{10-35}$$

(Any Hohmann transfer covers an angle of π radians; ω_E and ω_M are constant since we assume circular orbits.) By solving these equations, we get a solution for the total trip time T and the stay time T_S:

$$T = \frac{(W+1)2\pi - 2\omega_M T_H}{\omega_E - \omega_M} \tag{10-36}$$

$$T_S = T - 2T_H = \frac{(W+1)2\pi - 2\omega_E T_H}{\omega_E - \omega_M} \tag{10-37}$$

Obviously, the stay time has to be a positive number, so this condition defines the possible values for the parameter W for any combination of planets. Clearly, the fastest round trip can be achieved when W is chosen such that the stay time is smallest (in the case of a Hohmann trajectory, the transfer time is a given). Table 10-33 gives some numerical values for a number of back-and-forth trips from the Earth to some relevant planetary targets. Fig. 10-35 gives similar information for arbitrary distances between the target and the Sun. As expected, the curve for the one-way Hohmann transfer time shows a gradual increase with distance. The curves for stay time and total time, on the other hand, show regular jumps, which essentially represent integer increases of the parameter W, in order for the geometry to be favorable for the return leg (the stay time increases with $2\pi/(\omega_E - \omega_M)$ for each step in W). For targets at distances larger than about 2 AU, it is clear that the Earth needs to make more revolutions in order to make the round trip possible (for example, the minimum value for the parameter W increases to 5 for a trip to Jupiter). Clearly, objects that have a semi-major axis which approaches 1 AU (e.g., near-Earth asteroids) would be difficult targets for a Hohmann approach since their angular velocity is comparable to that of the Earth, and so the stay time would become quite large, awaiting a favorable return geometry; consequences for the total trip time are obvious.



Live Calc

Fig. 10-35. Hohmann Transfer Times, Stay Times, and Round Trip Times for Missions Between the Earth and Targets at Arbitrary Distances.

As shown in Wertz [2003], one may opt to go for a longer stay time while maintaining the total trip time at nearly the same values as shown in Table 10-33. However, in such a case, the transfer time itself has to be shortened. Obviously, this results in a non-Hohmann

Live Calc

Table 10-33. Hohmann Transfer Times, Stay Times, and Round Trip Times for Missions Between the Earth and Neighboring Planets.

Target	Mean Orbit Radius* [AU]	Angular Motion [rad/s]	Hohmann Transfer Time [yrs]	Stay Time [yrs]	Total Trip Time [yrs]	W [–]
Mercury	0.387	8.267×10^{-7}	0.289	0.183	0.761	–1
Venus	0.723	3.236×10^{-7}	0.400	1.279	2.078	–1
Mars	1.524	1.059×10^{-7}	0.709	1.244	2.661	1
Jupiter	5.203	1.667×10^{-8}	2.731	0.587	6.050	5
Saturn	9.537	6.760×10^{-9}	6.047	0.939	13.032	12

* rounded off to 3 digits behind the dot; see appendices for full value.

Table 10-34. Four Scenarios for a Mission to Mars, Plus their Main Characteristics. In all cases, the vehicle leaves the Earth somewhere in the time frame 2013–2015. The planet positions have been represented using JPL's DE200 ephemerides [Standish, 1982]. All scenarios start in a parking orbit around Earth at 185 km altitude. Scenario 1 does not enter into an orbit around Mars, whereas scenarios 2–4 enter in an orbit at 500 km around Mars. Scenarios 3 and 4 return to Earth, where scenario 4 ends up at the orbit of the ISS at 340 km altitude.

Scenario	ΔV [km/s]	Total ΔV [km/s]	Dates	Total Travel Time [days]
Mars Flyby	3.616 (Earth dep.)	3.616	Jan 2, 2014 (Earth dep.) Nov. 28, 2014 (Mars arr.)	330
Mars Orbit Insertion	3.649 (Earth dep.) 2.341 (Mars arr.)	5.989	Dec 4, 2013 (Earth dep.) Sep 24, 2014 (Mars arr.)	294
Mars Orbit Insertion, Return to Earth (direct landing)	3.649 (Earth dep.) 2.341 (Mars arr.) 1.920 (Mars dep.)	7.999	Dec 4, 2013 (Earth dep.) Sep 24, 2014 (Mars arr.) Dec 13, 2015 (Mars dep.) Oct 8, 2016 (Earth arr.)	1039
Mars Orbit Insertion, Return to Earth (rendezvous with ISS)	3.649 (Earth dep.) 2.341 (Mars arr.) 2.041 (Mars dep.) 3.987 (Earth arr.)	12.018	Dec 4, 2013 (Earth dep.) Sep 24, 2014 (Mars arr.) Dec 25, 2015 (Mars dep.) Aug 12, 2016 (Earth arr.)	982

transfer, at the expense of larger ΔV requirements. Another option is to extend the stay time by adding the synodic period; in that case one would simply wait until the departure geometry is favorable again.

As for the departure and arrival epochs (for both the outbound leg and the inbound leg), the same formulas given for one-way Hohmann trips can be used (see Table 10-32).

Lambert Approach

The numerical approach, as presented on the book website is more flexible since it uses a (first order) succession of 2-body formulations, and can involve direct trajectories, or multiple legs. Again, the basis for this approach is a succession of Lambert problems plus an optimization technique. The recipe is, in essence, the one that was proposed already for one-way missions: assume a range of epochs for departure from the Earth, possible GA's, arrival epoch at the target, epoch of departure from that target, possible GA's again, and arrival date at the Earth; compute the parameters of the various legs, including propellant usage, stay time, and so on; and optimize. By defining an objective function (i.e., the factor in the optimization that can be quantified and that needs to be minimized or maximized) that not only includes the required propellant expenditure but also the total trip time, one can end up at a trajectory design that performs (slightly) worse in terms of propellant usage, but does better in terms of time-of-flight. Do not forget: time-of-flight has consequences for degradation, radiation hazards, operational costs, and other characteristics.

As an example, Table 10-34 gives an overview of 4 fundamentally different missions to Mars, plus the main characteristics of each option. The first 2 concepts are one-way missions, and the last 2 are round trips. Clearly, bringing a sample back to the Earth does not exactly come for free (higher ΔV, longer time-of-flight). The situation becomes even more demanding when the return vehicle has to rendezvous in Earth orbit, rather than (for instance) plunge into the ocean directly (as was the case with the Apollo missions).

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11 Cost Estimating

Henry Apgar, ***MCR Technologies***

11.1 Introduction to Cost Estimating
Groundrules and Assumptions; Types of Estimates; Elements of Cost; Phases of the Estimate; Cost Estimating Process Steps

11.2 Estimating Tools
Top-Down Estimating Process Using Models, CERs, and Factors; Unmanned Space Vehicle Cost Model (USCM8); Small Spacecraft Cost Model (SSCM); QuickCost; NASA Instrument Cost Model (NICM); Software Cost Estimating Model—COCOMO 81; Facility Costs; Launch Costs; Additional Cost Factors; Operations Costs; Cost-Risk Assessment Model; Cost Trade Studies

11.3 Other Considerations in the Cost Estimate
Cost Realism; Cost Improvement; Inflation; Cost Spreading; Technology Readiness Levels; Establishing Cost Estimate Requirements; Documenting the Final Estimate

11.4 Example Space Mission Cost Estimates—*John Collins*
FireSat II; Supplemental Communications System (SCS); Summary

Cost estimating is the iterative process of calculating the expected cost of a space mission. Just as the space systems engineer predicts performance parameters for his evolving space system design, he can also predict development, production, and operations costs. Since any prediction entails some uncertainty, cost predictions need to be stated in risk terms, e.g., within uncertainty bounds or with an associated confidence level. Just as space systems engineers need to understand technical and program parameters (capacity, reliability, operating life) which drive design, so do they need to understand the engineering and program requirements (size, complexity, technology) which drive cost. Typical cost drivers are: size (weight in kilograms for hardware and lines of code for software), complexity (choice of material, power, bandwidth), availability of the technology (technology readiness level, degree of design completion), and schedule (time available for development).

The scope and depth of the cost estimating process depends on the intended purpose of the estimate. A rough order magnitude (ROM) estimate for a preliminary space system design will entail some opinion and judgment, but we seek estimating credibility to support design decisions and so we base our estimates on known facts and calculated expected cost values. A proposal price estimate depends on a well-crafted and documented prediction of future cost and the associated cost-risks. An engineering trade study, on the other hand, may require only a top-level understanding of the difference in cost between design alternatives. A cross-check estimate should require less time, less information, and fewer resources.

Technically, all estimates are based on *analogy*. That is, what will be the differences or application of changes from an earlier design or mission for which we already know and understand the relevant costs? The estimating process can be systematized by developing statistical *cost estimating relationships* (CERs) with known cost data from those earlier designs. CERs are equations which relate the **dependent** variable (e.g., dollars, labor or hours) to one or more **independent** variables (size, complexity, resources, and time). An aggregate of CERs is often called a *parametric cost model*. Similar to the application of engineering design models, cost models emulate the design process, production activities, and operations of space systems.

Since a space mission cost model emulates the materials and activities consumed during the space mission development, production, and operation, we strive to assure proper application of the chosen model to the space mission being designed. Section 11.2 in this chapter provides several parametric models, each applicable to a different phase (development, production, operations) of the program. The aggregate of costs from all program phases is popularly known as the *life cycle cost*.

Some help in understanding these terms and their relationship to space mission analysis and design may be found in NASA [2010] and ISPA [2007]. A list of cost-related acronyms is given in Table 11-1.

Cost estimates support the engineering and design process. Awareness of most likely cost, concern with cost targets from program inception, and sensitivity to the relative cost between design alternatives contribute to the acceptance and utility of the space system design. *Most likely cost* is the same as the mode of all likely cost outcomes and is the value which would occur most often in a Monte Carlo simulation of final cost.

Cost estimating has assumed a more proactive role in space systems development with the availability of parametric cost estimating tools, early awareness of system

Table 11-1. Cost Estimating Acronyms.

AUC	Average Unit Cost	*MM*	Man Months
BOM	Bill of Materials (basis for bottoms-up estimate)	*NICM*	NASA Instrument Cost Model
CAIV	Cost As an Independent Variable, an affordability initiative	*O&M*	Operating and Maintenance, a program phase, alternately called Operations and Support (O&S)
CADRE	Cost Analysis Data Requirements	*O&S*	Operating and Support, a program phase, alternately called Operations & Maintenance (O&M)
CARD	Cost Analysis Requirements Description	*OTS*	Off-the-shelf, designating a component which requires no development effort
CBS	Cost Breakdown Structure, variation of WBS	*PME*	Prime Mission Equipment (hardware and software), what most CERs will estimate
CER	Cost Estimating Relationship, building block of the parametric cost estimating model	*ROI*	Return on Investment
COCOMO	Cost Construction Model	*ROM*	Rough Order Magnitude, a type of cost estimate, typically when little information is available
COTS	Commercial off-the-shelf	*SEE*	Standard Error of the Estimate (of regression in deriving a CER)
DSI	Delivered Source Instructions (similar to SLOC) (software)	*SLOC*	Source Lines of Code, input parameter for most software estimating models
DTC	Design to Cost, an affordability initiative	*SSCM*	Small Satellite Cost Model
ESLOC	Equivalent Source Lines of Code, captures the labor impact of modifying (rather than developing) code	*T1*	See TFU
FTE	Full Time Equivalent (one person working full time for one year)	*TFU*	Theoretical First Unit; the starting point for production "learning," sometimes known as T1
ICE	Independent Cost Estimate	*TRL*	Technology Readiness Level, a measure of the state of development—a cost driver
KPP	Key Performance Parameter (traded against cost in trade studies)	*USCM8*	Unmanned Spacecraft Cost Model, version 8
LOOS	Launch and orbit operations support	*WBS*	Work Breakdown Structure
MDAL	Master Data and Assumptions List	*WLC*	Whole Life Cost (similar to Life Cycle Cost)
MUPE	Minimum unbiased percentage errors (method of developing CERs)		

cost drivers, and responsibility of the engineer to design affordable products. Engineers are especially involved in the application of *affordability initiatives* such as Design to Cost (DTC) and Cost as an Independent Variable (CAIV), where product design converges on a cost goal. Section 11.2.12 identifies the CAIV initiative as a process for reducing cost. Such initiatives effectively support lower cost space missions through better understanding of the life cycle cost drivers. Careful evaluation of Return on Investment (ROI) analyses and cost/performance trade studies are more effective than simply trying to make the system "cheaper."

The "laws of estimating" are different philosophically from the "laws of nature." For estimating, there can be no absolute right answer to the question, "How much will it cost?" With estimating, we are predicting a future event in a world of fiscal, political, and technological uncertainty.

11.1 Introduction to Cost Estimating

11.1.1 Groundrules and Assumptions

Our first step is to establish the estimating groundrules and assumptions which can be extracted from the mission description and the development plan.

Estimating groundrules are statements about the form and content of the estimate (e.g., *"...costs include a competitive development by two contractors up through system test, and are stated in 2012 dollars"*). Groundrules also establish the scope of the estimate, distinguishing specifically between costs that are included and those excluded from the analysis. For example, an estimate may not address operating costs at this time, or it may not include the cost of establishing new test facilities.

Estimating assumptions are suppositions about what will happen at some future time. Assumptions can be established globally for the system or program (e.g., laser technology that meets the requirement will be available before the start of system development; contractor A will be the system designer; all software will be written in C++); or they can be established for the specific cost elements (e.g., a VLSI chip will be used to implement the function; five test units will be needed for test evaluation; a software development lab will be established). Assumptions may change throughout the estimating process; thus, careful recording and tracking of assumptions and their changes to cost estimate versions are necessary. Assumptions help to identify significant cost drivers and indirectly the choice of a parametric cost estimating model.

Examples of estimating groundrules and assumptions can be seen in Sec. 11.4, particularly Tables 11-34 through 11-39, where we estimate the FireSat II and SCS system costs.

11.1.2 Types of Estimates

We typically develop our first ROM estimate before we can expect (1) a good *bill of materials* (BOM—a list of all materials, parts, and components) and (2) a description of development, production, and operations activities by labor classification. This is an estimating process commonly called a "bottoms-up" or "detailed cost estimating" process. Instead, we must rely on broad design concepts and subsystem-level design parameters, also known as cost drivers. This process is commonly called a "top-down" or "parametric cost estimating" process.

The parametric cost estimating process is supported by cost estimating tools described in this chapter, including cost estimating models and normalized historic databases. Models are comprised of CERs, which are statistically-based cost-predicting algorithms derived from these databases.

Models typically focus on specific cost elements, such as spacecraft hardware, payload software, or launch vehicle operations. CERs typically are derived at the subsystem or component level and relate cost to one or more cost drivers, such as weight, power, and amount of additional design required. Apgar [2009] and Fox [2008] provide an introduction to the selection of cost drivers for space mission CERs.

The distinction between non-recurring and recurring costs helps to choose the appropriate model or CER, as well as to correctly place (or map) cost predictions within the project work breakdown structure (WBS). CERs predict non-recurring cost or recurring costs, or sometimes both.

It is important to ascertain that model and CER definitions are consistent with WBS definitions. The following distinction is typically made between non-recurring and recurring costs:

- ***Non-recurring costs*** are associated with the labor and material associated with designing, developing, fabricating, and testing a space vehicle qualification test model plus program-peculiar ground support equipment (GSE). This is often identified as the prototype approach and does not produce a flight unit. For those programs that use the alternative *protoflight* approach, the qualification test unit will later be refurbished to become the first flight unit. This requires an additional 30% (approximate) non-recurring cost for the refurbishment.
- ***Recurring costs*** are associated with the labor and material of fabricating, manufacturing, integrating, assembling, and testing of follow-on space vehicle flight hardware plus the effort associated with launch and orbital operations in support of the program. Cost databases typically accumulate recurring program costs in total, rather than by specific production units. As a result, historical recurring cost data must be adjusted to reflect a theoretical first unit (TFU) cost, or T1, for the purpose of developing the recurring CERs. By default, a cumulative average learning curve of 95% is typically applied, both in CER development and cost prediction. (See Sec. 11.3.2 on Cost Improvement for more information on learning rate.) For the prototype approach, the T1 then is the first flight unit; for the protoflight approach, the T1 is actually the second flight unit.*

11.1.3 Elements of Cost

Elements of cost are essentially the hardware, software, and operations elements of the space system, plus necessary management and administrative elements (wrap factors) as identified by the project WBS. Collectively, then, the WBS elements describe the entire project and provide a disciplined structure for developing the cost estimate, budgeting the project, and tracking costs as the project proceeds.†

To facilitate matching CERs to WBS elements, and for instructional purpose, we have adopted an ad-hoc WBS numbering system for all models, CERs, and factors described in SME-SMAD. The engineer will most likely encounter a project-specific WBS early in the development process. The SME-SMAD example facilitates a standard cost estimating model.

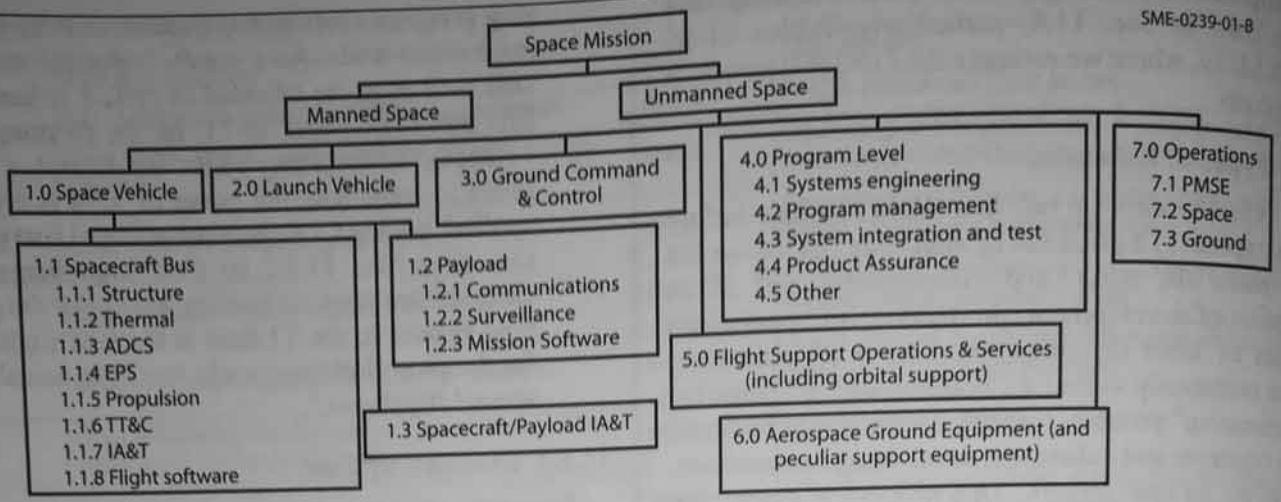
The estimator typically obtains the WBS from the system designer. The estimator must be able to translate, or map, between similar WBS structures, since the industry has not achieved the goal of a standard WBS. The estimator may find slightly different WBS structures for:

- Legacy programs which are the source of historical cost data, useful for analogy estimating.
- Special purpose cost estimating models, such as USCM8, which are based on a pre-determined WBS.
- The new program which he must estimate; in SME-SMAD, the WBS for FireSat II and SCS will be provided.

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* It may seem unusual to identify costs for a single spacecraft, in the prototype approach, as *recurring* costs; however, the convention has been established for cost models that the manufacturing costs beyond the qualification test model are generally regarded as recurring. In multiple-unit production, the first unit is often identified as the theoretical first unit (TFU or T1) cost. An example of this convention is seen in Sec. 11.2.2. On the other hand, Sec. 11.2.5 describes a single CER model for predicting the total cost of development and production of one protoflight unit. The engineer must determine the definitions and bounds of a set of CERs before using them.

† This product-oriented (versus function-oriented) WBS structure is similar to an expanded version called the cost breakdown structure (CBS); the difference is that the CBS includes additional elements for profit, cost of money, and budget reserve. For this chapter, we will use only the term WBS.



1.0 Space Vehicle—the housing or platform for carrying a payload and other mission-oriented *prime mission equipment* (PME) into space.

1.1 Spacecraft Bus—the space vehicle less the payload.

1.1.1 Structure—basic structural items including enclosures, deployable components, supporting structure, launch vehicle interface.

1.1.2 Thermal—structure and devices for maintaining temperature control; passive devices include paints, coatings and blankets, and space radiator; active devices include pumped-loop systems, heaters controlled by thermostats, mechanical devices, and refrigerators; passive systems cost less than active systems.

1.1.3 Attitude Determination and Control (ADCS)—hardware and software which stabilize and orient spacecraft using sensors and actuators; primary cost drivers are the pointing accuracy, control stability, and maneuvering requirements.

1.1.4 Electrical Power Subsystem (EPS)—hardware includes solar arrays, batteries, wiring harness, and power management electronics.

1.1.5 Propulsion—hardware and software which provides thrust for attitude control and orbit correction; orbit insertion may require an apogee kick motor (AKM).

1.1.6 Tracking, Telemetry, and Command (TT&C); Typically Includes Data Handling (DH)—consists of command/telemetry electronics, onboard computers, transponders, transmitters, receivers, data storage components, antennae, and associated electronics.

1.1.7 Integration, Assembly, and Test (IA&T)—the vehicle IA&T identifies the cost of integrating, assembling, and testing all space subsystems into an operable spacecraft bus. This does not include the costs of integrating the bus with the payload.

1.1.8 Flight Software—computer program for controlling the spacecraft.

1.2 Payload—the software and hardware associated with conducting the space mission.

1.2.1 Communications Payload—consists of hardware similar to the TT&C.

1.2.2 Surveillance Payload (sensors)—includes optics.

1.2.3 Mission Software—computer program for controlling the payload.

1.3 Integration of Bus with Payload—planning and executing the interface, including engineering, testing, and checkout; consideration of signal and power connections.

2.0 Launch Vehicle—typically a negotiated price for placing the spacecraft into orbit. The costs associated with mating the spacecraft with the launch vehicle, performing integrated system test and checkout, and tracking launch vehicle performance during ascent are (included in WBS 5.0):

3.0 Ground Command & Control—the ground hardware and software necessary for communicating between controlling and tracking facilities, monitoring the health and status of space vehicles, commanding the space vehicle's hardware, and adjusting the space vehicle's orbit.

4.0 Program Level—prime contractor and subcontractor costs for systems engineering, program management, systems integration and test, and product assurance; usually predicted as a percentage of prime mission equipment (PME) costs; normally does not include government costs or other customer oversight.

4.1 Systems Engineering (Sometimes Combined with Program Management)—the technical effort of directing and controlling an integrated engineering effort to include e.g., requirements analysis, program planning, reliability, maintainability, or human factors; excludes design engineering.

4.2 Program Management—the business and administrative planning, organizing, directing, coordinating, and controlling the program.

4.3 System Integration and Test (and Acceptance)—the application of prototype or production hardware to validate engineering performance data.

4.4 Product Assurance—the quality control function.

5.0 Flight Support Operations and Services (Sometimes Called Launch and Orbital Operations Support [LOOS])—these are contractor costs associated with prelaunch planning, trajectory analysis, launch site support, launch vehicle integration with spacecraft, and initial on-orbit operations for the first 30 days after launch.

6.0 Aerospace Ground Equipment (AGE), Ground Support Equipment (GSE), Peculiar Support Equipment (PSE)—test and support equipment for integration, assembly, and acceptance testing of spacecraft before launch; excludes common support equipment.

7.0 Operations—the on-going steady state support to operate the space mission to include mission operations, ground operations, sustaining engineering, flight hardware spares and maintenance, program/project management and integration, and communication services.

Fig. 11-1. SME-SMAD Space Project WBS Provides the Estimate Framework. Model descriptions and discrete CERs throughout this chapter will apply this WBS numbering scheme.

Table 11-2. SME-SMAD Space Program WBS and Reference to Suggested Cost Estimating Tools in SME-SMAD. The estimator typically needs to apply CERs and factors from a suite of cost prediction models because there are few models available which can estimate the entire project.

	Model/Database/Factors and where to find them (section number) in SME-SMAD									
SME-SMAD WBS Element (from Figure 11-1)	Top –down Estimating 11.1	USCM8 11.2.2	SSCM05 11.2.3	QuickCost 11.2.4	NICM 11.2.5	COCOMO 11.2.6	Ground 11.2.7	Launch 11.2.8	Factors 11.2.9	Operations 11.2.10
1.0 Spacecraft	N, R			N+R						
1.1 Spacecraft Bus		N, R	N+R							
1.1.1 Structure		N, R	N+R							
1.1.2 Thermal		N, R	N+R							
1.1.3 ADCS		N, R	N+R							
1.1.4 EPS	N	N, R	N+R							
1.1.5 Propulsion	N, R	N, R	N+R							
1.1.6 TT&C		N, R								
1.1.7 IA&T		N, R								
1.1.8 Flight Software	N					N, R				
1.2 Payload			N+R		N+R					
1.2.1 Communications		N, R								
1.2.2 Surveillance	N, R									
1.2.3 Mission Software						N, R				
1.3 Spacecraft IA&T		N, R	N+R							
2.0 Launch Vehicle								R		
3.0 Ground C&C	N, R						N+R		R	
4.0 Program Level		N, R	N+R		N+R				N, R	
5.0 Flight Support Ops		N, R	N+R					R		
6.0 AGE			N+R							
7.0 Operations										R

Notes from the Table:
N = Non-recurring CERs
R = Recurring CERs
N+R = Hybrid CERs defining both non-recurring and recurring costs in a single algorithm

The key for successful mapping of different WBS structures, and to assure configuration control for your estimate, is to establish a WBS definition for each WBS element. A top-level unmanned space WBS example is shown in Fig. 11-1, with the unique SME-SMAD WBS number system to facilitate correlation between WBS, hardware definitions, and CERs. Note that certain elements have been omitted from this example, such as: reentry vehicle, launch operations and mission support, system test, training, spares, et al. A more comprehensive guide is DoD [2005], which provides a detailed WBS with element definitions. Hardware and software estimates are generally combined under the appropriate subsystem title; in some program WBSs, software elements may be grouped separately from hardware elements.

The SME-SMAD Space Mission WBS elements are defined in Fig. 11-1. These definitions are consistent with Chaps. 3 and 4. Individual program definitions and specific CER definitions may differ based on peculiar requirements or specialized data collection; this presents an estimating challenge to the engineer to ensure the applied CER reasonably estimates the corresponding WBS element.

The space systems engineer relies upon one or more estimating tools (models, databases, factors) and then ensures these tools are compatible with the project WBS structure and its definitions. Table 11-2 provides a starting point for selecting appropriate tools described in this chapter. Apgar [2004] provides a guide to the development and review process for space missions parametric cost prediction models.

11.1.4 Phases of the Estimate

Program phases of the estimate should be defined by the program manager. The aggregate of all costs for all phases is typically identified as *life cycle cost (LCC)*, or

Table 11-3. Cost Estimating Process Steps. This is a relevant guide to the application of sequential steps described in this chapter, as appropriate to predicting future costs

Cost Estimating Process Steps	Where Discussed
1. Establish the estimate purpose, including the estimating groundrules and assumptions.	11.1.1
2. Establish the work breakdown structure (WBS) to identify elements of cost for all program phases (development, production, O&M). Determine the estimate detail and the lowest WBS level for which estimates are required.	11.1.3
3. Apply an estimating method and tool set for each lowest-level WBS element for each phase, based on availability of cost driver information. The choice of estimating method may depend upon the understanding of the cost drivers and the availability of information defining them. Products of this step are "most-likely estimates" for: • Hardware Development including prototype units. (11.2.2–11.2.5) • Software Development and Maintenance. (11.2.6) • Hardware Production of subsequent flight units, based on spacecraft first unit cost (T1) and cost improvement (learning curve) theory. (11.2.2–11.2.5, 11.3.2) • Operations and Support, for the space launch and facilities. (11.2.7–11.2.10) • Program-level elements (e.g., management, engineering), for each phase. (11.2.9) • Availability of off-the-shelf (OTS) hardware and computer software, for which purchase price may be used in lieu of the CER prediction. (11.2.1) • Operations Cost (11.2.10)	11.2
4. Establish a range of estimates around each element's "most-likely estimate" and then roll-up the element estimates to the top-level most-likely estimate with a confidence level. Consider likelihood of future cost growth. The result should be the estimate cumulative probability versus the range of top-level estimate values (an S-Curve), as generated by a Monte Carlo simulation. An alternative product would be an estimate range for each subsystem or system.	11.2.11
5. Inflate (or deflate) the estimate to reflect dollars in another base year, necessary for comparative cost trade studies and for budget purposes.	11.3.3
6. Spread the estimate of cost (allocate the dollars) by year—for the purpose of budgeting.	11.3.4
7. Document the estimate result and the estimating process to satisfy the estimate purpose. Consider credibility and affordability issues. Plan for subsequent performance/cost trade studies, affordability initiatives, and estimate updates.	11.3.1, 11.3.2, 11.3.7

sometimes *whole life cost (WLC)*. LCC includes all costs from system planning through end-of-life. The estimator needs to be sensitive to the three generic cost phases (common to DoD and NASA) comprising LCC since each phase generally depends on a unique cost prediction model. There are few models which can estimate more than one phase. The three generic phases and their associated costs are:

- **Development** costs which include design, analysis, and test of breadboards, brassboards, prototypes, and qualification units, comprising hardware and software elements. These are generally *non-recurring costs* and are determined on a program basis.
- **Production** costs which include manufacturing of hardware components and assembling them into flight units and then launching them. The estimation process is typically characterized by estimating the *theoretical first unit (TFU) cost* and then extrapolating the cost of additional units by assuming a human learning rate or cost improvement factor.
- **Operations** costs, alternatively identified as Operations and Support (O&S) costs, which include ongoing operations and maintenance, spacecraft replacement units, software updates, and ultimate system salvage. For unmanned missions, most of these costs are consumed by the ground station. These are generally recurring costs and are determined on an annual basis.

Table 11-3, Fig. 11-1◀, Eq. 11-0

11.1.5 Cost Estimating Process Steps

A recommended set of steps to execute this estimating process is shown in Table 11-3, with reference to detailed explanations elsewhere in this chapter. Variations on this process may be necessary to accommodate unique system requirements, peculiar estimating models, and availability of cost driver information. Your support to defining the low-cost and low-risk space mission depends upon your ability to accurately predict the development, production, and operations cost of any proposed space architecture. This process will organize your efforts.

11.2 Estimating Tools

11.2.1 Top-Down Estimating Process Using Models, CERs, and Factors

The parametric cost estimating model is the most common example of top-down estimating and is the preferred estimating tool for space system estimates in the pre-design phase of the program. Properly developed models, with statistically-sound CERs derived from representative and validated cost data, are expected to provide credible estimates for several decades, assuming there are no revolutionary technology changes. Even with evolutionary changes in the way certain space components are designed and built, the CERs can be adjusted if one understands the technology.

SME-SMAD provides convenient, and sometimes simplified, versions of contemporary parametric cost estimating models, all keyed to the SME-SMAD WBS described in Sec. 11.1.3 and all converted to a common parameter set (same year dollars and application of metric-system measurements). You may find same-name CERs from different models which provide slightly different estimates for the same WBS element; this is normal when the model CERs have been derived from different sets of historic data and, possibly, with different definitions (e.g., development estimates including protoflight hardware).

There are three categories of parametric models, identified below, and the user needs to assess their relative advantages in application and investment of training time.

1. **Publicly-available special purpose models** are generally developed by government agencies, federally-funded research centers, cost analysis support contractors, and public universities because of their likely access to primary (and often proprietary) source cost data, e.g. prime contractors. Such models typically are built around a complete WBS—such that the estimator can develop a full program acquisition cost estimate. The models' software and user manuals are typically available at no cost to qualified users. Examples include USCM8, NICM, and COCOMO 81 and these three models are described in this chapter (Sec. 11.2) and on the SME-SMAD website.

2. **Publicly-available general purpose models** are developed by private companies who provide software, training, and consulting services to licensed users. These users, then, are expected to calibrate these general purpose models to their specific product experience and program WBS. Model examples include PRICE and SEER models; these models are not discussed in SME-SMAD but the reader is encouraged to investigate their application to his needs.

3. **Private special purpose models** are generally developed by contractors for their own use since they are typically based on proprietary data for the purpose of generating proposal cost estimates and evaluating product design alternatives. These models are not identified or discussed in SME-SMAD.

There is no single space mission parametric cost model which constitutes such a wide range of CERs that it can provide estimates for the full life cycle WBS or even for multiple technologies within the single category of spacecraft payloads. To develop a full life cycle system estimate, one must apply multiple models and then aggregate the estimates. Refer to Sec. 11.2.11 for guidance on aggregating probabilistic estimates using a Monte Carlo approach. Although there is some statistical inaccuracy involved, the user could arithmetically sum the CER predictions of "most likely cost" (aka the point estimate) to achieve the system or mission-level "most likely cost."

The inclusion (or exclusion) of models in SME-SMAD does not imply endorsement (or lack of endorsement) of any particular model. The descriptions have been streamlined and annotated to fit the format and purpose of the SME-SMAD. Interested readers are encouraged to seek the full version of each model from the web sites or published reports identified in the references. Models in SME-SMAD were chosen primarily on their representing a class of models (hardware, software, or facilities).

The models presented here consistently predict costs either in (a) calendar year 2010 US Dollars or in (b) labor hours, which is a proxy for cost. To convert estimates to "other year" dollars, refer to Sec. 11.3.3 for the appropriate inflation adjustment factor.

We use the standard cost notations for CERs where $K means thousands of dollars and $M means millions of dollars. All size variables are in the metric (meters, kilograms) system. Some CERs include the cost of program level costs and some do not; the reader is cautioned to assure he understands a CER's definition before using it.

We have numbered individual CERs and have matched these numbers to the appropriate WBS element as previously shown in Fig. 11-1 to facilitate organizing the total mission cost estimate. This will prove useful where the model developers assigned inconsistent names to the CERs, such as aerospace ground equipment (AGE) vs. ground support equipment (GSE). Some models provide alternate CERs, with a different set of cost drivers, for the same WBS element, providing a choice to the engineer when choosing among cost driver metrics.

Good estimating practice considers model predictions to be starting points for an estimate. That is, the engineer still needs judgment to distinguish between disparate model predictions, and whether the model input metrics properly define his design, and whether the design assumptions (regarding technology, for example) match the model assumptions. Adjustments can be made, for example, for: very much or very little design legacy, quality excursions, and off-shore production. These models are based on cost data from international (not limited to US) developers and manufacturers and from US Government sources.

Cost estimating models are aggregates of CERs, which are statistically-based equations that relate cost to one or more non-cost drivers. Figure 11-2 depicts a database comparing Electrical Power Subsystem (EPS) development cost to the spacecraft bus dry weight, based on actual cost and technical data. The best fit statistical CER for this data set would be a non-linear (note the logarithmic axes) expression of the form $Y = aX^b$, where Y is cost (either dollars or labor hours) and X is weight or some other cost driver. Traditionally, weight of the spacecraft or one of its subsystems has proven to be a good predictor of cost.

Table 11-3◀, Fig. 11-1◀, Eq. 11-0

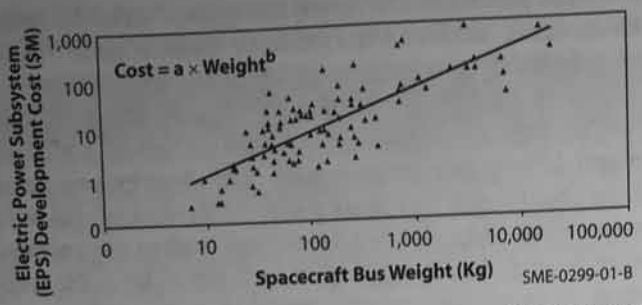


Fig. 11-2. Example of a Development Cost vs. Weight Database. This dataset provides the basis for deriving a statistical CER.

Factors can be simplified versions of CERs where the cost driver is typically a previous estimate, such as total hardware cost. Examples are shown in Table 11-4, where the amount of development effort required to complete the development can be applied as a scaling factor to the full-scale development estimate.

Table 11-4. Development-Completion Factors Useful in Development Estimates. This table provides factors for adjusting the full-development cost estimate to identify only the cost to complete the development.

Electronics Effort Previously Completed	Development Completion Factor; 1 minus this factor is then the multiplier of the full-scale estimate
Block Diagram Complete	0.10
Partitioning Complete	0.20
Schematic Design Complete	0.50
Parts Selection Complete	0.55
Breadboard Constructed	0.60
Breadboard Tested	0.70
Drawings Signed Off	0.85
Qualification Test Completed	1.00

Source: [PRICE Systems, 1999]

For example, if the total development is initially estimated to cost $1M, then the estimate to complete the development after breadboard testing could be predicted as $1M × (1–0.70) = $300K.

Factors provide the manager with a quick check on estimates and also provide engineers the basis for a rough order of magnitude (ROM) estimate before the cost drivers have been identified and quantified.

Table 11-5 provides a range of unmanned and manned spacecraft non-recurring (development) and recurring (Theoretical First Unit TFU or T1) hardware costs per kilogram of dry weight. Weight is not the only cost driver for our SME-SMAD cost models but, over time, weight has proven a reliable predictor of costs.

Table 11-6 provides typical software productivity metrics in (delivered) source lines of code (SLOC) per man-month.

Another set of spacecraft factors at the **subsystem** level is shown in Table 11-7, along with typical spacecraft bus weights for engineers who are not sure how much a typical subsystem weighs.

Table 11-5, Fig. 11-2, Eq. 11-0

Table 11-5. Development and Production Factors for Space Vehicles and Major Subsystems. These simple factors provide the basis for top-level rough-order-magnitude (ROM) estimates for SME-SMAD WBS elements.

SME-SMAD WBS Element	Develop. Cost (in FY2010$K) per kg	T1 (TFU) (in FY2010 $K per kg)
WBS 1.0 Earth Orbital Space Vehicle (typical)	311	71
• WBS 1.0—Communication Space Vehicle	429	90
• WBS 1.0—Mapping/ Meteorological	277	52
• WBS 1.0—Observatory	129	33
• WBS 1.0—Positioning/ Navigation	475	55
• WBS 1.0—Reconnaissance	402	82
• WBS 1.0—Science	238	63
WBS 1.0 —Planetary Space Vehicle (typical)	1,167	161
• WBS 1.0—Inner Planetary	1,022	84
• WBS 1.0—Outer Planetary	1,056	171
• WBS 1.0—Planetary Lander	2,390	317
• WBS 1.0—Probe	550	69
• WBS 1.0—Crewed Space Vehicle	698	55
WBS 1.1.5—Engines	1,500	33
WBS 1.1.5—Liquid Rocket Stage	176	12
WBS 1.1.5—Solid Rocket	156	8

For example, the predicted non-recurring development cost for a small communications spacecraft, weighing 175 kg, should be 429 × 175 = $75M (in 2010 dollars), while the recurring T1 production cost should be 90 × 175 = $16M (in FY2010 dollars).
All factors assume space vehicle or rocket engine dry mass (no fuel).

11.2.2 Unmanned Space Vehicle Cost Model (USCM8)

The Unmanned Space Vehicle Cost Model, version 8, (USCM8) was developed by Tecolote Research for the US Air Force, Space and Missile Systems Center (SMC/FMP), [Tecolote Research, 2002]. The CERs were derived, using statistical regression techniques, from 44 satellites (23 military, 12 NASA, and 9 commercial) to support parametric cost estimates of unmanned, earth-orbiting space vehicles with a communications (mission equipment) payload. The CERs were derived to represent minimum unbiased percentage errors (MUPE).

Each CER is consistent with engineering expectations. This chapter provides only the subsystem CERs; component CERs are also available from the model's website. An update to this model (version 9) is scheduled for release in 2011.

In addition to CERs for large spacecraft development (non-recurring cost) and T1 production (recurring cost), the model provides a set of non-recurring and recurring CERs for estimating aerospace ground equipment (AGE), launch operations & orbital support (LOOS) and a communications (mission equipment) payload. CERs for other payloads (navigation or sensor payloads, for

Table 11-6. Productivity Factors for Software Development Cost in Man-months (MM). Note the references to the SME-SMAD WBS; this table must next be converted to dollars using applicable labor rates.

SME-SMAD WBS Element	Avg. Prod. in Source Lines of Code (SLOC) per man-month (MM) (a)	Typical Application
WBS 1.1. Flight Software	111 SLOC/MM	For controlling spacecraft
WBS 3.0 Ground Software	340 SLOC/MM	For ground operations
WBS 3.0 Instrument Operations	362 SLOC/MM	For controlling sensors
WBS 3.0 Image Processing	161 SLOC/MM	Highly algorithmic

(a) Based on 152 hours/MM and includes requirements analysis, design, coding, integration and test, QA, CM, and task management.

For example, the predicted development of 10,000 SLOC of flight software (WBS 1.1.8) should require 10,000/111 or 90 man-months of labor. This can be converted to 13,770 man-hours of labor, considering the typical labor month has 152 billable hours.

Table 11-7. Estimating Factors for Spacecraft Subsystems. Using this table, in FY2010 dollars, the predicted T1 cost for a small spacecraft electrical power system weighing 175 kg is: 25.5 × 175 = $4,462 K (in 2010 dollars)

SME-SMAD WBS Element (Component)	T1 (TFU) Prod. Cost in FY2010 $K/kg	Typical Comp. Weight in Kilograms (kg)	Typical T1 (TFU) Prod. Cost in FY2010 $K
WBS 1.0 Spacecraft Bus			
1.1.1 Structure	22.8	205	4,674
1.1.2 Thermal Control	22.8	40	912
1.1.3 ADCS	165.9	65	10,784
1.1.4 EPS	25.5	251	6,400
1.1.5 Propulsion	44.1	75	3,308
1.1.6 TT&C	82.5	45	3,712
1.1.6 C&DH	97.8	46	4,500

example) can be obtained from payload-specific models, as described in Sec. 11.2.5. However, there is some additional risk in aggregating estimates from multiple models, so assure the groundrules and assumptions are consistent.

For this model, non-recurring cost CERs predict the cost of: design and development, manufacturing and test of one space vehicle qualification model, and acquisition of program peculiar support equipment. Recurring cost CERs predict the cost of: fabrication, manufacturing, integration, assembly, and test of the space vehicle flight hardware. Recurring cost CERs also include efforts associated with the LOOS.

The model estimates are for contractor and subcontractor costs including fee (cost to the government) but does not include government program office costs. We will need to determine the design values for subsystem weight and other input parameters.

We can arithmetically sum the most likely costs from each element, although a more precise estimate will be obtained by aggregating the cost ranges from each element using a Monte Carlo simulation, as explained in Sec. 11.2.11.

The subsystem non-recurring CERs are presented in Table 11-8. These non-recurring costs are presumed to include one qualification (prototype) unit plus a typical amount of software development. Many estimators, however, prefer to estimate software development and maintenance costs independently from hardware, as described in Sec. 11.2.6. These development cost estimates assume a new design and not a modified design. If there is substantial design legacy, and the effort is more of a modification than a new design, the estimate needs to be reduced, probably by the amount of legacy or development effort which need not be repeated. For a starting point, consider the design completion adjustment factors previously presented in Table 11-4.

The cost driver input range, as used by the CER developer, appears in the table. The estimator is cautioned when using a cost driver value outside this range (more than 25% above or below) without considering scope and technology differences. For example, a significantly lighter component may cost *more* per pound due to more expensive materials or additional labor to reduce its weight, thereby altering the normal cost per pound relationship.

The Standard Error of the Estimate (SEE) of the resulting CER prediction, also shown in the table, is based on a statistical analysis of the actual data and the resulting CER (SEE) may be expressed as a percent or absolute value. For each MUPE regression exercise, the model's SEE (in %) is calculated as a measure of how well the CER function fits the data. The SEE is the root-mean square of the percentage error residuals about the regression line, corrected for degrees of freedom and is calculated as shown here in Eq. (11-1):

$$SEE_{\%} = \sqrt{\frac{1}{n-p}\sum_{i-1}^{n}\left[\frac{y_i - f(x_i)}{f(x_i)}\right]^2} \qquad (11\text{-}1)$$

where n = number of pairs of data points and p = number of parameters in CER, and x_i and y_i are the independent and dependent variables (cost value) for the ith data point, respectively.

The tables include program level (administrative, management, and systems engineering) CERs based on "wrap factors," which are percentage multipliers of the non-recurring or recurring costs previously estimated. These costs are not included in the hardware prime mission equipment (PME), CERs and, therefore, must be calculated separately and added to the aggregate total.

USCM8 makes a distinction for recurring CERs between standard and non-standard (unique) spacecraft bus costs. The standard bus distinction refers to an existing bus which can be produced directly on an existing

Table 11-8. USCM8 Non-recurring Subsystem CERs in FY2010 Thousand s of Dollars. These CERs predict the cost of development plus one qualification unit. See Tables 11-34 and 11-37 for application of these CERs.

SME-SMAD WBS Element (Non-recurring subsystem)	CER Y = non-recurring cost in FY2010 thousands of dollars for development plus one qualification unit.	Cost Driver(s)	Cost Driver Input Range	SEE
1.1 Spacecraft				
1.1 Spacecraft Bus (alternate CER when no component information is available)	$Y = 108\ X1$	X1 = Spacecraft Weight (kg)	114–5,127 kg	47%
1.1.1/1.1.2 Structure and Thermal Control	$Y = 646\ X1^{0.684}$	X1 = Structure + Thermal Weight (kg)	59–501 kg	22%
1.1.3 Attitude Determination & Control System (ADCS)	$Y = 324\ X1$	X1 = ADCS Weight (kg)	35–524 kg	44%
1.1.4 Electrical Power System (EPS)	$Y = 64.3\ X1$	X1 = EPS Weight (kg)	47–1,065 kg	41%
1.1.5 Propulsion (Reaction Control)	$Y = 20.0\ X1^{0.485}$	X1 = Total RCS tank volume (cubic centimeters)	Not given	35%
1.1.6 Telemetry, Tracking, & Command (TT&C)	$Y = 26{,}916$	Y = Average TT&C Cost (since there is no statistical CER for this element)	*CER* based on S-Band telemetry	Not given
1.2 Payload				
1.2 Communications Payload (based on weight and number of channels)	$Y = 339\ X1 + 5{,}127\ X2$	X1 = Communications Subsystem Weight (kg) X2 = Number of Communication Channels	160–395 kg 2–32 channels	40%
1.2 Communications Payload (alternate CER based on weight alone)	$Y = 618\ X1$	X1 = Communications Subsystem Weight (kg)	160–395 kg	38%
1.3 Spacecraft Integration, Assembly, and Test				
1.3 Integration, Assembly, & Test (of bus and payload into spacecraft)	$Y = 0.195\ X1$	X1 = Spacecraft Bus + Payload Non-recurring Cost ($K)	3,600–545,000 $K	42%
4.0 Program Level				
4.0 Program Level (for a Communications Satellite)	$Y = 0.236\ X1$	X1 = Space Vehicle and IA&T Non-recurring Cost ($K)	7,850–353,804 $K	23%
4.0 Program Level (for an other than Communications Satellite)	$Y = 0.357\ X1$	X1 = Space Vehicle and IA&T Non-recurring Cost ($K)	7,850–353,804 $K	50%
6.0 Aerospace Ground Equipment (AGE)				
6.0 Aerospace Ground Equipment (AGE)	$Y = 0.432\ X1^{0.907} \times 2.244^{X2}$	X1 = Spacecraft Bus Non-recurring Cost ($K); X2 = 0 for comm sats and X2 = 1 for non-comm sats	7,850–353,804 $K	37%

product line without extensive engineering development. The non-standard (unique) bus (applied in this chapter) refers to a bus requiring substantial development, or that requires new production tooling, or for which a product line does not exist. Both versions of CERs appear on the USCM8 website. There is no similar distinction for non-recurring CERs.

The subsystem recurring production T1 or TFU (theoretical first unit cost) CERs are presented in Table 11-9. Estimates using these CERs may be assumed to be the delivered cost for the first flight unit after the test validation model. The cost of each additional similar unit may be extrapolated using learning curve theory, as explained in Sec. 11.3.2.

Table 11-10 shows estimating examples as a step-by-step calculation of several subsystem CERs from Tables 11-8 and 11-9. Input parameter values (cost drivers) are assumed for each example and are based on a space vehicle design within the scope of the database. For this example, we assume a 95% cumulative average learning curve to be consistent with the derivation of the USCM8 CERs. We will assume a space vehicle flight unit quantity of five; therefore, the average unit cost is $0.89 \times T1$ based on a factor from Table 11-30. All estimates are in 2010 dollars.

11.2.3 Small Satellite Cost Model (SSCM)

Over the past two decades, the level of functionality possible in small spacecraft increased dramatically due to the availability of space-compatible computational power. Smaller spacecraft are those weighing less than 500 kg, whereas larger spacecraft (better estimated with USCM8 CERs) typically weigh several thousand kg. Smaller spacecraft may be less expensive per kg than

Table 11-9. USCM8 Spacecraft Bus Recurring T1 CERs in FY2010 Thousands of Dollars. These CERs predict the manufacturing cost of the first flight unit. See Tables 11-34 and 11-37 for application of these CERs.

SME-SMAD WBS Element (Recurring subsystem T1)	CER Y = Recurring T1 cost in FY2010 thousands of dollars	Cost Drivers	Cost Driver Input Range	SEE
1.1 Spacecraft				
1.1 Spacecraft Bus (alternate CER when no component information is available)	$Y = 283.5\ X1^{0.716}$	X1 = Spacecraft Weight (kg)	288–7,398 kg	21%
1.1.1/1.1.2 Structure and Thermal Control (a)	$Y = 22.6\ X1$	X1 = Structure + Thermal Weight (kg)	59–501 kg	21%
1.1.3 Attitude Determination & Control System (ADCS)	$Y = 795\ X1^{0.593}$	X1 = ADCS Weight (kg)	27–524 kg	36%
1.1.4 Electrical Power Supply (EPS)	$Y = 32.4\ X1$	X1 = EPS Weight (kg)	111–1,479 kg	31%
1.1.5 Propulsion Apogee Kick Motor (AKM)	$Y = 29\ X1 + 0.024\ X2$	X1 = AKM Weight (kg) X2 = Burn-time (seconds)	81–966 kg	22%
1.1.6 Telemetry, Tracking, & Command (TT&C)	$Y = 883.7\ X1^{0.491} \times 1.13^{X2}$	X1 = TT&C weight (kg) X2 = Geosynchronous Transfer Orbit (1 = yes; 0 = no)	12–76 kg for S-band	18%
1.2 Payload				
1.2 Communications Payload	$Y = 189\ X1$	X1 = Communications Payload Weight (kg)	38–928 kg	39%
1.3 Spacecraft Integration, Assembly, and Test				
1.3 Integration, Assembly, & Test (IA&T) of bus and payload into space vehicle	$Y = 0.124\ X1$	X1 = Spacecraft Bus + Payload Recurring Cost ($K)	35,367–142,044 $K	34%
4.0 Program Level				
4.0 Program Level (for a Communication Satellite)	$Y = 0.234\ X1$	X1 = Space Vehicle (Spacecraft Bus + Payload + IA&T) Recurring Cost ($K)	13,287–268,225 $K	12%
4. Program Level (for an other than communication satellite)	$Y = 0.320\ X1$	X1 = Spacecraft (Spacecraft Bus + Payload + IA&T) Recurring Cost ($K)	13,287–268,225 $K	40%
5.0 Flight Support				
5.0 Launch Operations & Orbital Support (LOOS)	Y = 5,850	Y= Average LOOS cost in $K	Not given	Not given

(a) Current version of USCM8 provides a combined structure/thermal CER.

larger spacecraft due to shortened development times by using standard designs, lower launch costs by using smaller launch vehicles, and cheaper production cost due to smaller size if their component complexities are similar. The Aerospace Corporation [1996], developed a parametric cost model for predicting development and first unit cost for smaller Earth-orbiting and near-planetary spacecraft. SSCM does not distinguish between non-recurring and recurring cost, as is done in USCM8, and so these CERs predict the aggregate of non-recurring and recurring costs. SSCM CERs are based on 53 individual satellites, of which most are smaller than 100 kg. These CERs represent MUPE and are presented in Table 11-11. This is not the latest version of SSCM CERs (since that version of the model is not available for public release); these are the same CERs in Wertz [1999]. Government agencies may request release of the latest version; for more information, go to Aerospace Corporation's website.

Another model suitable for estimating small spacecraft, although not described in SME-SMAD, is the NASA—Air Force Cost Model (NAFCOM) [SAIC, 2010].

In Eq. (11-2), we provide a single example, similar to multiple USCM8 examples, for the aggregate of development and one protoflight unit in 2010 dollars for the thermal control system.

11.2.4 QuickCost

A more simple approach to predicting cost for an unmanned robotic space vehicle was developed for NASA projects in 2010 from historic cost data. The single-CER QuickCost model, version 4.2, was created by Hamaker [2011], the former Director of the Hq NASA

Table 11-10. USCM8 Spacecraft Bus Non-recurring and Recurring Estimate Examples, Keyed to SME-SMAD WBS Elements and Estimating Examples.

1.1.1/1.1.2 STRUCTURE/THERMAL SUBSYSTEM

Non-recurring $Y = 646\ X1^{0.684}$

where

X1 = Structure + Thermal Weight = 175 kg

$Y = 646 \times 175^{0.684}$

Y = $22,104 K

Recurring T1 $Y = 22.6\ X1$

where

X1 = Structure Weight + Thermal Weight = 175 kg

$Y = 22.6 \times 175$

Y = $3,955 K

The average unit cost for a lot of five units = 0.89 × 3995 = $3,556 K

1.1.3 ATTITUDE DETERMINATION AND CONTROL SUBSYSTEM

Non-recurring $Y = 324\ X1$

where

X1 = ADCS Weight = 280 kg

$Y = 324 \times 280$

Y = $90,720 K

Recurring T1 $Y = 795\ X1^{0.593}$

where

X1 = ADCS Weight = 280 kg

$Y = 795 \times 280^{0.593}$

Y = $22,466 K

The average unit cost for a lot of five units = 0.89 × 22,462 = $19,991 K

1.1.4 ELECTRICAL POWER SUBSYSTEM

Non-recurring $Y = 64.3\ X1$

where

X1 = EPS Weight = 556 kg

$Y = 64.3 \times 556$

Y = $35,751 K

Recurring T1 $Y = 32.4\ X1$

where

X1 = EPS Weight = 556 kg

$Y = 32.4 \times 556$

Y = $18,014 K

The average unit cost for a lot of five units = 0.89 × 18,014 = $16,032 K

1.1.6 TELEMETRY, TRACKING AND COMMAND SUBSYSTEM

Non-recurring Y = $26,916 K

Recurring T1 $Y = 883.7\ X1^{0.491} \times 1.13^{X2}$

where

X1 = TT&C Weight = 109 kg

X2 = Orbit (1 = Geo Orbit, 0 = Non-Geo Orbit)

$Y = 883.7 \times 109^{0.491} \times 1.13^{1}$

Y = $9,993 K

The average unit cost for a lot of five units = 0.89 × 9,967 = $8,871 K

1.2 COMMUNICATIONS PAYLOAD

Non-recurring $Y = (339\ X1) + (5{,}127\ X2)$

where

X1 = Communications Payload Weight = 226 kg

X2 = Number of Channels = 17

$Y = (339 \times 226) + (5{,}127 \times 17)$

$Y = 76{,}614 + 87{,}159$

Y = $163,773 K

Recurring T1 $Y = 189\ X1$

where

X1 = Communications Payload Weight = 226 kg

$Y = 189 \times 226$

Y = $42,714 K

The average unit cost for a lot of five units = 0.89 × 42,714 = $38,015 K

1.3 INTEGRATION, ASSEMBLY, AND TEST (IA&T)

Non-recurring $Y = 0.195\ X1$

where

X1 = Spacecraft Total Non-recurring Cost = $196,600 K

$Y = 0.195 \times 196{,}600$

Y = $38,317 K

Recurring T1 $Y = 0.124\ X1$

where

X1 = Space Vehicle First Unit Cost = $127,650 K

$Y = 0.124 \times 127{,}650$

Y = $15,829 K

The average unit cost for a lot of five units = 0.89 × 15,829 = $14,088 K

4.0 PROGRAM LEVEL for Communications Satellite

Non-recurring $Y = 0.236\ X1$

where

X1 = Space Vehicle Total Non-recurring Cost = $398,900 K

$Y = 0.236 \times 398{,}900$

Y = $94,140 K

Recurring T1 $Y = 0.234\ X1$

where

X1 = Space Vehicle Total Recurring Cost = $272,600 K

$Y = 0.234 \times 272{,}600$

Y = $63,788 K

5.0 LAUNCH OIPERATIONS & ORBITAL SUPPORT (LOOS) for Communications Satellite

Recurring T1 Y = $5,850 K

6.0 AEROSPACE GROUND EQUIPMENT (AGE) for Communications Satellite

Non-recurring $Y = 0.421\ X1^{0.907} \times 2.244^{X2}$

where

X1 = Spacecraft Total Non-recurring Cost = $196,600

X2 = Mission Type (0 = Comm Sat, 1 = Non-Comm Sat)

$Y = 0.421 \times (196{,}600^{0.907}) \times (2.244^{0})$

$Y = 0.421 \times 63{,}282 \times 1$

Y = $26,642 K

Table 11-11. SSCM Earth Orbiting Total Non-recurring Cost (development plus one protoflight unit) CERs in FY2010 Thousands of Dollars. Useful for spacecraft weighing less than 500 kg. It is presumed that these CERs include the cost of contractor program management, systems engineering, product assurance, and I&T. See Tables 11-35 and 11-38 for application of these CERs.

SME-SMAD WBS Element	CER Y = total non-recurring cost of development plus one protoflight flight unit in FY10 $K	Cost Driver(s)	Cost Driver Input Range	Standard Error of Estimate (absolute) FY10 $
1.1 Spacecraft				
1.1 Spacecraft Bus (alternate CER when no component information is available)	$Y = 1{,}064 + 35.5\,X^{1.261}$	X = Spacecraft Bus Dry Weight (kg)	20–400 kg	3,696
1.1.1 Structure	$Y = 407 + 19.3\,X \times \ln(X)$	X = Structure Weight (kg)	5–100 kg	1,097
1.1.2 Thermal Control	$Y = 335 + 5.7\,X^2$	X = Thermal Control Weight (kg)	5–12 kg	119
1.1.3 Attitude Determination & Control System (ADCS)	$Y = 1{,}850 + 11.7\,X^2$	X = ADCS Dry Weight (kg)	1–25 kg	1,113
1.1.4 Electrical Power Supply (EPS)	$Y = 1{,}261 + 539\,X^{0.72}$	X = EPS Weight (kg)	7–70 kg	910
1.1.5 Propulsion (Reaction Control)	$Y = 89 + 3.0\,X^{1.261}$	X = Spacecraft Bus Dry Weight (kg)	20–400 kg	310
1.1.6a Telemetry, Tracking, & Command (TT&C)	$Y = 486 + 55.5\,X^{1.35}$	X = TT&C Weight (kg)	3–30 kg	629
1.1.6b Command & Data Handling (CD&H)	$Y = 658 + 75\,X^{1.35}$	X = Command & Data Handling Weight (kg)	3–30 kg	854
1.2 Payload				
1.2 Payload	$Y = 0.4\,X$	X = Spacecraft Bus Total Cost ($K)	2,600–69,000 ($K)	
1.3 Spacecraft Integration, Assembly, and Test				
1.3 Integration, Assembly, & Test (IA&T)	$Y = 0.139\,X$	X = Spacecraft Bus Total Cost ($K)	2,600–69,000 ($K)	
4.0 Program Level				
4.0 Program Level	$Y = 0.229\,X$	X = Spacecraft Bus Total Cost ($K)	2,600–69,000 ($K)	
5.0 Flight Support				
5.0 Launch & Orbital Operations Support (LOOS)	$Y = 0.061\,X$	X = Spacecraft Bus Total Cost ($K)	2,600–69,000 ($K)	
6.0 Aerospace Ground Equipment				
6.0 Ground Support Equipment (GSE)	$Y = 0.066\,X$	X = Spacecraft Bus Total Cost ($K)	2,600–69,000 ($K)	

1.1.2 THERMAL CONTROL SUBSYSTEM

Non-recurring $Y = 335 + 5.7X^2$

where

X1 = Thermal Control Weight = 12 kg

$Y = 335 + 5.7 \times 12^2$

$Y = \$1{,}156\ K$ (11-2)

Cost Analysis Division, for budgetary exercises and top-level trade studies. The CER is based on a relatively large number of cost drivers, which are likely to be available to system planners and systems engineers early in the design process.

As shown in Table 11-12, the CER predicts, in 2010 dollars, full program cost (contractor plus government) for development plus the first flight unit. See notes below the table for full description of the costs predicted and for some useful factors which can be applied to other models.

An example of using QuickCost to predict the cost of an automated scientific satellite is shown in Table 11-13. The shaded column recommends an appropriate cost driver value.

The table of the QuickCost input and output includes a cell giving the regression model result of $268 million in 2010 dollars (which is the median cost or 50% confidence level).

11.2.5 *NASA Instrument Cost Model (NICM)*

The NASA Instrument Cost Model (NICM), Version IIIC, was developed by the Jet Propulsion Laboratory in 2010 from 159 data points obtained from instrument contractors [Habib-agahi, 2010]. The model predicts the cost of development plus one flight unit of each of five payload instrument types. The five alternative subsystem-level CERs are shown in Table 11-14. The costs include contractor fee. Select one of these CERs for the payload subsystem estimate, depending on the technology (e.g.,

Table 11-12. QuickCost Non-recurring plus Recurring (T1) CER in FY2010 Dollars. Note the reference to the SME-SMAD WBS. See Tables 11-13, 11-36, and 11-39 for application of these CERs.

SME-SMAD WBS Element	CER and Cost Drivers Y = total cost of development plus one protoflight flight unit in 2010 dollars		Cost Driver Data Range	Standard Error of Estimate (percentage)
1.0 Space Vehicle for Unmanned Robotic Mission	$Y = 2.829 \times (\text{Dry Mass}^{0.457}) \times (\text{Power}^{0.157}) \times (2.718^{(0.171 \times \text{Data \%})}) \times (2.718^{(0.00209 \times \text{Life})}) \times (2.718^{(1.52 \times \text{New})}) \times (2.718^{(0.258 \times \text{Planetary})}) \times 1/(2.718^{(0.0145 \times (\text{Year}-1960))}) \times (2.718^{(0.467 \times \text{InstrComp\%})}) \times 1/(2.718^{(0.237 \times \text{Team})})$			41%
	Dry Mass	Dry mass of spacecraft bus and instruments in kg	76 to 14,475 kg	
	Power	LEO equivalent beginning of life power (BOLP) in watts	90 to 10,000 Watts	
	Data%	Data Rate Percentile (fraction) relative to the state-of-the-art at Authority to proceed (ATP) (enter 0.5 for median data rate missions, <0.5 for lower data rates, >0.5 for higher data rates)	0 to 100%	
	Life	Advertised Design Life (in months)—advertised design life excluding extended operations	6 to 180 months	
	New	Percent New (fraction): 0.2-0.3 = Simple Mod, 0.3–0.7 Extensive Mod, 0.7–1.0=New, >1.0 for New Technology	28% to 130%	
	Planetary? (0 =no, 1=yes)	0 for earth orbital, 1 for planetary	0 and 1	
	Year	ATP date in 4 digit calendar year minus 1960	1961 to 2005	
	InstrComp%	Instrument complexity percentile (fraction)—relative to "average" instrument complexity (enter 0.5 for median complexity, <0.5 for lower complexity >0.5 for higher complexity)	0% to 100%	
	Team	Team Experience: 1 = Unfamiliar, 2 = Mixed, 3 = Normal, 4 = Extensive	1 to 4	

Notes to the Table: 1. CER provides cost in millions of 2010 dollars including Phase B through Phase C/D (NASA program phases). 2. Cost includes all contractor cost through fee (10%) and NASA full cost support (~17%). 3. Cost includes spacecraft bus and instruments DDT&E plus TFU (first flight unit). 4. Includes all flight and ground software cost. 5. Cost excludes launch services (add $150M to $250M typical if EELV Delta IV or Atlas V variants used). 6. Add 2% for Phase A if desired (rough rule of thumb). 7. Add 9% for Ground Station if desired (e.g., rough rule of thumb for the cost of retrofitting existing facility, terminals, software). 8. Add 5% per year for each year of Mission Operations and Data Analysis (MO&DA) (rough rule of thumb).

Table 11-13. Example of Using QuickCost Model with Typical Input Parameters and Risk-Adjusted Predictions.

Cost Driver:	Example Value	Cost Driver Range
Dry Mass (kg)	664	kg mass of total spacecraft bus and instruments
Power (watts)	653	In LEO equivalent beginning of life watts
Data Rate Percentile (fraction)	0.50	Relative to the state-of-the-art at ATP (enter 0.5 for median data rate missions, <0.5 for lower data rates, >0.5 for higher data rates)
Design Life (months)	36	Advertised design life excluding extended operations
Percent New (fraction)	0.68	0.2–0.3=Simple Mod, 0.3–0.7=Extensive Mod, 0.7–1.0=New, >1.0 for New Technology
Planetary? (0=no, 1=yes)	0	0 for Earth orbital, 1 for planetary
ATP Date Year	1987	Authority to Proceed date in 4 digit Calendar Year
Instrument Complexity Percentile (fraction)	0.50	Relative to "average" instrument complexity (enter 0.5 for median complexity, <0.5 for lower complexity, >0.5 for higher complexity
Team Experience	2.0	1=Unfamiliar, 2=Mixed, 3=Normal, 4=Extensive
Year Dollar of Output	2010	Calendar Year entered as 4 digits (20XX)
Regression Model Result	$268	DDT&E + TFU (Phases B/C/D) in Millions Dollars including fee, including full cost for full cost years

planetary optical, Earth orbiting optical, microwave, particles, or fields).

NICM is where the CER predicts the development and single-unit fabrication without management and systems engineering "wrap" factors. These additional costs can be calculated from the Wrap CERs in Table 11-15. It is likely that these same wrap factors could also be used to predict program-level costs associated with other models.

Example of using NICM CERs:

- 1.2.2 Optical planetary instrument payload:
  - Sensor Cost = $Y = 328 \times M^{0.426} \times P^{0.414} \times DL^{0.375}$
  - Where M = instrument mass = 20 kg
  - P = power = 30 Watts
  - And, DL = Design Life = 24 months
  - $Y = 328 \times 20^{0.426} \times 30^{0.414} \times 24^{0.375}$
  - $Y = 328 \times 3.58 \times 4.09 \times 3.29 = 15{,}685$
  - Therefore, estimated PME cost = \$15,685K in FY10\$.
- 4.1 Management Cost (Wrap Factor)
  - $Y = 0.07124 \times S^{1.0317} = 0.07124 \times 15{,}685^{1.0317} = 1{,}518$
  - Therefore, estimated management cost = \$1,518K in FY10\$.

Similarly we can calculate the estimated cost for the other wrap factors (systems engineering, product assurance, and I&T).

Live Calc

Table 11-14. NICM Non-recurring Hardware Development Plus One Protoflight Unit Cost for Remote Sensing Instrument Payload in FY2010 Thousands of Dollars. These are alternative instrument (non-communications) payloads; select one CER from the table. See Table 11-34 (item 1.2) for application of CER 1.2.2.

SMAD WBS Element	CER Y in FY2010 thousands of dollars for development plus production of one protoflight unit	Cost Drivers [plus Data Range]
1.2.2 Optical Planetary (for instruments visiting planets other than Earth) Payload (e.g. cameras, spectrometers, interferometers)	$Y = 328 \times M^{0.426} \times P^{0.414} \times DL^{0.375}$ $R^2 = 0.76$ SEE = 39%	M = Instrument Total Mass in kilograms (kg). [1–75 kg] P = Maximum Instrument Power in watts (W). [1–75 W] DL = Design Life in months [10–150 months]
1.2.2 Optical Earth-Orbiting (instruments on spacecraft in geocentric orbits) Payload (e.g., cameras, spectrometers, interferometers)	$Y = 1{,}163 \times M^{0.328} \times P^{0.357} \times DR^{0.092}$ $R^2 = 0.89$ SEE = 35%	M = Instrument Total Mass in kilograms (kg). [10–350 kg] P = Maximum Instrument Power in watts (W) [0. 5–400 W] DR = Total Data Rate in kilo bits per second (kbps). [0.1–30,000 kpbs]
1.2.2 Active and Passive Microwave Payload (e.g., radars, altimeters, scatter meters,sounders, GPS receivers)	$Y = 23{,}620 \times M^{0.284} \times P^{0.325} \times DR^{0.090} \times T^{-1.296}$ $R^2 = 0.88$ SEE = 37%	M = Instrument Total Mass in kilograms (kg). [10–50 kg] P = Maximum Instrument Power in watts (W). [10–600 W] DR = Total Data Rate in kilo bits per second (kbps). [0.1–1,000 kbps] T = Instrument Technology Readiness Level (TRL) [4–9]
1.2.2 Particles Payload (e.g., plasma detectors, plasma wave detectors)	$Y = 980 \times M^{0.327} \times P^{0.525} \times DL^{0.171}$ $R^2 = 0.65$ SEE = 29%	M = Instrument Total Mass in kilograms (kg). [1–40 kg] P = Maximum Instrument Power in watts (W). [1–40 W] DL = Design Life in months. [10–150 months]
1.2.2 Fields Payload (e.g., electric field detectors, magnetic field detectors)	$Y = 1{,}130 \times M^{0.184} \times P^{0.238} \times DL^{0.274}$ $R^2 = 0.87$ SEE = 28%	M = Instrument Total Mass in kilograms (kg). [0.1–35 kg] P = Maximum Instrument Power in watts (W). [0.1–25 W] DL = Design Life in months. [1–100 months]

Table 11-15. Estimate Wrap Factors for NICM in Same Currency as Hardware Cost Estimate from Table 11-14. Note the references to the SME-SMAD WBS.

SMAD WBS Element	CER Y in FY2010 thousands of dollars for program-level cost elements	Cost Drivers
4.2 Management	$Y = 0.07124 \times S^{1.0317}$ $R^2 = 0.85$.	S = Sensor Cost from CER in Table 11-14 (WBS 1.2.2)
4.1 Systems Engineering	$Y = 0.4931 \times S^{0.8645}$ $R^2 = 0.75$	S = Sensor Cost from CER in Table 11-14 (WBS 1.2.2)
4.4 Product Assurance	$Y = 0.1427 \times S^{0.9422}$ $R^2 = 0.91$	S = Sensor Cost from CER in Table 11-14 (WBS 1.2.2)
4.3 Integration and Test (I&T)	$Y = 0.1457 \times S^{1.0071}$ $R^2 = 0.87$	S = Sensor Cost from CER in Table 11-14 (WBS 1.2.2)

11.2.6 Software Cost Estimating Model—COCOMO 81

Since software development costs have increasingly become a percentage of lifetime space mission costs, it is no longer acceptable to predict them as a percent of hardware costs. To satisfy the need of the estimator to accurately predict software costs, unique software cost estimating models have been developed. The Cost Construction Model (COCOMO81) was first described in Boehm [1981]. The model was originally based on 63 completed software development programs. A major refinement to the model, identified as COCOMO II, was subsequently described in a sequel book [Boehm, 2000]. These and other software estimating tools are maintained by the University of Southern California (USC) Center for Systems and Software Engineering. More information, including software versions of the suite of COCOMO models, can be found on the USC website.

The software estimating sub-process is shown in Fig. 11-3 and described in the following sections.

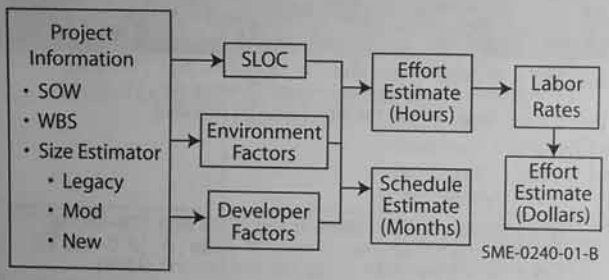


Fig. 11-3. Software Estimating Process.

This chapter describes COCOMO 81,* presented as a set of effort and schedule CERs and factors, and usable on a hand-held calculator. These CERs support the prediction of the most likely (point estimate) effort (cost) and time (schedule) required for a software product developed under the traditional waterfall development process. Other software models (but with more complex algorithms) emulate other development processes, such as the agile or incremental process.

Effort Estimate

Software development effort for flight software and for mission software subsystems, as well as software for each hardware component, in man-months, can be predicted with Eq. (11-3). Note that the result in man-months must then be multiplied by the applicable monthly labor rate to establish the year-currency estimate cost.

To use this equation to predict the development effort, you must first predict the number of SLOC and assign a rating to each of the Effort Adjustment Factors (EAFs).

BASIC COCOMO 81 Development Effort (Man Months) Equation

SME-SMAD WBS 1.1.8 and 1.2.3

$$\textbf{Effort (MM)} = 3.312\,(\textbf{KSLOC})^{1.2} \times \pi\textbf{EAF} \quad (11\text{-}3)$$

Where:

Effort = cost in man-months of full development cycle, including requirements validation, architecture design, detail design, coding, and testing (then transformed to monthly currency cost when multiplied by monthly labor rate).

KSLOC = thousands of delivered source lines of code (SLOC)—note that the original model reference identified this cost driver by a similar parameter: delivered source instructions (DSI). This parameter can represent (a) all new code or (b) a mix of new and reused code with a weighted metric called "equivalent SLOC." (ESLOC).

πEAF = cumulative effort adjsutment factor (EAF_1 x EAF_2 x EAF_3 x ... EAF_n). (See Table 11-17 for defining EAF)

The primary cost driver in Eq. (11-3) is software **size**, measured in thousands of delivered source lines of code (SLOC). A simple definition of SLOC is: a software statement that must be ***designed***, ***coded***, and ***tested.*** When counting SLOC, we should consider all programmed instructions including those contributed by pre-processors, compilers, and assemblers. This includes job control language, format statements, and data declarations but excluding comment lines, blank lines, unmodified utility software, and non-delivered support software such as test drivers.

There are alternate ways to characterize software size, such as the number of function points (FP), operating points (OP), and number of requirements. Using these would require different CERs.

The first challenge, then, is to establish the SLOC metric. During the mission design phase, this size metric is typically not known with any degree of certainty. Several approaches will work:

- The most direct approach is based on analogy to similar software development programs, considering software function, application, and quality. Table 11-16 displays some typical space software SLOC values from other programs, but we need to be careful about using such information from someone else's database. Note the calculation for equivalent SLOC (ESLOC), as explained later.
- Expert opinion is useful if more than one expert can be engaged.
- A structured Delphi approach, using multiple experts and possibly a pair-wise comparison.

Note that the software effort exponent in Eq. (11-3) is greater than unity, thereby driving this non-linear relationship to increasing man-months faster than the software module size increases and displaying a cost per SLOC increasing with size. Contrast this behavior to

* We chose to represent available software models with this one example, COCOMO 81, because it is time-tested, easy to apply with a hand-held calculator, and requires a smaller number of Effort Adjustment Factors (EAFs) than some of the more recent models, such as COCOMO II.

Table 11-15◀, Fig. 11-3, Eq. 11-3

Table 11-16. Example of Sizing (SLOC) and Calculation of ESLOC for Space Flight Software Components. Note how the calculated ESLOC (40,545) in this example is less than the sum of new SLOC and Reused SLOC; it is the ESLOC value (using the weighting factors from Table 11-8) that is used for the software development effort CER Eq. (11-3). This real-life example is from space probe software programmed in C.

Flight Software Component	New SLOC (C Language)	Reused SLOC	% Redesign	% Reimplementation	% Retest	Equivalent SLOC (ESLOC)
Executive	5,000	5,000	30%	30%	75%	7,175
Communications	2,000	8,000	20%	20%	50%	4,320
Command & Data Handling	3,000	22,000	25%	25%	50%	10,150
Power Control	1,000	4,000	20%	20%	50%	2,160
Attitude & Control	2,000	18,000	30%	30%	50%	8,480
Thermal Control	1,000	9,000	20%	20%	50%	3,610
X-Band Uplink & Downlink	1,000	10,000	20%	20%	75%	4,650
TOTAL	15,000	76,000				40,545
Example: Communications—2,000 + 8,000 (20% x 0.4 + 20% x 0.3 + 50% x 0.3) = 4,320						

Table 11-17. COCOMO 81 Effort Adjustment Factors (EAF). Only four of the more common COCOMO 81 EAFs are described here.

Project Attribute	Effort Adjustment Factor (EAF) Based on Expected Impact for each Project Attribute					
	Very Low	Low	Nominal	High	Very High	Extra High
Required Software Reliability (RELY)	(0.75)—slight inconvenience	(0.88)—easily recoverable losses	(1.00)—recoverable losses	(1.15)—high financial loss	(1.40)—risk to human life	No factor
Product Complexity (CPLX) (note 1)	(0.70)—simple expressions, simple statements, simple arrays	(0.85)—moderate level expressions, no edits, no intermediate files	(1.00)—simple nesting, standard math and statistical routines	(1.15)—high nesting, basic numerical analysis special purpose routines	(1.30)—partial differential equations, communications handling, command processing	(1.65)—multiple resource scheduling, highly coupled and dynamic relationships
Programmer Capability (PCAP) (note 2)	(1.42)—4 months experience (15th percentile)	(1.17)—1 year experience (35th percentile)	(1.00)—3 years experience (55th percentile)	(0.86)—6 years experience (75th percentile)	(0.70)—12+ years experience (90th percentile)	No factor
Required Develop. Sched. (SCHED) (note 3)	(1.23)—75% of nominal time	(1.08)—85% of nominal time	(1.00)—100% of nominal time	(1.04)—130% of nominal time	(1.10)—160% of nominal time	No factor

NOTES: 1. Product complexity factors shown are an extrapolation from expanded tables in the COCOOM 81 Reference Manual which describe multiple sub factors for the complexity attribute. 2. Lesser experience typically requires a longer time to develop the product; sometimes this increase in man-months is offset by lower labor costs; the reference to "nominal time" would be the shortest time for an experienced team. 3. Note how adjust factors increase for both compressed and extended development schedule requirements, thus confirming that the COCOMO default effort estimates are based on a minimum effort schedule; some software model default effort estimates are alternatively based on a minimum schedule effort.

most *hardware* CERs where corresponding components are usually less than unity, illustrating economy of scale where cost per pound generally decreases with size. The constant coefficient (3.312) was derived from the data used to develop the CER.

For example, the nominal effort (assuming a nominal cumulative adjustment factor) to develop 10,000 new SLOC would be predicted by Eq. (11-3) as: $3.312\ (10)^{1.2} = 52.5$ man-months. If the cost per man-month (including salary and fringe benefits) were $25,000, then the monetary cost would be $52.5 \times 25{,}000 = \$1{,}312\text{K}$, or about $131 per SLOC. Note that the predicted cost to develop twice as many SLOC, i.e., 20,000, would be about $148 per SLOC—reflecting the "diseconomy of scale" associated with software development.

For better estimating accuracy and closer emulation to the software development process, COCOMO 81 relies on the cumulative effect of fifteen environment and developer EAFs which, when multiplied together, identified by π**EAF** in Eq. (11-3), provide a scaling or adjustment to the **nominal** effort equation.

Each adjustment factor is described by a series of rating levels that expresses the impact on development effort. These ratings are categorized between Very Low and Extra High, with a nominal value requiring no adjustment. Table 11-17 summarizes four of the avail-

able fifteen attributes, with a corresponding numerical value to be used in calculating π**EAF**. (Note, for example, that a very high software reliability attribute would increase the nominal effort development estimate to account for: the more detailed test plans and procedures; higher quality assurance, configuration management, and documentation standards; extensive stress tests; and IV and V interfaces required for human space flight.) In the effort example, we saw that 10,000 new SLOC could be developed in 52.5 man-months if all the adjustment factors were nominal (value of 1.0). If, however, the software were required for a manned space mission, where the required software reliability was very high and we were able to assign programmers with an average of 6-years of experience, the predicted effort may increase to $3.312\ (10)^{1.2} \times 1.4 \times 0.86 = 63.2$ man-months.

Schedule Estimate

Software development schedule duration in months may now be predicted, from the prior man-months prediction of effort, using Eq. (11-4), the *Basic COCOMO Schedule Equation*. To use this equation, you must first predict the development effort in man-months. Note how the development duration does not increase as fast as effort, suggesting that multiple software developers can work together in parallel activities, rather than in serial activities.

SME-SMAD WBS 1.1.8 and 1.2.3

Duration (in months) = 4.376 (MM)$^{0.32}$

Where:

Duration = Schedule time, in calendar months, to complete software development, including requirements analysis, coding, assembly, testing, and documentation

MM = effort in man-months derived from Eq. (11-3)

(11-4)

In the first effort example, we saw that 10,000 new SLOC could be developed for a cost of 52.5 man-months if all the adjustment factors were nominal. The predicted development time could then be calculated as: $4.376 \times (52.5)^{0.32} = 15.5$ months. This equates to an average staff loading of 3.4 persons. Is this reasonable as a sanity check on our estimate?

Estimates Based on Equivalent SLOC

Not every software development requires the development of all new SLOC; and, some allowance must be made for the integration and testing effort of legacy (reused and modified) SLOC. When legacy software is reused or modified, and often combined with new software, the software estimator typically calculates an "equivalent number" of SLOC (ESLOC), based on the activities (design, implementation, and test) planned for each software module. Then, he applies the calculated ESLOC parameter to the effort CERs as substitutes for the SLOC parameter (or KESLOC in this example).

Table 11-18, Fig. 11-3◀, Eq. 11-4

Table 11-18. ESLOC Weighting Factors for Reused and Modified SLOC.

Redesign	Reimplementation	Retest
0.4	0.3	0.3

COCOMO recommends the weighting factors to the SLOC count as shown in Table 11-18 as a starting formula, but actual weighting factors depend on the organization's experience, the type of software being implemented, and the cost prediction model used.

Commercial off-the-shelf (COTS) products are not included in this equivalent SLOC process because the end user typically has no idea how many SLOC characterize a typical commercial software product. However, the estimator needs to determine the purchase price and needs to estimate the cost of integration software. He also needs to be aware of any auto-generated code as well as code to be ported from another software system.

As an example, calculate the ESLOC for the aggregate of the following software packages:

- 10,000 new SLOC
- 6,000 SLOC reused as is; but, requires 80% retest
- 3,000 SLOC modified; requires 25% redesign, 50% reimplementation, and 100% retest

Using these weighting factors, one should calculate ESLOC to be 13 KESLOC, not 19 KESLOC, which would be the arithmetic total, as shown in Table 11-19. The difference is due to the reduced development effort associated with reusing some of the SLOC.

Software Maintenance Effort

For post-launch ground operations, the software maintenance cost can contribute 50% to 75% of the life cycle software cost. The annual effort to maintain software can be predicted either by the appropriate level-of-effort (LOE) of manpower assigned to operate and support the software or by a parametric approach based on the number of SLOC to be modified each year. This can be considered a recurring cost under SME-SMAD WBS element 1.1.8, 1.2.3, 7.2, and 7.3.

LOE predictions can be based on the number of SLOC which can be managed and maintained by one person. The metrics are subjective and depend upon the following considerations:

- What is included?—failures only or new applications, also
- Who does the maintenance?—users or separate software development facility

Refer to Sec. 11.2.10 for more information.

A parametric approach, which seems more practical, can be based on a productivity ratio defining the number of SLOC modifications per man-month of dedicated maintenance effort. COCOMO defines this metric as Annual Change Traffic (ACT) which is the ratio of

Table 11-19. Solution to ESLOC Example.

Software Source	Original KSLOC	KESLOC			
		Redesign	Reimplementation	Retest	Subtotal
New	10	n/a	n/a	n/a	10.00
Reused—As Is	6	n/a	n/a	6 KSLOC × 80% retest × 0.3 = 1.44	1.44
Modified	3	3 KSLOC × 25% redesign × 0.4 = 0.30	3 KSLOC × 50% reimplementation × 0.3 = 0.45	3 KSLOC × 100% retest × 0.3 = 0.90	0.30 + 0.45 + 0.90 = 1.65
TOTAL	19.00				13.09

SLOC modified each year to SLOC delivered and ranges from 0.05 for mathematical and logical operations to 0.15 for more complex, non-real-time software. This metric works best after the initial software "bugs" have been repaired.

So, for example, if our previous example of 10,000 lines of complex software were to be maintained, the "steady state" annual maintenance requirement might cost 15% per year of the 19.0 man-months or 2.85 MM per year—equivalent to 0.24 persons to maintain the software. How does this prediction compare with the LOE of 25,000 to 50,000 SLOC per man-year?

More Information

More information on using COCOMO 81 (Intermediate version) may be found in Boehm [1981] and in the USC COCOMO 81 Reference Manual. Some useful space software estimating rules of thumb are shown in Table 11-20, but there are wide variations in these metrics.

Table 11-20. Software Development Estimating Rules of Thumb in 2010 Dollars per SLOC. These factors may need adjustment if your labor rates are unusually low or high.

Space Platform	FY10$/SLOC
Manned Flight	$800
Unmanned Flight	$550
Ground	$300

11.2.7 Facility Costs

Facility costs are not normally included in the prediction of space program costs and so do not appear in the SME-SMAD Project WBS. However, facility costs could be important to the project planner who is concerned with the impact on facility planning and consequential trade studies.

Facility construction cost information may be found in the much-used Air Force Construction Cost Handbook [USAF, 2003]. Extrapolations from this historical data are typically made on the basis of size and geographic location. Portions of the database, useful for facility estimates, are shown in Table 11-21. This data is useful for sizing (typical facility size), factor development (range of costs per square foot), and analogy estimating (total cost). Space engineers should scale and extrapolate predicted costs from these actual costs.

Typical location cost-adjustment factors are shown in Table 11-22.

Table 11-21. Space-Related Ground Facility Cost Factors. This data must also be adjusted for geographic location if other than Washington DC as shown in Table 11-22.

Facility Type	Typical Size (m^2)	Avg. Const. Cost in Wash DC per m^2 in 2010 $
Communications Facility	474	$2,170
Control Tower	465	$8,346
Hazardous Material Storage	3,716	$1,557
High Tech Laboratory	3,716	$3,993
Missile Assembly	929	$3,341
High Bay Assembly	3,250	$2,622
Headquarters Building	3,700	$2,410
Satellite Communications Center	550	$6,471

Table 11-22. Facility Cost Environmental Adjustment Factor (Multipliers) for Facility Costs. These are examples only; a complete list may be found in USAF [2003].

Location	Adjust Factor	Location	Adjust Factor
Marshall Center, AL	0.85	*Germany*	1.22
Cape Canaveral, FL	0.97	*Fort Irwin, CA*	1.27
Washington, DC	1.00	*Australia*	1.30
JPL, CA	1.13	*Japan*	1.62
Vandenberg, CA	1.19	*Hawaii*	1.70

For example, a high bay assembly building of 4,000 square meters, to be built at Cape Canaveral, would likely cost: 4,000 × 2622 × 0.97 = $10.173 M in 2010 dollars.

11.2.8 Launch Costs

The recurring cost of space launch is best predicted from previous cost history. The space mission engineer should select one or more candidate launch vehicles, based on capacity payload and current availability, and then negotiate the launch price. Estimators may approximate this cost from historical cost information such as presented in Table 11-23.

Selecting the appropriate launch vehicle, as described in Chap. 26, is conditioned by the required orbit (LEO or

Table 11-23. Historical Launch Vehicle Costs for Predicting SME-SMAD WBS 2.0 Cost. Launch costs are typically negotiated; these factors are best used for planning and worst case budgeting.

Mass Class (a)	Vehicle Name	Country of Origin	Capacity Payload in kg to Orbit LEO (b)	GTO (c)	Average Launch Cost (FY2010 $K) (d)	Launch Cost in FY2010 $K per kg Placed into Orbit (e) LEO (b)	GTO (c)
			Small				
Small	***Athena 2***	USA	2,065	590	$32,688	$15.8	$55.4
Small	***Cosmos***	Russia	1,500	N/A	$18,387	$12.1	N/A
Small	***Pegasus XL***	USA	443	N/A	$18,454	$41.5	N/A
Small	***Falcon 1e***	USA	1,010	N/A	$10,900	$10.8	N/A
Small	***Minotaur IV***	USA	1,650	N/A	$22,000	$13.3	N/A
Small	***Rockot***	Russia	1,850	N/A	$18,454	$9.9	N/A
Small	***Shtil***	Russia	430	N/A	$272	$0.7	N/A
Small	***START***	Russia	632	N/A	$10,215	$15.9	N/A
Small	***Taurus***	USA	1,380	448	$25,878	$18.8	$57.7
			Medium and Intermediate				
Inter.	***Ariane 44L***	Europe	10,200	4,790	$153,225	$15.0	$32.0
Inter.	***Atlas 2AS***	USA	8,618	3,719	$132,795	$15.4	$35.7
Medium	***Delta 2***	USA	5,144	1,800	$74,910	$14.6	$41.7
Medium	***Dnepr***	Russia	4,400	N/A	$20,430	$4.6	N/A
Inter.	***Falcon 9***	USA	10,450	4,540	$56,750	$5.4	$12.5
Medium	***Long March 2C***	China	3,200	1,000	$30,645	$9.5	$30.6
Inter.	***Long March 2E***	China	9,200	3,370	$68,100	$7.4	$20.2
Inter.	***Soyuz***	Russia	7,000	1,350	$51,075	$7.4	$37.9
			Heavy				
Heavy	***Ariane 4G***	Europe	18,000	6,800	$224,730	$12.5	$33.1
Heavy	***Atlas 5***	USA	20,050	8,200	$172,000	$8.6	$25.3
Heavy	***Delta 4 Heavy***	USA	22,560	13,130	$215,000	$9.5	$16.4
Heavy	***Long March 3B***	China	13,600	5,200	$81,720	$6.0	$15.7
Heavy	***Proton***	Russia	19,760	4,630	$115,770	$5.9	$25.1
Heavy	***Space Shuttle***	USA	28,803	5,900	$408,600	$14.2	$69.2
Heavy	***Zenit 2***	Ukraine	13,740	N/A	$57,885	$4.2	N/A
Heavy	***Zenit 3 SL***	Multinational	15,876	5,250	$115,770	$7.2	$22.1

NOTES: (a) The FAA Office of Commercial Space Transportation has divided all vehicles into four mass classes: small, medium, intermediate, and heavy, based on their payload capacities in pounds to LEO (<5,000, <12,000, <25,000, >25,000); in the table, all weights have been converted to kg. (b) Low Earth Orbit (c) Geosynchronous Transfer Orbit; these costs do not include the additional cost of apogee kick motors or other payload injection means. (d) These costs (in 2010 thousands of dollars) include the launch vehicle and related launch services. Where necessary, the costs were converted to dollars from other currencies. (e) Assuming maximum use of the payload weight capacity in the calculations, these are minimum values.

GTO), the total payload weight (with allowance for weight growth), country of manufacture (see large cost differences between Western and non-Western countries), and trajectory (usually dependent on designated launch site), and whether the mission is to be manned or unmanned. This table illustrates the sensitivity of launch cost to mass class, orbit, and country of origin (Western vs non-Western). From a cost estimator's point of view, it is interesting to note that cost per kg decreases with capacity, thereby confirming our general cost presumption about economies of scale.

Note, also, that this cost information has been derived from multiple sources [Futron Corporation, 2002; AIAA, 2004; NASA, 2006], and may not reflect the recent competitiveness observed in the launch vehicle industry where price drops of 20% or more have occurred over the past few years.

The discussion up to this point has focused on the recurring cost of launch systems, which is what most users will be interested in for costing a particular mission. Additionally, TRANSCOST is a parametric cost model for estimating the development (non-recurring) cost of space launch systems and is currently in version 8.1 [Koelle, 2010]. This tool is more appropriate for use by new launch vehicle designers. (See also Sec. 10.3.3.)

11.2.9 Additional Cost Factors

Certain cost factors or wrap factors, representative of those activities not directly associated with "touch labor" or with specific hardware or software elements, are typically used to estimate "adders" or wrap factors to the prime mission equipment (PME) costs—both non-recurring and recurring. PME are the hardware and software "deliverables."

Typical factors are shown in Table 11-24. The estimator might consider the range of factors (minimum, maximum) for each element; for example, a higher project management factor would be appropriate for small projects or for projects which require coordination among several team members or subcontractors. As with

Table 11-24. Cost Factor Percentages to Calculate Wrap Factors. These are additions to PME costs, where PME is total cost of spacecraft and payload costs. Consider the range of factors and the reasons for not always choosing the average factor. Not all of these factors would apply to every estimate.

SME-SMAD WBS Element	Minimum Factor	Average Factor	Maximum Factor	Reasons for Using the Maximum Factor
3.0 Annual Operations and Support for Ground Station	1%	5%	27%	Many subsystems and platforms, contractor support
4.1 System Engineering	15%	20%	35%	Frequently changing requirements, multiple interfaces
4.2 Project Management	10%	15%	35%	Small project; multiple team members or working locations
4.3 System Integration & Test	1%	15%	33%	Many components, conflicting requirements, changing baseline
4.4 Product Assurance	1%	3%	6%	Anticipated short life, harsh operating environment
4.5 Configuration Management	1%	4%	20%	Detailed control level, multiple users
4.5 Contractor (or subcontractor) Fee	10%	12%	15%	Contractor assumes project risk, such as with fixed price contract
4.5 Data Management	1%	2%	5%	Multiple data clients, multiple IT platforms
4.5 Development Support Facility	1%	4%	11%	Real time support, multiple operating locations
4.5 Hardware/Software Integration	10%	16%	24%	When hardware or software components are still being developed
4.5 Integrated Logistics	1%	6%	15%	Multiple operating platforms, multiple operating agencies
4.5 Safety & Mission Assurance (S&MA)	7%	7%	8%	When human life is at risk
4.5 Site Activation	1%	3%	6%	Remote area, crowded environment, harsh climate

most factors or rules of thumb, these are "starting points" for the estimate and should be tempered with judgment, other information, or other methods (e.g., level of effort) of estimating. It is common to evaluate these factor-derived costs by converting the dollars to man-months of labor to see if the estimate is reasonable.

For example, if the development cost for a one-year project were estimated to be $2 million without project management and the average project management factor of 15% were used to estimate a cost of $300K, then it would be reasonable to ask if one full-time senior project manager at $300K per year would be adequate. The real challenge with this level-of-effort approach comes when we estimate the cost for fractions of persons.

11.2.10 Operations Costs

Mission operations costs are integral to total life cycle cost but are too often overlooked during the development phase of the project. A discussion of operations cost drivers may be found in Larson [2006]. Understanding the cost impact of mission requirements, logistics, and concept of operations may require investigation of the following parameter sets and alternatives:

- Defense/Commercial/Experimental/Communications/Microsatellite
- Manned/Unmanned
- US/Russian/European/Japanese Cooperation
- Communication Architecture and Criticality
- Earth/Interplanetary/Lunar Orbits
- Number of Missions/Spacecraft/Orbits
- Number of Ground Stations/Sites
- Levels of Automation and Staffing (Space and Ground Segments)

Estimating Process

Consistent with the estimating process defined in Sec. 11.1.5, we begin with establishing the operations phase ground rules and assumptions; these will be explained in Chap. 29, particularly Table 29-1. Consider, especially, the duration and complexity of the mission, as well the staffing levels. The long-term steady-state labor costs to operate the space system are best predicted by a level-of-effort (LOE) analysis; that is, by planning for the number of operators required, their labor costs, and their schedule of support (full-time versus part-time) to the operations phase.

Next, we expand the SME-SMAD WBS Element 7.0, Operations, as previously identified in Fig. 11-1, into the following sub elements (or a similar subset), consistent with the level of information available and the desired detail of the estimate:

7.1 Program Management and System Engineering (PMSE)

7.2 Space Segment
- 7.2.1 Payload Software
- 7.2.2 Bus Software
- 7.2.2 Communication System

7.3 Ground Segment
- 7.3.1 Mission Operations
  - Monitor and Control
  - Anomaly resolution

- Data retrieval, processing, dispersion
- Rehearsals/Training
- Disposal/Termination

7.3.2 Ground Software
- Work Stations
- Simulation Test-beds

7.3.3 Ground Hardware
- Mission Control Center
- Antennas/Reference Stations

7.3.4 Facilities

Finally, we must apply cost factors, parametric cost estimating CERs, or historic analogies to each WBS sub element, considering the following definitions of these Operations WBS sub elements:

WBS Element 7.1, Program Management and System Engineering (PMSE) includes business management, business development, data management, operations, logistics/support analysis/planning, scheduling, and office administration. PMSE may be estimated either as a level of effort (number of personnel required per year) or as a factor of total operations costs. A general rule of thumb, depending on mission duration and complexity, is to predict PMSE at 10% to 20% of the annual space and ground operations costs.

WBS Element 7.2, Space Segment, includes maintaining the software of the space vehicle. Software maintenance activities include code corrections, code adaptations, code perfections, and occasional enhancements. The three major software subsystems to be maintained are the payload (mission software), the space vehicle (bus software), and the communication software. The bus software communicates the vehicle's position, movements, and status while the payload software communicates valuable data and information related to the mission. Software sustainment and maintenance may be estimated as a level of effort (number of personnel required per year).

WBS Element 7.3, Ground Segment, includes most of the costs for space mission operations. The key components are the mission operations costs, ground software sustainment costs, ground hardware maintenance costs, and the facilities.

WBS Element 7.3.1, Mission Operations, includes the cost of operating personnel, who are responsible for:

- Monitor and Control—the number of personnel and hours required to monitor space vehicle, including orbital corrections, scheduled systems checks, shut downs, and planned space software upgrades and corrections.
- Anomaly Resolution—extra personnel to resolve complex issues outside the planned or typical orbital adjustments.
- Data Retrieval, Processing, Dispersion—the man hours needed to retrieve, process, format, and disperse data to and from the satellite; this includes both the payload and the bus data and may be dispersed to multiple end users or teams.
- Rehearsals/Training—planned rehearsals and training for multiple missions in accordance with launch schedules and key milestones.
- Disposal/Termination—personnel hours for planning and terminating.

WBS Element 7.3.2, Ground Software, includes communication with the space vehicle from the ground. Costs include simulation and test-bed software to analyze new software, proposed orbits, and anomaly resolutions. Ground software maintenance includes corrections, adaptive code, perfections, and some enhancements. Ground software may include some related hardware elements such as work stations and displays to monitor and control the space systems. A system test-bed may include hardware to simulate software that mimics all aspects of the space system and mission. However, the cost of these hardware elements should have been included in the program development cost estimate.

WBS Element 7.3.3, Ground Hardware, includes monitoring and control stations, reference stations, antennas, networks, and mobile units required for mission operations. However, the cost of these hardware elements should be included in the program acquisition cost estimate. A general rule of thumb predicts annual ground hardware maintenance cost at 7% to 20% of the hardware acquisition (production phase) costs.

WBS Element 7.3.4, Facilities, include structures that house operations equipment and personnel required to perform space systems operations. Annual dollars per square meter factors may be applied to predict rent, maintenance, taxes, insurance, and fees. If the operations center is not charged to this project, then ignore this cost element. See Sec. 11.2.7 for more information.

Estimating Models

NASA developed several parametric cost prediction models which can be used as the primary basis, or as a cross-check, for operations costs. One of these models is the Exploration Architectures Operations Cost Model (ExAOCM) [Shishko, 2006] for estimating lunar outpost operations costs. This model traces its heritage to NASA's Model Evaluation of Space Station Operations Cost (MESSOC), which dealt with the costs of mature operations for a LEO space station. Another model, TRANSCOST [Koelle, 2010], is a commercially available model of space transportation costs. A description and application of these models is beyond the scope of SME-SMAD.*

ExACOM was developed by the Jet Propulsion Laboratory (JPL) under NASA sponsorship for the Constellation Program. It runs in MS Excel and can predict operations cost by fiscal year in accordance with the program WBS. Access to the model is controlled by Hq NASA, Cost Analysis Division.

Operations costs have not received as much attention

Table 11-24◀, Fig. 11-3◀, Eq. 11-4◀

* Some of the material in Sec. 11.2.10 was contributed by Robert Shishko, Ph.D., Principal Systems Engineer/ Economist; Jet Propulsion Laboratory, California Institute of Technology.

and detailed analysis within the space costing community as have development and production costs. However, operations costs are often a substantial contributor to total life cycle costs, especially for human spaceflight programs. Early system design decisions can exert a tremendous influence on operating costs and, therefore, we must ensure that the operating cost drivers have been properly identified and the operating costs of all alternatives have been considered in the selection of the preferred alternative.

To estimate operations costs, we follow the same steps as for development and production costs. We must first establish an operations WBS, understand the operations concept, select the proper cost model or other estimating method, and develop the cost estimate. Mission operating costs are inherently time-based, and must be spread over time and inflated.

Operations costs include mission operations, ground operations, training, sustaining engineering, sustainment of the mission control center, flight hardware spares and maintenance, program/project management and integration, and communication services.

In addition to parametric modeling and LOE analysis, discrete event simulation has been used with some success to estimate the workforce and costs associated with aspects of ground operations. For each sub-function, we develop an equation relating the program/project's approach to the number of personnel needed and then to cost. With L as the length of crew training in years, a simple equation for the cost of trainers in year t is:

Mission Ops Trainers Cost Estimate

Trainer Cost $= \alpha + \beta \times NC_t$

$$NC_t = \sum_{\tau=t+1}^{\tau=t+L} \left(Number\ of\ crewed\ flights_\tau \right)$$

Where:

α = fixed trainer cost for training curriculum maintenance and updates, participation in procedure development, mission planning, training system engineering studies

β = variable trainer cost per crew in training.

NC_t = number of crews in training

(11-5)

The parameters, α and β, are usually obtained using expert judgment and are based on the program/project's operations concept for crew training.

A more sophisticated algorithm would model the dynamic flow of flight crews through the training system, taking into account the type of training (e.g., classroom lectures, flight simulators), the number of trainers for each training activity, the need for refresher training, the unique training requirements of each mission, and the constraints on the availability of trainers and training assets.

Estimating Example

Here we present an example of a **WBS 7.0 Operations** cost prediction, using the principles and information previously presented plus some representative analogy information. This example is for a low to moderately complex scientific reconnaissance mission, similar to FireSat II using readily available information.

Operational staffing levels must be determined for each year of operations, and are typically lower for pre-launch years, peaking during the more active data collection and processing years, and lower again during the maturing years. A convenient analogy data point would be the number of operators employed for a similar program. However, adjustments should be considered for pertinent cost drivers, such as: whether an interactive payload operation will be supported, the degree of automation to be acquired, the number of spacecraft to be supported, whether the team also controls the ground antenna, and whether some of the support can be provided by an external organization.

Table 11-25 provides an example of a relatively large and dedicated support team responsible for a complex mission experiencing frequent contact with the spacecraft, operating 24 hours per day and 7 days per week, with a separate engineering support staff. The term, full time equivalent (FTE), is a metric for a full-time worker employed for one year. Obviously, the operating plan cannot realistically predict each operator will be engaged in his function, and his function alone, for exactly 2,000 hours per year, and so we assume that most staff are cross-trained to be able to back-up other operators.

Less complex missions are easier to automate, leading to smaller staffs which do not work as many shifts. Smaller staffs typically will expect more cross-training between the staff and might also engage an existing and separate organization for engineering (payload) support which can support additional missions, thereby requiring less than half the number of assigned FTEs each year. Members of smaller staffs, however, need to be more experienced (more expensive) so that they can support multiple functions.

Once we have predicted the operations staffing level, we must next determine the annual cost for each FTE, considering salary, overhead (typically 75–150%), administrative costs, contractor fee (if any), and other support costs (such as travel). Labor cost estimates may be different for contractors, government, and university staff, due to variations in the accounting methods. Salaries vary by location. Overhead rates can vary significantly when staff members are provided pre-arranged working space and tools. For reference, consider Table 11-26 which provides typical FY2010 FTE costs, for equivalent mission operations staff (which are less expensive than for development engineers and development technicians).

Since the earlier discussion of software estimating in Sec. 11.2.6 focused primarily on development phase costs, we need a new set of factors relevant to operations phase software maintenance costs. Consider factors to characterize the number of software lines of code (SLOC) which can be maintained by one person for one year (one FTE), as shown in Table 11-27. Note, however, that it is unlikely we could employ a fraction of a

Table 11-25. Typical Highly Complex Mission Operations Staffing Levels. [AIAA, 2003].

Staff Position	FTE	Shift (Hours/Days)	Comments
Operations Staff			
Operations Manager	1	8/5	
Flight Operations Supervisor	1	8/5	Cross-trained for Spacecraft and Payload Controller position.
Spacecraft Controller	4	24/7	
Payload Controller	4	24/7	
Ground Controller	4	24/7	
Spacecraft Operations Engineer	2	8/5	Cross-trained for Spacecraft Controller and Payload Analyst position.
Payload Analyst	1	8/5	Cross-trained for Spacecraft Analyst job
Mission Planner	2	8/7	Share Orbit Analysis
Data Analyst	2	8/7	
Orbit Analyst	1	8/5	Share Mission Planning
Operations Engineer	2	8/5	
Engineering Support Staff			
Engineering Manager	1	8/5	
Ground Systems Engineer	1	8/5	
Systems Engineer	1	8/5	
Technician	1	8/5	
Testbed/Simulator Engineer	2	8/7	
Flight Software Engineer	1	8/5	
Ground Software Engineer	2	8/5	
System Database Administrator	1	8/5	

Table 11-26. Annual Labor Costs for One FTE, Considering Salary and Overhead. This information was adjusted from Department of Labor statistics for similar engineering and technician skills and adjusted for 150% overhead (excluding administrative costs, contractor fee, and travel). (Data from various sources.)

Labor Category	Typical Annual Salary in FY2010 Dollars	Typical FTE Cost in FY2010 Dollars
Engineers and Managers: *Spacecraft controllers, payload controllers, operations engineers, software engineers, systems engineers*	$80K	$200K
Technicians: *Database administrators, data analysts,*	$60K	$150K

software engineer so we need to round up or down to the best number of whole FTE (unless we can share staff with another mission).

Next, we demonstrate the operations cost model with an example, using some of the factors and analogy data already presented. Table 11-28 takes us through the steps for calculating the most likely cost prediction for each sub element of WBS 7.0 Operations.

Table 11-29 summarizes our cost prediction for a single mission year; the total cost will be the aggregate prediction for all mission years. The number of operating staff typically will be different for each year and the number of SLOC to be maintained may change from one year to the next. This estimate summary is based on the following assumptions:

- Total FTE staff consists of 8 engineers & 4 technicians, none experienced in software maintenance.
- Software to be maintained:
  - Space—100,000 SLOC.
  - Ground—25,000 SLOC (the amount to be adapted from a much larger ground software package intended to manage multiple missions).
- Hardware acquisition cost was $1,400K.
- Facility lease of 1,000 square meters; annual local rent will be $1.25K per square meter.

11.2.11 Cost-Risk Assessment Model

Until now, we assumed that component and subsystem cost predictions may be arithmetically summed to equal the mission system cost. Statistically, however, the arithmetic sum of element most likely cost predictions, typically understate the system most likely cost prediction.

This simplified arithmetic summation process is often necessary when there is no single CER which effectively predicts the total cost. However, there are some statistical errors in this aggregation process since the cost uncertainty spread is unique to each CER and we may fail to capture the uncertainty metrics.

A more rigorous method, not further described in

Table 11-26, Fig. 11-3◀, Eq. 11-5◀

Table 11-27. Range of Software Maintenance Effort Estimating Factors for Space Mission Operations (SLOC Maintained per Person-Year). [Galorath Incorporated, 2011].

	Low	Most Likely	High	Typical Functions
Flight Software—Unmanned	7,800	16,000	25,600	Command & Control, Communications, Signal Processing, Diagnostics
Flight Software—Manned	5,400	12,000	19,400	Command & Control, Communications, Signal Processing, Diagnostics
Ground Control Software	17,000	28,200	39,600	Ground Command & Control, Mission Planning, Data Handling
Ground Support Software	18,800	30,700	44,700	Offline Testing, Simulation, General Support

Table 11-28. Mission Operations Cost Prediction Model.



WBS Element 7.0 Operations	Cost Category	Assumptions, Factors, Calculations	Annual Cost ($K 2010)
7.1 PMSE	Labor	X% of the other Ops Costs (10% to 20%)	A = X% × (B+C+D+E+F)
7.2 Space Segment Software Maintenance	Labor	$N_{SLOC,FSW}/16{,}000 \times FTE_{ENG}$	B
7.3 Ground Segment			
7.3.1 Mission Operations	Labor	$N_{ENG} \times FTE_{ENG} + N_{TECH} \times FTE_{Tech}$	C
7.3.2 Ground Segment Software Maintenance	Labor	$N_{SLOC,GSW}/28{,}200 \times FTE_{ENG}$	D
7.3.3 Ground Hardware Maintenance	Labor	Y% of H/W Acquisition Cost (7% to 20%)	E
7.3.4 Facilities	Facility Lease	Z m^2 × W (price/m^2 floor space)	F
TOTAL ANNUAL OPERATIONS PHASE COST			T_{ann} = A+B+C+D+E+F
TOTAL MISSION OPERATIONS COST =	$N_{years} \times T_{ann}$		

Table 11-29. Example of Predicting Annual Unmanned Space Mission Operations Cost in 2010 Dollars Using Level-of-Effort Estimating Techniques.

WBS Element 7.0 Operations	Cost Category	Assumptions, Factors, Calculations	Annual Cost ($K 2010)
7.1 PMSE	Labor	15% of other costs (as high as 20%)	742
7.2 Space Segment Software Maintenance	Labor	100,000 SLOC/16,000 ~ 6 FTE @ $200K	1,200
7.3 Ground Segment			
7.3.1 Mission Operations	Labor	8 engineers @ $200K + 4 technicians @ $150K	2,200
7.3.2 Ground Segment Software Maintenance	Labor	25,000 SLOC/28,200 ~ 1 FTE @ $200K	200
7.3.3 Ground Hardware Maintenance	Labor	7% of $1,400K	98
7.3.4 Facilities	Facility Lease	1,000 square meters @ $1.25K per square meter	1,250
TOTAL ANNUAL OPERATIONS PHASE COST			5,690

SME-SMAD, requires simulation software which can be incorporated into a Microsoft Excel spreadsheet. Such software incorporates a Monte Carlo summation achieved through running multiple summation iterations, each considering the cost range for every element. Cost ranges are typically determined by the uncertainty in the input parameter of each CER, such as the uncertainty of weight growth when weight is the principle cost driver. If there is some expectation that the design weight may grow, for example, the CER could be applied individually to estimate the lowest possible value, the most-likely value, and the highest possible value of weight in order to derive a range of estimates.

The process works like this:

1. Calculate a range of costs for each estimate, emulated by a triangular distribution as shown in Fig. 11-4. Lower and Upper Bound estimates

depend on optimistic and pessimistic assumptions about one or more cost drivers, thereby requiring three estimates for each element. By definition, the mode (most likely cost) estimate represents the result which would occur the greatest number of times in a multi-event simulation.

2. Enter the three costs for each WBS element into the Monte Carlo simulation spreadsheet, likely an add-on to MS Excel such as @-Risk or Risk Plus and run the aggregating program many times (several thousand would be appropriate).
3. Plot the probability and then the cumulative probability of the resulting system-level estimate.

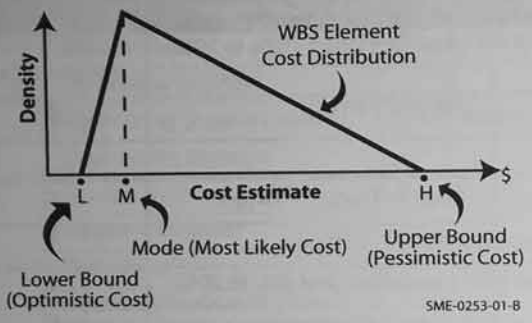


Fig. 11-4. Assumed Triangulation of the Cost Range for Each WBS Element.

After the element most-likely cost estimates have been aggregated through a Monte Carlo simulation approach, the final estimate looks like the plots in Fig. 11-5 after 10,000 trials.

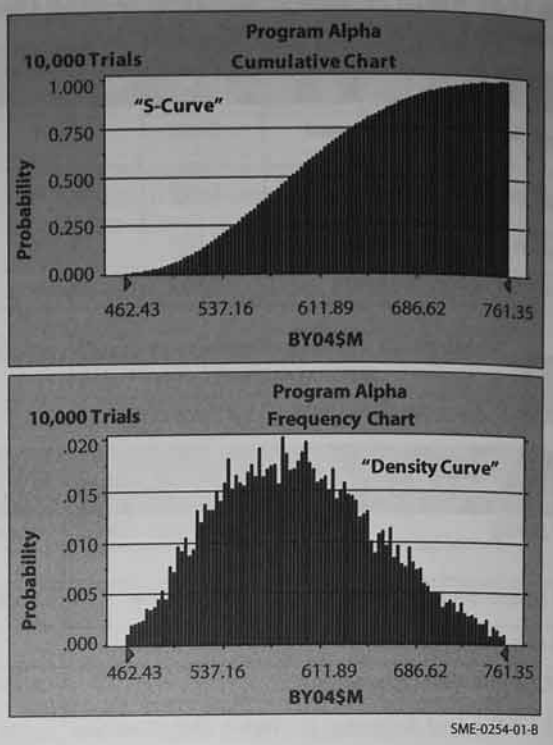
From the "S-Curve," we can determine that there is a 50% probability that the true cost will not exceed $593M and that there is an 80% probability that the true cost will not exceed $651M. If we were budgeting at 50% confidence and funded at the 80% confidence, we would enjoy a $58M management reserve to spend on risk mitigation to reduce our cost risk.

This process, as applied to cost estimating, is described in several publications, including Garvey [1999], Tecolote Research [2007], Book [2008], and Book [1999].

An alternate approach to describing the uncertainty of a cost estimate is to state the range of expected costs, corresponding roughly to the lower bound and upper bound defined in Fig. 11-4.

11.2.12 Cost Trade Studies

A primary purpose for developing a space system cost estimate is to support engineering trade studies, often in support of affordability initiatives. A trade study might help the system architect determine, for example, the most affordable option or the best value option among design alternatives. Figure 11-6 illustrates a performance/cost trade where the vertical scale indicates the range of acceptable key performance parameters (KPP)



Percentile	Value
10%	516.81
20%	538.98
30%	557.85
40%	575.48
50%	592.72
60%	609.70
70%	629.19
80%	650.97
90%	683.01

Statistics	Value
Trials	10,000
Mean	596.40
Median	592.72
Mode	---
Standard Deviation	63.18
Range Minimum	450.19
Range Maximum	796.68

Fig. 11-5. What a System-Level Risk-Adjusted Cost Estimate Looks Like, Showing the Instant Probability and the Cumulative Probability. In this figure, you can see the system-level mode value is $593 and the range extends between $517M (at the 10% confidence level) and $683M (at 90% confidence level).

such as communications bandwidth or service life, and the horizontal scale the range of cost. Boundaries (dotted lines) on the vertical scale represent the threshold (minimum required) KPP and the more expensive objective value representing the most performance the user is willing to buy.

Boundaries (solid line) on the horizontal scale represent the maximum cost the user can afford and the lower cost which typically represents his budget.

Of the four design alternatives, only alternatives 1, 2, and 3 satisfy the cost requirement and only alternatives 2, 3, and 4 can satisfy the performance requirement. Likely, alternative 3 would be judged the most effective choice because it lies within both ranges and it costs only a little more than alternative 2 but offers significantly more performance—good ROI.

Refinements of this approach is promoted by the US DoD as Design to Cost (DTC) and Cost as an Independent Variable (CAIV) affordability initiatives. An overview description of CAIV can be found in Apgar [2001]. Comparative cost estimates are a useful product while parametric models are the tool of choice.

Table 11-29◀, Fig. 11-5, Eq. 11-5◀

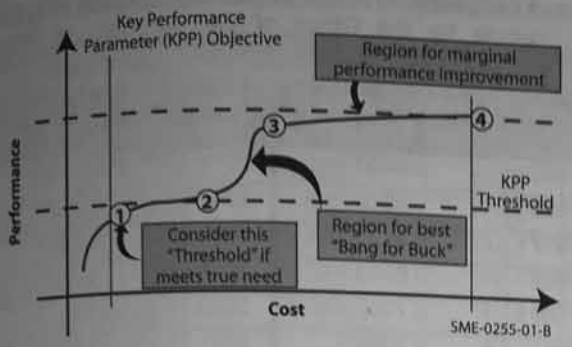


Fig. 11-6. Performance/Cost Trade Space.

11.3 Other Considerations in the Cost Estimate

11.3.1 Cost Realism

Cost realism is not the same as accuracy or precision. Realism refers more to the perception of the estimate, i.e., how likely is the estimate to closely predict the future event. This issue is more than statistical correctness; it also describes the inherent quality of the prediction.

Cost realism occurs when the estimator uses commonly accepted estimating procedures, applies proven estimating tools which have been calibrated to the product or service being estimated, and when he demonstrates estimating cross-checks and "second opinions." Cost realism results when an independent cost estimate (ICE) compares well with the primary estimate. Shown in Fig. 11-7, multiple risk adjusted and overlapping estimates give the manager or customer a more comfortable feeling about the estimate and thereby assigns a high degree of cost realism to the estimate.

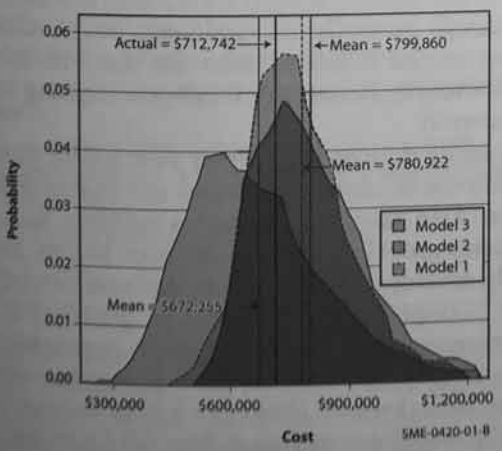


Fig. 11-7. Realism Improves When Using Multiple Cost Models and Then Comparing Results. This figure demonstrates the plotting of cost-risk distributions (as described in Sec. 11.2.10) from multiple models in overlapping manner comparison.

11.3.2 Cost Improvement

Cost improvement refers to the observable phenomenon by which workers become more efficient and productive as they continuously produce more units of a nearly-identical design and technology with a similar production process. Essentially, the second unit tends to go faster and cheaper than the theoretical first unit (T1), particularly with higher complexity products requiring a substantial labor (less automation) contribution. The original concept, learning theory, was developed in the airplane industry in the 1930s with observations on high-quantity production lines.

Cost improvement is also identified as *cost progress*, *experience curves*, or *learning curves*. These terms are somewhat interchangeable but the user must be careful to determine if the improvement rate refers to the entire subsystem or just the labor portion.

Cost improvement has been adapted to the space industry but has actually been quantified in only a few programs. Its main application has been with those parametric models, such as the recurring CERs of USCM8, which predicts the T1 cost. The T1 then must be factored ("down the learning curve") to predict average unit cost (AUC) for quantities beyond the first. Cost improvement rates (learning factors) are usually derived by fitting a curve through historical lot costs and then projecting lot costs backward to determine T1, as shown in Fig. 11-8. When starting with a different T1, indicating a modified product or a changed production process, another learning curve may then be used to apportion the costs to any follow-on unit.

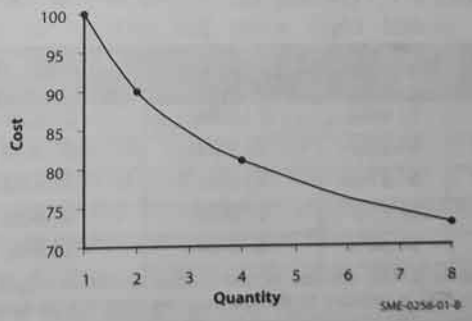


Fig. 11-8. Learning Rate is Determined Empirically.

Spacecraft production today is not as efficient as airplane production, primarily because each satellite tends to differ from its predecessor, because spacecraft typically are one-of-a-kind products, and because there are typically many changes from one spacecraft to the next in a series. Therefore, cost improvement adjustments must be applied with reasonable care. Further discussion on the selection and application of learning theory to space mission products is found in Fox [2008].

The basic cost improvement formula, essentially a CER based on quantity and learning rate as cost drivers, is of the form shown in Eq. (11-6). This is the classic learning curve theory.

For example, when T1 = $1.00M (as determined by our recurring cost model), the expected learning curve slope is 95%, and the number of units (lot size) being built at the same time is 5, we can calculate three distinct costs:

Table 11-29◀, Fig. 11-8, Eq. 11-5◀

Total Lot Cost = $T1 \times N^{(1+\ln S/\ln 2)}$

Where:

T1 = the theoretical first unit production cost (e.g., as predicted by recurring CER in Table 11-8)

N = quantity (lot size)

S = learning curve slope in % or decimal form, (e.g., 95% OR 0.095; an empirical parameter)

(11-6)

- *Total lot cost* = $T1 \times N^{(1 + \ln S/\ln 2)} = 1 \times 5^{(1 + \ln 0.95 / \ln 2)} = 1 \times 5^{[1 + (-0.0513 / 0.6931)]} = 1 \times 5^{(1 - 0.0740)} = 1 \times 5^{0.9260} = 4.438$ = \$4.44M (for 5 units)
- Average Unit Cost = \$4.44M / 5 units = \$0.89M
- Unit (marginal) Cost for 5th unit = $Total_5 - Total_4 = (1 \times 5^{0.9260}) - (1 \times 4^{0.9260}) = 4.438 - 3.610 =$ \$0.83M

Although the estimator needs to assess whether the learning effect is to be applied to the entire product (element) or only to the labor component of that product, USCM8 assumes the rate is applied to the entire product.

Making similar calculations for 95% learning rate, for each unit numbered 1 through 5, we can develop a convenient set of data (Table 11-30), which is useful for quick learning curve calculations.

Table 11-30. Cost Improvement Factors Due to 95% Learning Rate.

Unit #	Total Lot Cost (a)	Avg Unit Cost (AUC) (b)	Unit (marginal) Cost (c)
1	\$1.00M	\$1.00M	\$1.00M
2	\$1.90M	\$0.95M	\$0.90M
3	\$2.77M	\$0.92M	\$0.87M
4	\$3.61M	\$0.90M	\$0.84M
5	\$4.44M	\$0.89M	\$0.83M

(a) From Eq. (11-6). (b) Total lot cost divided by number of units; note how the AUC is reduced by the learning rate whenever N doubles.(c) Difference in total lot cost between N units and N-1 units

11.3.3 Inflation

For consistency in referring to cost estimates, and to minimize confusion in the review of cost estimates, costs should be stated in constant-year dollars. Trade studies and cost-benefit analyses should also be developed in constant-year dollars. Then-year dollars (or real inflated dollars) may be necessary for funding analysis or budget analysis. Inflation adjustments are necessary in the following situations:

- When aggregating subsystem or component estimates from multiple models and the model CERs predict cost in different "years"
- When incorporating historic cost data from multiple sources, in different years, before assembling them into a normalized database as the basis for developing CERs

Table 11-31, Fig. 11-8◀, Eq. 11-6

Table 11-31. Inflation Factors Relative to the Year 2012 Based on Projects by the Office of the Secretary of Defense (Mar. 2011). See text for discussion.

Fiscal Year	Inflation Factor to Base Year 2012	Fiscal Year	Inflation Factor to Base Year 2012
1995	0.6408	2016	1.0857
1996	0.6558	2017	1.1089
1997	0.6723	2018	1.1326
1998	0.6913	2019	1.1568
1999	0.7102	2020	1.1815
2000	0.7285	2021	1.2067
2001	0.7516	2022	1.2324
2002	0.7717	2023	1.2588
2003	0.7979	2024	1.2856
2004	0.8240	2025	1.3131
2005	0.8583	2026	1.3411
2006	0.8867	2027	1.3698
2007	0.9108	2028	1.3990
2008	0.9375	2029	1.4289
2009	0.9501	2030	1.4594
2010	0.9701	2031	1.4906
2011	0.9850	2032	1.5224
2012	1.0000	2033	1.5549
2013	1.0201	2034	1.5881
2014	1.0413	2035	1.6221
2015	1.0630	2036	1.6567

Beyond 2012, the rates shown are based on an extrapolated constant rate of inflation. New projections are available annually.

- When applying aggregated estimates to a single program estimate in a different year
- When transforming costs from constant-year to then-year and vice versa

Although generally not significant, you should notice whether annual inflation tables are based on calendar-year (starting in January) or fiscal-year (starting in some other month).

To adjust from one to another year dollars, or to convert between constant-year dollars and then-year dollars, use one of the many inflation tables available. Such tables sometimes provide multiple conversion factors, each tailored to development, production, and operations dollars.

An example conversion table in government fiscal year (running from October 1 through September 30) is shown in Table 11-31. This table allows conversion of historic data (back to 1985) to FY2010 and conversion from the model output year-dollars (all CERs and tables in this chapter predict cost in FY2010 dollars) to the preferred common year for analysis purposes.

Updates to this table may be found on the SME-SMAD website. For example:

- To convert historic data in FY2000 dollars to FY2010 dollars, multiply by 1.332
- To convert an estimate in FY2010 dollars to FY2015 dollars, divide by 0.913

Table 11-32. Example of Cost Spreading on Both Base-Year and Then-Year Costs.

Year of Expenditure	Percentage of Funding Each Year	Base Year Cost	Cumulative Cost (BY)	Inflation Factor	Then Year Cost	Cumulative Cost (TY)
2007	8.54%	85.4	85.4	1.084	92.6	92.6
2008	16.42%	164.2	249.7	1.107	181.8	274.4
2009	18.68%	186.8	436.5	1.130	211.1	485.5
2010	17.05%	170.5	607.0	1.154	196.7	682.2
2011	13.47%	134.7	741.6	1.178	158.6	840.8
2012	9.59%	95.9	837.5	1.203	115.4	956.2
2013	6.39%	63.9	901.4	1.228	78.5	1,034.7
2014	4.17%	41.7	943.1	1.254	52.3	1,087.0
2015	2.83%	28.3	971.4	1.280	36.2	1,123.2
2016	2.10%	21.0	992.4	1.307	27.5	1,150.6
2017	0.76%	7.6	1,000.0	1.335	10.1	1,160.8
		1,000.0	1,000.0		1,160.8	1,160.8

11.3.4 Cost Spreading

A cost estimate phasing (spread by year) model was developed by Burgess [2004] using a Weibull distribution function. Full explanation of this model is more complex than appropriate for SME-SMAD. A simplified version of this model is shown in Eq. (11-7), with default values for the Weibull function; normally, these parameters are based on calibration to an organization's expenditure experience, including the percent funding expended at specific schedule phase points. Analysts are advised to query the original paper, which is cleared for public release and can be downloaded from the Defense of Defense Cost Analysis Symposium (DoDCAS) website.

$$E(t) = d[Rt + 1 - e^{(-\alpha t^{\beta})}]$$

$$d = \text{Estimate}/[(R + 1 - e^{(-\alpha)}]/\text{Total Cost}$$

Where:

$E(t)$ = that portion of the total cost estimate apportioned to a cumulative portion of the program (through one or more years)

$R = 0.00148 \times$ (program duration in months)

t = the ratio of cumulative time (years) divided by total time (years)

$\alpha = -0.414 + 0.0729 \times$ (number of units) $+ 0.0488 \times$ (months duration) $+ 0.0145 \times$ percent of estimate which is non-recurring

$\beta = 1.71$ (the Weibull shape parameter)

(11-7)

Since total cost is estimated in the user-specified base year (BY) dollar value, the Phasing Model first spreads the expenditures in base year dollars and then applies the appropriate inflation factor to adjust the base year effort value to then-year (TY) dollars.

An example of spreading a \$1M estimate over eleven years starting in 2007, in both base year and then year costs with influence of inflation, is shown in Table 11-33.

11.3.5 Technology Readiness Levels

Another consideration in predicting development cost is the magnitude of the engineering effort to progress from a lower Technology Readiness Level (TRL) to a higher one. TRLs are defined in Chap. 1 and Sec. 1.3.4.

The cost prediction models in this chapter can generally predict the engineering and development costs to progress from level 6 to level 9. For planned development efforts with a greater or lesser "spread," e.g., starting from a TRL level other than level 6, the engineer may need to adjust his cost model prediction accordingly.

The key to successful development cost estimating is to calibrate this list to your real space program experience.

There have been numerous studies to characterize the difference in cost due to the maturity of technology, instrument maturity and space flight history. Most recently, Jet Propulsion Laboratory, [JPL, 2009] performed an assessment for the JIMO program and included several TRL correction factors. Relevant for this effort are those contributed by Smoker [2005], Malone [2011] and Hamaker J. [2008]. Table 11-33 presents a set of model factors which can be applied to correct the parametric model estimates to the appropriate TRL.

Note that the default case is when we assume the hardware to be developed is at TRL 6—then there is no adjustment required to the result from a development CER. If, however, the technology starting level is less than TRL 6, indicating more than typical engineering development is required, then the adjustment is greater than 1.0.

As an example of estimate correction based on TRL: An engineering development estimate from a reputable cost model predicts the engineering cost will be \$2.0M.

Table 11-33. TRL Adjustment Factors for Development Estimates.

TRL	Factor	TRL	Factor
1	4.0	6	1.0
2	3.0	7	0.8
3	2.0	8	0.7
4	1.5	9	0.5
5	1.3		

Table 11-33, Fig. 11-8◀, Eq. 11-7

However, since the starting point for the development is judged to be TRL 4 (because the hardware development has not progressed beyond the breadboard stage) and the goal is TRL 9 (mission ready), the multiplier is 1.5 and the corrected estimate will be $2.0 \times 1.5 = \$3.0\text{M}$.

11.3.6 Establishing Cost Estimate Requirements

The US DoD Cost Analysis Requirements Description (CARD), the NASA Cost Analysis Data Requirements (CADRe) document, and the UK MoD Master Data and Assumptions List (MDAL) are to the space systems cost estimator what the system specification is to the space system architect. These documents typically provide the quantitative and narrative description of the technical, physical, and performance features pertinent to costing the system being acquired and of the acquisition program-necessary parameters that serve as the basis of estimate (BoE) for cost modelers.

A typical cost estimating requirements document would include:

1. System Overview
2. Risk
3. Operational Concept
4. Quantity Requirements
5. Manpower Requirements
6. Activity Rates
7. Milestone Schedule
8. Acquisition Strategy/Plan
9. Development Plan
10. Facilities
11. Track to Prior CARD
12. Contractor Cost Data Reporting Plan

Even if your project does not formalize the system requirements in time for you to develop your estimate, you should be seeking similar information in order to determine the scope of your estimate, which estimating tools are most suitable, and what cost driving parameters you must obtain.

11.3.7 Documenting the Final Estimate

Documenting the estimate is just as important as the engineering notebook which documents the evolving hardware and software design. It is important to provide a "thread" to demonstrate how each sequential estimate was created and what caused the estimate to change over time. A suggested outline of the risk-adjusted space mission estimate description would include the following topics:

1. Executive Summary: A high-level summary of the estimate, summarizing purpose, scope and result of estimating cost and schedule
2. System Description: Technical information in sufficient detail to understand and evaluate the technical cost drivers
3. Acquisition Strategy: A description of the acquisition and deployment cost drivers
4. Groundrules and Assumptions: Aspects of the analysis framework that will have significant impact if they are changed
5. Estimating Methodologies: An overview of the primary and secondary methods for deriving the cost estimates by WBS element
6. Detailed Cost Estimate Derivation: A complete description of the calculation of the cost estimate in sufficient detail to allow replication; often in the form of bases of estimate (BoEs)
7. Uncertainty Analysis: A detailed description and quantification of uncertainties in the system's cost and schedule estimates
8. Conclusions and Recommendations: A summary of key insights and important findings, including a discussion on affordability

Table 11-33◀, Fig. 11-8◀, Eq. 11-7◀

11.4 Example Space Mission Cost Estimates

John Collins, *Microcosm*

In this section, we will apply the space mission cost models presented earlier in the chapter, to both the FireSat II and SCS missions, and estimate full mission cost for each, including costs for: spacecraft bus non-recurring and recurring cost for the appropriate number of flight units (2 units for FireSat II and 3 units for SCS), payload development, program level costs, IA&T, ground support equipment, launch and orbital ops support, launch vehicle, and flight software. The cost models to be applied for each mission are USCM8, SSCM, NICM, and QuickCost. Comparisons will be made of the costs generated by each model and an estimate will be selected based on which model best approximates the mission class represented by FireSat II and SCS.

The purpose of presenting the results in this way is to both show how to apply each of these mission cost models as well as the value of creating several cost estimates using different cost models. For each example cost estimate given, all input parameters that are used in calculating the cost estimates will be explicitly shown. The cost numbers are generated using the CERs presented for each cost model earlier in the chapter, namely: USCM8 CERs presented in Tables 11-8 and 11-9; SSCM CERs presented in Table 11-11, and the QuickCost CER presented in Table 11-12. Estimated costs are given, along with cost uncertainty for each cost element. The QuickCost model generates a conglomerate single cost estimate and uncertainty based on numerous inputs and uses a single CER to compute cost.

11.4.1 FireSat II

We now apply the various cost models presented in this chapter to the FireSat II mission, to assess the applicability of the various models to FireSat II and evaluate which model may provide the best cost estimate based on FireSat II mission and spacecraft characteristics. First, a traditional large spacecraft cost model will be employed to

Table 11-34. FireSat II Space, Ground, and Launch Segment Costs in FY10$K. Cost estimates based on the CERs in Tables 11-8 and 11-9 using the USCM8 Cost Model and in Table 11-14 for the NASA Instrument Cost Model (NICM). Input Data from FireSat II Spacecraft Mass and Power budgets is from Table 14-28.

Cost Component	CER Input Parameter	Value	Units	RDT&E (NRE) Cost (FY10$K)	1st Unit Cost (FY10$K)	2nd Unit Cost (FY10$K)	Total Cost (FY10$K)	Std Error ($K)
1.1 Spacecraft Bus								
1.1.1 Structure and Thermal**	mass	29.1	kg	6,480	657	591	7,728	1,658
1.1.3 ADCS**	mass	11.6	kg	3,767	3,405	3,065	10,237	3,987
1.1.4 Electrical Power System**	mass	20.9	kg	1,346	678	610	2,634	951
1.1.5 Propulsion**	N/A†							
1.1.6 TT&C and Data Handling**	mass	8.1	kg	26,916	1,680	1,512	30,108	5,419
1.1.7 Integration, Assembly, & Test (IA&T)	S/C Bus and P/L NRE Cost	73,722	FY10$K	14,376	9,142	8,227	31,745	11,943
1.1.9 Flight Software‡	source lines of code	38,100	N/A	20,955	0	0	20,955	6,287
Spacecraft Bus Total Cost				73,840	15,562	14,005	103,407	15,201
1.2 Payload					NRE + 1st Unit Cost			
1.2.2 IR Sensor*		mass(kg)	pwr(W)	Data (kbps)				
1.2.2.1 Fabrication	mass, power, total data rate	30.2	65.0	100	24,120	21,708	45,929	16,040
1.2.2.2 Management	sensor cost	24,120	FY10$K		2,366	2,129	4,496	899
1.2.2.3 Systems Engineering	sensor cost	24,120	FY10$K		3,030	2,727	5,758	1,439
1.2.2.4 Product Assurance	sensor cost	24,120	FY10$K		1,921	1,729	3,650	730
1.2.2.5 Integration and Test	sensor cost	24,120	FY10$K		3,775	3,398	7,173	1,793
IR Sensor Total Cost					35,212	31,691	67,006	16,245
2. Launch Segment	2 Minotaur Launches	50,000	FY10$K	0	25,000	25,000	50,000	5,000
4. Program Level	S/C Bus, P/L, & IA&T NRE Cost	88,098	FY10$K	36,473	28,191	25,372	90,036	29,587
5. Launch and Orbital Ops Support (LOOS)		5,850	FY10$K	0	5,850	5,265	11,115	3,335
6. Ground Support Equipment (GSE)	S/C Bus NRE Cost	38,509	FY10$K	13,627	0	0	13,627	5,042
Total Space Segment Cost to Contractor				123,940	78,966	71,070	273,975	
10% Contractor Fee				12,394	7,897	7,107	27,398	
Total Space Segment Cost to Government				136,334	86,863	78,177	301,373	
Total Cost of Deployment							**351,373**	**37,842**

* IR Sensor CER (1.2.2) is from the NASA Instrument Cost Model (NICM), see Table 11-14. **Spacecraft bus subsystem masses shown include a fraction of the spacecraft mass margin of 14 kg. † FireSat II propulsion system is taken into account in the ADCS CER (1.1.3). The 13.1 kg mass value for ADCS mass includes propulsion system hardware mass. The typical USCM CER for propulsion systems are for reaction control systems and apogee kick motors, not standard spacecraft propulsion systems. ‡ Software cost CER (1.1.9) taken from Table 11-20, unmanned flight CER.

understand what a typical cost estimate for this type of mission would be if traditional procurement and system development practices were employed in developing and launching two FireSat II spacecraft. Table 11-34 depicts a mission cost estimate using the USCM8 cost model (Sec. 11.2.2) for all mission elements except the payload, and the NASA Instrument Cost Model (Sec. 11.2.5) for the IR sensor payload. The total mission cost estimate for FireSat II, with 2 spacecraft, is $351.4 M (FY2010), with an uncertainty of $37.8 M. This is not unusual for a traditional military surveillance spacecraft procurement, and actually is less than traditional costs only because the spacecraft here are much smaller than traditional surveillance spacecraft with IR sensors. This example estimate will serve as the "traditional large mission" cost estimate.

Table 11-35 shows the FireSat II mission cost as estimated using the Small Satellite Cost Model (SSCM) from the Aerospace Corp, as described in Sec. 11.2.3. This full mission cost model is geared toward missions like FireSat II that are small spacecraft (under 400 kg spacecraft dry mass) and have relatively rapid development timelines. The cost that SSCM yields is $116 M total, with an uncertainty of $9.5 M, including an allocation of $50 M for launching both spacecraft on two separate Minotaur launches. This total mission cost is 33.3% of that estimated in Table 11-34 with the USCM8 and NICM models.

Table 11-36 displays a third cost estimate for the FireSat II mission utilizing the QuickCost cost model from NASA, showing a total mission cost of $186 M with a standard error of $56 M. This estimate is 61.3% higher than the SSCM estimate shown in Table 11-35, but is understandably larger because the QuickCost model has much larger spacecraft in its database than does the SSCM database. The QuickCost estimate is 53.7% of the USCM8/NICM cost estimate for FireSat II.

Table 11-34, Fig. 11-8◀, Eq. 11-7◀

Live Calc

Table 11-35. FireSat II Space, Ground, and Launch Segment Costs in FY10$K. Cost estimates based on the CERs in Table 11-11 using the SSCM Cost Model that includes cost estimates for both the spacecraft bus and payload, program level costs, launch and orbital operations support, and ground support equipment. Input Data from FireSat II Spacecraft Mass and Power budgets is from Table 14-28.

Cost Component	CER Input Parameter	Value	Units	RDT&E plus 1st Unit Cost (FY10$K)	2nd Unit Cost (FY10$K)	Total Cost (FY10$K)	Std Error ($K)
1.1 Spacecraft Bus							
1.1.1 Structure*	mass	26.7	kg	2,103	1,893	3,996	2,084
1.1.2 Thermal*		2.3	kg	366	329	695	226
1.1.3 ADCS*	mass	7.0	kg	2,419	2,178	4,597	2,115
1.1.4 Electrical Power System*	mass	20.9	kg	3,553	3,198	6,752	1,729
1.1.5 Propulsion*	spacecraft bus dry mass	69.8	kg	723	651	1,373	589
1.1.6a TT&C*	mass	3.5	kg	786	707	1,493	1,195
1.1.6b Command & Data Handling	mass	4.7	kg	1,255	1,130	2,385	1,623
1.1.7 Integration, Assembly, & Test (IA&T)	spacecraft bus total cost	11,206	FY10$K	1,558	1,402	2,960	888
1.1.9 Flight Software†	source lines of code	38,100	N/A	20,955	0	20,955	6,287
Spacecraft Bus Total Cost				33,718	11,488	45,206	7,522
1.2 Payload							
1.2.2 IR Sensor	spcrft bus total cost	11,206	FY10$K	4,482	4,034	8,517	2,555
2. Launch Segment	2 Minotaur Launches	50,000	FY10$K	25,000	25,000	50,000	5,000
4. Program Level	spcrft bus total cost	11,206	FY10$K	2,566	2,310	4,872	1,463
5. Launch & Orbital Ops Support (LOOS)	spcrft bus total cost	11,206	FY10$K	684	615	1,299	390
6. Ground Support Equipment (GSE)	spcrft bus total cost	11,206	FY10$K	740	666	1,405	422
Total Space Segment Cost to Contractor				41,507	18,497	60,004	
10% Contractor Fee				4,151	1,850	6,000	
Total Space Segment Cost to Government				45,658	20,346	66,004	
Total Cost of Deployment						**116,004**	**9,517**

* Spacecraft bus subsystem masses shown include a fraction of the spacecraft mass margin of 14 kg.
† Software cost CER (1.1.9) taken from Table 11-20, unmanned flight CER.

Live Calc

Table 11-36. FireSat II Space, Ground, and Launch Segment Costs in FY10$K. Cost estimates based on the single QuickCost CER in Table 11-12. All FireSat II inputs are within the QuickCost model ranges of applicability for the various inputs, although at the low end. Input Data from FireSat II Spacecraft Mass and Power budgets is from Table 14-28.

Parameter	Applicable Range	Value	RDT&E plus 1st Unit Cost (FY10$K)	2nd Unit Cost (FY10$K)	Total Cost (FY10$K)	Std Error ($K)
Dry Mass of Bus and Payload (kg)	76 to 14,475 kg	100	71,833	64,649	136,482	55,958
Power (LEO BOL, W)	90 to 10,000 W	183				
Data% (Data rate percentile [fraction] relative to state-of-the-art ATP)	0 to 100%	50%				
Life (Advertised design life, months)	6 to 180 mo	36				
New (% New [fraction], 0.2–0.3 simple mod, 0.3–0.7 extensive mod, 0.7–1.0 New, >1.0 new tech)	20% to 130%	90%				
Planetary (0 = no, 1 = yes)	0 or 1	0				
Year (authority to proceed date in 4 digit calendar year)	1961 to 2025	2011				
InstrComp% (Instrument complexity percentile [fraction], relative to avg. instrument complexity)	0 to 100%	50%				
Team (Team Experience, 1 = Unfamiliar, 2 = Mixed, 3 = Normal, 4 = Extensive)	1 to 4	3				
Launch Cost (FY10$K)—Assume 2 Minotaur Launches	2 Launches –$50M (FY10)	50,000			50,000	5,000
Total Mission Cost					**186,482**	**56,181**

Of the three separate cost estimates for FireSat II discussed in this section, the SSCM estimate in Table 11-35 is the estimate that should be closest to the actual mission costs, based on the type of missions in the SSCM database and similarity with the FireSat II mission.

11.4.2 Supplemental Communications System (SCS)

This section describes 3 cost estimates for the SCS, consisting of three 200 kg spacecraft launched into a 20,500 km circular, equatorial orbit with a mission lifetime of 5 years. Table 11-37 shows a SCS mission cost estimate using the USCM8 cost model, which includes a cost estimate for both the spacecraft bus and communications payload. The total mission cost is $441.5 M (FY2010) with uncertainty of $55.6 M. This will serve as the SCS traditional large space mission type cost estimate for purposes of comparison with other mission cost models.

Table 11-38 shows the SCS mission cost as estimated using the SSCM. The cost that SSCM yields is $165.7 M total with an uncertainty of $11.3M, including an allocation of $50 M for launching all three spacecraft on a single Falcon 9 launch. This total mission cost is 37.5% of that estimated in Table 11-37 with the USCM8 model. This percentage is fairly close to the previous results for the FireSat II mission, comparing the USCM8 and SSCM model results together. For both FireSat II and SCS examples, SSCM yields a mission cost estimate that is about 1/3 of the mission cost generated by USCM8. Again, this reflects the new type of procurement and spacecraft development and test processes employed in the small satellite arena vs. the much more complex and more costly processes employed traditionally for larger spacecraft

Table 11-39 displays a third cost estimate for the SCS mission utilizing the QuickCost cost model from NASA, showing a total mission cost of $435.8 M with a standard error of $158.3 M. This estimate is fairly close to the USCM8 mission cost from Table 11-37, and may reflect the fact that the QuickCost database does not include many communications satellite spacecraft, if any, as NASA's mission database is primarily scientific spacecraft with science instrument payloads. For the same spacecraft dry mass, a science mission will generally yield a higher cost than a communications satellite would, and so the cost generated by the QuickCost model is understandably higher than what would be expected for a communications spacecraft.

Just as concluded in Sec. 11.4.1 for FireSat II, of the three separate SCS mission cost estimates discussed in this section, the SSCM estimate in Table 11-38 is the

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Table 11-37. SCS Space-Ground, and Launch Segment Costs in FY10$K. Cost estimates based on the CERs in Tables 11-8 and 11-9 using the USCM8 Cost Model, which includes a CER for the communications payload. Input Data from SCS Spacecraft Mass budget is from Table 14-26.

Cost Component	Parameter	Value	Units	RDT&E (NRE) Cost (FY10$K)	1st Unit Cost (FY10$K)	2nd & 3rd Unit Cost (FY10$K)	Total Cost (FY10$K)	Std Error ($K)
1.1 Spacecraft Bus								
1.1.1 Structure and Thermal*	mass	56.1	kg	10,164	1,269	2,240	13,674	2,973
1.1.3 ADCS*	mass	26.0	kg	8,421	5,486	9,688	23,595	9,168
1.1.4 Electrical Power System*	mass	34.3	kg	2,206	1,112	1,963	5,281	1,858
1.1.5 Propulsion*	N/A†							
1.1.6 TT&C and Data Handling*	mass	14.6	kg	26,916	2,235	3,946	33,096	5,957
1.1.7 Integration, Assembly, & Test (IA&T)	S/C Bus and P/L NRE Cost	85,406	FY10$K	16,654	10,590	18,700	45,944	16,953
1.1.9 Flight Software‡	source lines of code	22,918	N/A	12,605	0	0	12,605	3,781
Spacecraft Bus Total Cost				76,966	20,692	36,537	134,195	20,821
1.2 Payload								
1.2.1 Comm Payload	payload mass	61.0	kg	37,698	11,529	20,357	69,584	26,761
2. Launch Segment	1 Falcon 9 Lnch	50,000	FY10$K	0	0	50,000	50,000	5,000
4. Program Level	S/C Bus, P/L, & IA&T NRE Cost	102,060	FY10$K	42,253	32,659	57,668	132,580	42,193
5. Launch and Orbital Ops Support (LOOS)		5,850	FY10$K	0	5,850	10,330	16,180	4,854
6. Ground Support Equipment (GSE)	S/C Bus NRE Cost	47,708	FY10$K	16,550	0	0	16,550	6,123
Total Space Seg Cost to Contractor				173,467	64,880	114,562	352,909	
10% Contractor Fee				17,347	6,488	11,456	35,291	
Total Space Segment Cost to Government				190,814	71,368	126,018	388,200	
Total Cost of Deployment							**438,200**	**54,918**

*Spacecraft bus subsystem masses include a fraction of the spacecraft mass margin of 5 kg. † SCS propulsion system is taken into account in the ADCS CER (1.1.3). The 26.0 kg mass value for ADCS mass includes propulsion system HW mass. typical USCM CER for propulsion systems are for reaction control systems and apogee kick motors, not standard spacecraft propulsion systems. ‡ Software cost CER (1.1.9) taken from Table 11-20, unmanned flight CER.

Live Calc

Table 11-38. SCS Space, Ground, and Launch Segment Costs in FY10$K. Cost estimates based on the CERs in Table 11-11 using the SSCM Cost Model that includes cost estimates for both the spacecraft bus and payload, program level costs, launch and orbital operations support, and ground support equipment. Input Data from SCS Spacecraft Mass budget is from Table 14-26.

Cost Component	Parameter	Value	Units	RDT&E plus 1st Unit Cost (FY10$K)	2nd and 3rd Unit Cost (FY10$K)	Total Cost (FY10$K)	Std Error ($K)
1.1 Spacecraft Bus							
1.1.1 Structure*	mass	47.8	kg	3,977	7,022	10,999	3,034
1.1.2 Thermal*		8.3	kg	729	1,288	2,017	329
1.1.3 ADCS*	mass	12.5	kg	3,671	6,482	10,154	3,078
1.1.4 Electrical Power System*	mass	34.3	kg	5,611	9,908	15,519	2,517
1.1.5 Propulsion*	spacecraft bus dry mass	131.0	kg	1,492	2,634	4,126	857
1.1.6a TT&C*	mass	8.3	kg	1,455	2,569	4,024	1,740
1.1.6b Command & Data Handling	mass	6.2	kg	1,546	2,730	4,276	2,362
1.1.7 Integration, Assembly, & Test (IA&T)	spacecraft bus total cost	18,481	FY10$K	2,569	4,536	7,105	2,131
1.1.9 Flight Software†	source lines of code	22,918	N/A	12,605	0	12,605	3,781
Spacecraft Bus Total Cost				33,655	37,169	70,825	7,301
1.2 Payload							
1.2.2 Comm Payload	spcrft bus total cost	18,481	FY10$K	7,392	13,053	20,446	6,134
2. Launch Segment	1 Falcon 9 Launch	50,000	FY10$K	0	50,000	50,000	5,000
4. Program Level	spcrft bus total cost	18,481	FY10$K	4,232	7,473	11,705	3,512
5. Launch & Orbital Ops Support (LOOS)	spcrft bus total cost	18,481	FY10$K	1,227	1,991	3,118	935
6. Ground Support Equipment (GSE)	spcrft bus total cost	18,481	FY10$K	1,220	2,154	3,374	1,012
Total Space Segment Cost to Contractor				46,499	59,849	106,348	
10% Contractor Fee				4,650	5,985	10,635	
Total Space Segment Cost to Government				51,149	65,834	116,983	
Total Cost of Deployment						**166,983**	**11,409**

* Spacecraft bus subsystem masses shown include a fraction of the spacecraft mass margin of 5 kg. † Software cost CER (1.1.9) taken from Table 11-20, unmanned flight CER.

Live Calc

Table 11-39. SCS Space, Ground, and Launch Segment Costs in FY10$K. Cost estimates based on the single QuickCost CER in Table 11-12. All SCS inputs are within the QuickCost model ranges of applicability for the various inputs, although at the low end. Input Data from SCS Spacecraft Mass budget is from Table 14-26.

Parameter	Applicable Range	Value	RDT&E plus 1st Unit Cost (FY10$K)	2nd and 3rd Unit Cost (FY10$K)	Total Cost (FY10$K)	Std Error ($K)
Dry Mass of Bus and Payload (kg)	76 to 14,475 kg	192				
Power (LEO BOL, W)	90 to 10,000 W	614				
Data% (Data rate percentile [fraction] relative to state-of-the-art (ATP)	0 to 100%	50%				
Life (Advertised design life, months)	6 to 180 mo	60				
New (% New [fraction], 0.2–0.3 simple mod, 0.3–0.7 extensive mod, 0.7–1.0 New, >1.0 new tech)	20% to 130%	90%	139,489	246,303	385,792	158,175
Planetary (0 = no, 1 = yes)	0 or 1	0				
Year (authority to proceed date in 4 digit calendar year)	1961 to 2025	2012				
InstrComp% (Instrument complexity percentile [fraction], relative to avg. instrument complexity)	0 to 100%	80%				
Team (Team Experience, 1 = unfamiliar, 2 = Mixed, 3 = Normal, 4 = Extensive)	1 to 4	3				
Launch Cost (FY10$K)—Assume Falcon 9 Launch for 3 SCS Spacecraft	1 Launch–$50M (FY10)	50,000			50,000	5,000
Total Mission Cost					**435,792**	**158,254**

estimate that should be closest to the actual mission costs, based on the type of missions in the SSCM database and similarity with the SCS mission.

11.4.3 Summary

Table 11-40 summarizes all the cost estimates generated for both FireSat II and SCS, as discussed in Secs. 11.4.1 and 11.4.2. In each case, the SSCM yields a cost estimate that is more consistent with the class of mission for each. The other estimates with USCM8 and QuickCost yield estimates which are too high since the databases of missions for each model are generally larger, more expensive missions with much longer development cycles. Therefore, for the same input parameters, they yield total mission costs which are substantially higher than the SSCM model, which has a database of small missions much more consistent with the FireSat II/SCS mission class. It may even be the case that SSCM is not representing FireSat II or SCS as accurately as the cost estimator would like in terms of their relatively small sizes and very short development cycles. These actually are both on the low end of the mass range for spacecraft in the SSCM database. Carrying out more detailed bottoms-up cost estimates for each mission may yield more accurate mission costs which may be less than the estimates generated with the SSCM.

Table 11-40. Summary of Alternate Mission Cost Estimates for FireSat II and SCS Using Three Cost Models.

Cost Model	FireSat II	SCS
USCM8	$362.1 ± 39.2 M	$441.5 ± 55.6 M
SSCM	$120.5 ± 9.6 M	$165.7 ± 11.3 M
QuickCost	$194.3 ± 59.4 M	$435.8 ± 158.3 M

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12 Space System Financing and Space Law

12.1 Sources of Space Financing
Developing Business Strategies; Finance Source Summary; Government; Corporate

12.2 GAAP, Amortization, and Return on Investment (ROI)
Useful Parameters and Terms; Government Accounting; Effects of Program Lifetime; Role of Insurance; Price vs. Demand; Project Life Cycle

12.3 Law and Policy Considerations
Foundations of International Space Law; Outer Space Treaty; Liability Convention; Registration Convention and Rescue and Return Agreement; International Cooperation; US National Space Policy; Import and Export Controls and Technology Transfer Constraints; Import Restrictions; Commercial Space Launch Act of 1984; Satellite Communications Law; Intellectual Property Rights; Insurance; Remote Sensing; Space Debris; Environmental Concerns; Lasers, Particle Beams and Nuclear Power

12.1 Sources of Space Financing

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12.1.1 Developing Business Strategies

An accepted rule of thumb in the commercial arena for high risk investments is that they should return at least 18% and preferably 25% annually on invested capital [Jurist et al. 2005]. This means that an established investor will expect his or her investment to increase in value by that amount every year. This return must be built into any plan to attract investment capital as a means of financing a space business. Otherwise, the potential investment is not attractive.

Developing a business strategy is simple in concept but often difficult in its execution. The strategy must be converted into a plan to make money for investors in the business. That business plan is normally used to attract investment and is generally the initial iteration of a continuously evolving plan to implement the stated strategy.

The plan should delineate how to get from the current status quo to the desired end state in a step by step fashion. It must be clear and readily understood by management and potential investors. The reader of such a plan should be able to use it to visualize the day to day activities of the management team members. Details of creating an effective business plan are beyond the scope of this textbook.

12.1.2 Finance Source Summary

The next sections describe the various following sources and methods of funding, as well as associated considerations that may be relevant to space finance. They are summarized in the adjacent boxed section:

Potential Sources of Finance

Government
- Procurement
- Contracted R&D
- Small Business Innovation Research (SBIR)
- Small Business Administration (SBA) Loans
- Grants

Corporate
- Stock
- Funding with income stream
- Debt financing
- Public/private partnership financing
- Venture capital (VC)
  - Primarily for startups
  - Return expected within short time
  - Active participation

Angel Investors
- Individuals
- Family, friends, and fools (FFF)
- Similar to venture capital

Convertible Instruments and Debentures

Non-Profits
- Limited opportunities
- Grants

Prizes

12.1.3 Government

The US government is a major source of space funding either directly or indirectly. Governmental funding is typically negotiated in a procurement contract. This is an agreement between the government and a supplier or vendor to provide goods or services in exchange for government money. Multiple mechanisms used in these contracts include fixed price, cost plus a fixed fee, and cost plus a fixed percentage contract. Various hybrid arrangements can also be created. Contracted R&D

services are frequently sought by the US government. Basically, a statement of work to be performed, a budget, a reporting arrangement, and various performance and payment milestones to monitor progress are negotiated between the supplier and the funding agency and, depending on budget appropriations consistent with congressional actions, may be funded. Contracted R&D can also be performed by a subcontractor to a larger corporation.

The Office of Technology within the *Small Business Administration (SBA)* administers the Small Business Innovation Research (SBIR) Program and the Small Business Technology Transfer Research (STTR) Program. SBA does not directly fund projects funded under these programs.

The *Small Business Innovation Research (SBIR)* program is a set-aside program (2.5% of a government agency's extramural budget) for domestic small business concerns to engage in research and development that can be potentially commercialized or exploited by commercial interests as an addition to the US economy. If those ideas meet the research and research and development needs of the specific government agency, they may be funded. SBIR is a competitive 3-phase award system which provides qualified small business concerns with opportunities to propose innovative ideas:

- Phase I is a feasibility study to evaluate the scientific and technical merit of an idea. Awards are for up to six months and up to $150,000
- Phase II expands on the Phase I results and further pursues the development of the idea. Awards are for up to 2 years and up to $1,000,000
- Phase III is for the commercialization of the results of Phase II and requires the use of private sector or non-SBIR federal funding. Although US government agencies cannot fund Phase III SBIR projects, they may provide technical assistance

A small business is defined as a for-profit business located within the US and operating primarily within the US with no more than 500 employees. Non-profits are not eligible. The business must be owned and controlled by one or more US citizens or permanent resident aliens in the US. It may be an individual proprietorship, partnership, limited liability company, corporation, joint venture, association, trust, or cooperative. Joint ventures can have no more than 49% participation by foreign business entities.

A significant requirement is that the principal researcher's primary employment must be with the small business during the duration of the project. Innovations to be developed under SBIR cannot be patented or have patents pending. Part of an SBIR award may be subcontracted:

- For Phase I, a minimum of 2/3 of the research and analytical effort must be performed by the proposing firm
- For Phase II, a minimum of 1/2 of the research and analytical effort must be performed by the proposing firm

Small Business Technology Transfer Research (STTR) is different in that it specifically involves partnership of ideas and technologies between a for-profit business and a non-profit organization. All federal agencies with annual extramural R&D budgets exceeding $1 billion devote 0.3% of their extramural R&D budgets to STTR. Extramural R&D is that funded by a government agency but performed by the grantee—usually an academic or commercial organization. Projects funded under the STTR program consist of 3 phases:

- Phase I is a feasibility study that determines the scientific, technical, and commercial merit and feasibility of a selected concept. Phase I projects are competitively selected from proposals submitted against solicitations. Each solicitation contains topics seeking specific solutions to stated Federal government needs. The Phase I selection process is highly competitive; with perhaps 10% of submitted Phase I proposals receiving awards
- Phase II represents a major research and development effort, culminating in a well defined deliverable prototype—a technology, product, or service. The Phase II selection process is also highly competitive. Only successful Phase I contractors may submit Phase II proposals as there are no separate Phase II solicitations
- In Phase III, the small business or research institution is expected to obtain funding from the private sector and/or non-STTR government sources to develop the prototype into a viable product or service for sale in the government or private sector

The STTR program is open to any organized team consisting of a small business and a research institution. The goal is expansion of the public/private sector partnership to include joint venture opportunities for small business and the nation's premier non-profit research institutions by funding meritorious joint proposals. The most important role of STTR is to foster the innovation necessary to meet the scientific and technological challenges facing the US. The research institution may be any US non-profit research institution, federally funded research and development center, university, or college. The small business must perform at least 40% of the Phase I and Phase II work. The research institution must perform at least 30% of the Phase I and Phase II work. Up to 30% of the work may be subcontracted. The principal investigator must be primarily employed with either the small business or the research institution for the Phase I and II efforts. The Department of Defense generally issues 2 STTR solicitations each year.

The SBA oversees a wide variety of programs and services intended to support small businesses in addition to those described above.

The SBA does not fund grants to start or expand small businesses, but does offer a variety of loan programs. SBA grants are instead intended to expand or strengthen organizations that provide small business management, technical, or financial assistance and are usually awarded to state and local governments, intermediary lending institutions, and non-profit organizations. Because the conventional commercial loan market may not offer many small businesses the capital access needed to grow and remain strong, SBA manages multiple financial programs for small businesses. These include SBA-backed loans and government contracting opportunities. Additionally, SBA offers disaster assistance and a wide variety of training programs.

12.1.4 Corporate

Corporate funding allows a variety of approaches with various advantages and disadvantages. The advantages and disadvantages for R&D funding may differ from those for other purposes.

A corporation can issue stock at a reasonable market value. Stock purchases do not have to be paid back by the company like loans, but the buyers expect a return—either as dividends (money paid back to investors from corporate earnings) or as capital gains (increased value of stock over time). Issuance of public stock is regulated by the Securities and Exchange Commission (SEC) and state corporation commissions. The issuance of private stock for closely-held corporations is governed by a series of SEC and state regulations specific to private stock investments at both the federal and the state level. A private stock offering may need to qualify in every state involved in the business, serving as residences for potential investors, or where the offer is made. How the corporation is actually structured may also depend on which state will serve as the corporate home since states have differing rules, taxes and fees, often based on the number of outstanding shares and the classification of those shares.

Stock can come in the form of common stock, preferred stock, warrants, and even convertible preferred stock. Founders and key corporate employees can take advantage of Letter Stock, which is not registered and cannot be sold or transferred until it is registered. Registering stock is complicated and can happen only if certain other conditions exist. Rather than defining all the types and classifications of stock, it suffices to state that all forms of stock represent a fractional share in the company rather than debt. The equity represented by stock is at risk if the company fails. If the stock is classified as preferred or guaranteed, it has some asset protection behind it, mainly for dividend payments. It would have a slightly preferred status over common stock in a bankruptcy liquidation of the corporation

Corporate R&D can potentially be funded out of an internal revenue stream. However, this approach must compete with other potential uses of the revenues, such as paying off outstanding loans or paying shareholder dividends.

A corporation with adequate credit can simply borrow the money needed to finance a space project. Traditional loan sources frequently require securing the loan with corporate assets, or, in the case of smaller closely held corporations, personal guarantees of the shareholders. Unlike a stock issue, loans must be paid back. Principal amounts are carried as liabilities for the term of the loan and interest payments are treated as expenses. Generally, principal repayments are made out of pretax dollars.

Numerous combinations of public and private financing sources and mechanisms exist. They can be difficult to negotiate—especially if closely held corporations are involved.

Venture capitalists (VCs) exist to fund smaller companies in order to realize large gains as companies in which they invest grow and profit. Although specific conditions of investment vary, venture capitalists usually expect returns of at least 20% annually on their investments. Many small startup companies attempt to acquire funding for expansion from a VC source. A small company will assume that it can readily access this capital pool only if the corporate principals do not understand opportunity cost and the necessary foundation of venture capitalism. While VC pros may take financial positions in companies that they might not normally consider, a candidate space company must still face competition from an extremely large pool of competitive terrestrial opportunities. Furthermore, terrestrial investment opportunities have a long and established history of working with the VC industry. Space companies do not. Space projects generally take longer than the time horizon of a typical VC. That VC usually expects significant returns (18%/yr–25%/yr) on his or her investment within a 2–5 year time frame.

There is a high potential price to pay when accepting VC money and support. If things do not follow the business plan, or they head south, company managers may have their lives made miserable by the active involvement of the VC. This may even lead to the forced break up of the company. When accepting VC money and support, company management must willingly allow the participation of outside team members, a change in the company vision, and VC representation on the board of directors. In extreme circumstances, the company must accept replacement of management and the board of directors as well as other restrictive and possibly dramatic changes. How many NewSpace company management teams are open to VC management influence and takeover if the company does not perform as promised? How many management teams of space companies understand the true scarcity of capital due to lucrative alternative investments (opportunity costs) and otherwise competitive market forces among terrestrial business ventures competing for the same investment dollars? How many management teams really understand that it is not usually about the rocket science or space objective, rather it is mostly about the return on investment (ROI), payback period, risk assessment for both business and political risks, and internal rates of return (IRR)? These terms are defined in Sec. 12.2.

To focus on space and space objectives rather than the fundamentals that make a business venture attractive for capital acquisition is to delay NewSpace industrial development. Although wealthy players and benefactors are important, a handful of them with their private investments do not constitute an industrial development program.

Angel investors generally have a personal interest in the goals of the candidate company that transcends simply maximizing investment returns. They generally want to participate in the early formation and growth of a startup. In a NewSpace company, the typical angel is a "space cadet"—often times with more enthusiasm than sound business judgment. Angel investors often seek to place their funds in companies with a high "coolness factor" in order to gain bragging rights. Yet they expect eventual returns on their investments unless they are completely ego-driven. A typical example of the latter might be the owner of sports teams who earned his or her fortune in other venues. In a sense, angels are similar to VCs by typically wanting an eventual return, but they differ in their willingness to accept longer time horizons before seeing returns.

Often corporate startups, whether space-related or otherwise, begin with an individual. That individual frequently has significant technical training and a bright idea but often lacks business experience or understanding of the intricacies of running a new and growing business. Failure rates of these startups are correspondingly high.

Traditional funding sources first sought by an individual entrepreneur who lacks the means of financing his or her company from personal funds or personally guaranteed credit lines are from family, friends, and fools (FFF). After that, "outside investors" are commonly sought.

Convertible instruments can be used for corporate funding. One of several forms is the convertible loan, in which a funding source puts money into a company to fund a project. That money is treated contractually as a loan that can be converted into stock at some future date and at a negotiated price or simply treated as a loan depending on circumstances. Those circumstances must be spelled out in detail in the instrument documentation and agreed upon by all parties. In the US, a debenture is an unsecured corporate bond. Other countries may treat them differently. A debenture is a medium or long term debt incurred by borrowing money. In legal terms, a debenture is the document that creates and describes the debt. Debentures are usually transferable by the holder, who has no voting rights. Interest paid to debenture holders is charged against corporate profits (pre-tax dollars) rather than being treated as a deductable expense in the company's financial statements.

Non-profit corporations and foundations are treated differently than typical for-profit corporations by the government. Generally, they must have a charitable purpose or a charter to improve society in some sense. They frequently fund R&D for projects that are congruent with their specific societal interests.

Opportunities for standard for-profit companies to obtain funding from non-profits are very limited because of tax laws that limit the non-profit's ability to engage in businesses that are operated for profit.

Project funding from non-profits generally are in the form of grants. The non-profits or foundations generally require an application that explains the nature of the intended project, the methodology to be used, the goals of the project, plans for reporting progress, and a budget. If approved and funded, money is transferred to the grantee and the project can begin.

Ever since the Ansari X-Prize was offered and won by Scaled Composites of Mojave, California for creating SpaceShipOne and flying it to apogees of over 100 km under specified conditions, prizes seem to be proliferating in the NewSpace arena. They can be offered by private non-profits, (as was the X-Prize) by a foundation, or they can be offered by governmental agencies, such as NASA's Lunar Lander Challenge. This contest was redesignated the Northrop-Grumman Lunar Lander Challenge in deference to that corporation's partial sponsorship of the prizes. The NASA Office of the Chief Technologist has extended the Lunar Landing Challenge concept by issuing a set of "Centennial Challenges" with associated prizes. As a general rule, the prizes are offered to the first team (or first few teams) that succeed at some difficult task while complying with a specified set of preconditions. Recently, Masten Space Company of Mojave won the Lunar Lander Challenge. Prizes are not usually cost effective for the winner and are certainly not cost effective for the losers. The $10 million Ansari X-Prize reportedly cost $30 to 40 million to win. Winning is therefore more about ego-feeding and prestige than money, although winning a prize may increase a small company's attractiveness to potential angel investors.

12.2 GAAP, Amortization, and Return on Investment (ROI)

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12.2.1 Useful Parameters and Terms

General Accepted Accounting Practices (GAAP)

Private sector business accounting follows a set of rules that control and govern how corporate financial reports are made. The rules are set by the Financial Accounting Standards Board (FASB) [2010] and are recognized by both the Securities and Exchange Commission (SEC) and the American Institute of Certified Public Accountants (CPAs). These organizations can and do vary in accounting rules depending on the type of business. Special accounting rules may apply that are unique to a specific industry. These rules exist because financial activity can be reported in a way that can potentially impact the bottom line of a company, its

stock value, and its business operations by misleading the person reviewing the accounting statement. The rules are intended to minimize the potential for being misled.

The accounting rules are detailed in the GAAP [GASB, 2010]. This term has a unique meaning among accounting, business and auditing professionals. Frequently a financial report contains a professional opinion about the financial health of the company being reported on or being audited. The American Institute of CPAs' Code of Professional Conduct does not allow its members to provide an opinion on the financial health of a company unless the report meets GAAP. A corporate annual audit report or audited financial statement frequently contains a statement such as:

> **In our opinion, the financial statements referred to above present fairly, in all material respects, the consolidated financial position of ABC Company as of December 31, 2010 and the changes in their net assets and their cash flows for the year then ended are in conformity with accounting principles generally accepted in the United States of America.**

A company's accountants and auditors must be able to provide this "clean" statement in order to accurately reflect the financial health of the company for the reporting period in question. It is important to note that financial statements, accounting and audit rules may differ in different foreign countries, even if the parent corporation is an American company.

While management is not expected to know all the detailed accounting rules that are in play for the company and its industry sector, management must know that outside professional and legal rules govern the financial conduct and reporting for the company.

Engineers and other specialists within the company do not work in a vacuum sheltered from these rules and principles. It is possible, for example, that an engineering decision may need to be made based on how the consequence of the decision is financially reported by the company. Engineers and specialists should have some familiarity with the broad overlay of GAAP.

The rules for non-profit corporations and for government accounting related to reporting grants and contract work differ somewhat from GAAP. Relevant staff members may need to become familiar with more specialized accounting rules in addition to GAPP including rules that may affect foreign subsidiaries or operations.

GAAP rules allow uniform reporting of corporate financial conditions. Analytical comparisons of different entities require the use of some terms, tools, and methodologies that are also used by corporate management in fulfilling their duties. Several of these are discussed below.

Interest

Interest is the value of money expressed in percentage terms over time. If one invests a certain amount of cash and the investment is worth more cash at some later date, the difference is interest. With a loan, an individual is expected to pay interest back to the lender along with the original amount of the loan (principal). If a bank savings account pays interest of 3%/yr, a dollar placed into that account today will be worth $1.03 in 1 yr.

Dividends

Dividends are funds paid to investors in a corporation out of corporate profits. They are usually evaluated by investors in percentage terms relative to the original amount invested or as specified cash amounts per share of stock owned. The US tax code occasionally treats dividend income differently from interest income.

Capital Gains

If a company is profitable, some of the profits are paid to the US and state governments as corporate income taxes. Other portions of the profits may be paid to investors as dividends or to retire debts. If some of the profits are retained by the company, the company increases in value. That value is reflected by the potential sale of shares of stock for a higher price than their original purchase price. The difference in purchase price and sales price of stock is a *capital gain*. If the stock has been held for a long period of time, the gain is a long-term gain. If the stock was bought and sold over a short period of time, the gain is a short-term gain. The US tax code treats long-term capital gains differently from short-term capital gains at various times and defines long and short term differently according to the whims of the US Congress.

Payback Time

A simple way to evaluate a potential investment is to determine how long it takes to get money back from investment revenues. This is called the payback method. For example, if a company invests $1,000,000 and it takes 5 years to get it back, the payback is $200,000/yr. After 5 years, any additional returns shift to profit on the investment. In this example, management must decide if getting all the invested money back in 5 years is acceptable. The payback method ignores inflation, imputed interest, or the decreased value of money as time passes. Put another way, payback time ignores the time value of money impacting cash flow and investment return. It also ignores risk, and it ignores accounting rules that may prevent the complete classification of cash flows from the project as being treated as a return of the investment principal.

When value of an investment over time is considered, the concept of time value of money becomes compelling. The basic concept behind the time value of money is a proposition: "Would you rather have a dollar today or a dollar 10 years from today?" The rational choice is to take the dollar today because the real value of a dollar in 10 years is unknown and probably reduced given inflation. Next, what would the effective interest rate have to be to make the amount 10 years from now equivalent in value to the dollar today? This is discussed in detail in the following sections.

Table 12-0◀, Fig. 12-0, Eq. 12-0

Time value oriented investment tools are far more useful than calculations of simple payback time. Nevertheless, the payback method has its use, particularly for space investments because many of them are very long term. This is especially significant when comparing potential space investments to other terrestrial investment opportunities of similar risk. For example, knowing that you will not get your money back on an investment for 10 years, assuming the investment goes according to plan, tells you right away that you are engaged in something long term. If you are not prepared to invest and wait for your return over that period of time, you would typically not go any further in the analysis of that particular investment opportunity. Using the simple payback method is a very useful first step in evaluating the basics of any investment project or venture, but more sophisticated methods are used for those projects or ventures that pass the payback time test.

Net Present Value (NPV)

A commonly used analytical tool is NPV. NPV is based on a very simple concept: How much money does it take today (present value) to be equal to a specified amount at some time in the future (future value) if it were returning a specific rate of interest from now until that future date? That interest rate is called the discount rate and is the cost of money. NPV allows the measurement of a time series of cash flows in terms of the present value of the investment at an accepted discount rate.

Present value is defined by the following formula:

$$PV = FV / (1+i)^t \qquad (12\text{-}1)$$

In this formula, FV is the value of some cash in some future year or future value, i is the discount rate or effective interest rate, and t is the number of years between now and the specified future year. Restating this formula for FV will yield the future value of a current asset PV invested at an annual interest rate of i for t years. This formula is commonly used to show returns on savings accounts offered by banks. Interest rates may be fixed by contract as they often are in savings accounts, or they may fluctuate according to financial market conditions.

For example, if a person has a savings account that contractually pays interest at 3%/year and wants to have \$10,000 in the account in 10 years, the amount that is required to be put into the account today (PV) is $\$10{,}000/(1.0+0.03)^{10}$ or \$7,440.94.

In terms of a simple individual savings account, future value defines how money saved today will grow over time at a known return. In business terms, Net Present Value or NPV is the difference in the present value of future incomes less future expenses as applied to specific investments or projects.

There are several ways to calculate NPV, but a common formula is:

$$NPV = \sum_{t=1}^{Total\ years} \frac{R_t}{(1+i)^t} \qquad (12\text{-}2)$$

Table 12-0, Fig. 12-0, Eq. 12-2

In this formula, R_t is the net cash flow (revenues less expenses) in some single future year, i is the discount rate or effective interest rate, and t is the number of years between now and each specified future year. The total NPV is the sum of the present values calculated over all years under consideration.

Assigning an appropriate discount rate or cost of money is critical. NPV shows the value for future cash flows discounted back to present value using the cost of capital or some other parameter that represents the minimum acceptable return for the project or investment. NPV includes an assessment of risk built into discount rate. The riskier the venture, the more heavily the project is likely to be discounted, thus suggesting that larger returns are required in order to compensate for the risk.

Internal Rate of Return (IRR)

Internal Rate of Return or IRR is another common investment analysis tool. IRR shows the sum of the NPVs of all the cash flows over all years related to the proposed investment or project as zero based on a defined discount rate. IRR is used to select projects that have returns in excess of the cost of capital. When using IRR, one usually incorporates an anticipated cost of capital rather than some other discount rate. Risk is also factored into the discount rate. If the venture is extremely risky, the cost of capital will be much higher than were the investment less risky. Generally, in an investment, there is an initial outflow of money in the first year that is to be offset by the future net inflows discounted to the present year. The IRR is the effective interest or discount rate such that the present values of the outflows and the inflows, including capital, add up algebraically to zero.

The concept of IRR can be illustrated by a simple example. Assume a project to develop a sounding rocket will cost \$10,000,000 today and the discount rate or cost of money is 5%/yr for this prospective project. Then, assume that the net sales (revenues from sales less manufacturing and sales costs) are expected to be \$1,500,000/yr for 3 years, starting in the 2nd year, and \$2,000,000/yr for the rest of the 9 year project.

Table 12-1 shows the projected annual cash flows over the 9 year duration of the project, the PV of each annual cash flow at the 5%/year discount rate, and the NPV of each cash flow at the IRR of 8.503%/yr.

The initial cash outlay is returned in the 7th year of the project (payback time). The total net return on the project over its lifetime is \$4.5 million. The anticipated and significant effects of time are shown in the Present Value column. Assuming that the 5% discount rate approximates the market cost of money over the 9 year duration of the project, investing the \$10 million would return, if all goes well, a net gain of \$1,564,802 in beginning year dollars. According to the last column in the table, the total NPV over all years of the project is zeroed out at a discount rate of just over 8.5%. Therefore, the contemplated project is expected to return an IRR of a bit over 8.5% assuming accurate cash flow calculations. At this coarse level of analysis, the proposed project appears to

Table 12-1. Illustrative Sounding Rocket Project Cash flow Analysis.

Discount	0.05	per year	
IRR	0.085033816	per year	
Year	*Cash Flow*	*Present Value (PV)*	*PV at IRR*
1	$ (10,000,000)	$ (10,000,000)	$ (10,000,000)
2	$ 1,500,000	$ 1,428,571	$ 1,382,445
3	$ 1,500,000	$ 1,360,544	$ 1,274,104
4	$ 1,500,000	$ 1,295,756	$ 1,174,252
5	$ 2,000,000	$ 1,645,405	$ 1,442,969
6	$ 2,000,000	$ 1,567,052	$ 1,329,884
7	$ 2,000,000	$ 1,492,431	$ 1,225,661
8	$ 2,000,000	$ 1,421,363	$ 1,129,606
9	$ 2,000,000	$ 1,353,679	$ 1,041,079
Totals	$ 4,500,000	$ 1,564,802	$ 0

be a good deal since the IRR is greater than the discount rate unless one can find a better use for the $10 million. That is, an alternative with a higher IRR. Deliberately forgoing a project with a higher IRR for one with a lower IRR causes an *opportunity cost* to be incurred.

Return on Investment (ROI)

ROI is the annualized return or earnings expressed as a percentage of the total investment. For a simple savings account, ROI would be the stated interest rate to be paid on the account (less fees charged to the account, if any). Several tools are available to determine the value of a series of cash flows for a business. This facilitates evaluation of a potential investment and the likely ROI. Using the various ROI tools also allows comparison of likely ROI to the cost of money.

Earnings Before Interest and Taxes (EBIT)

EBIT allows analysis of profitability before applying interest and tax expenses to the gross earnings. Tax rates, accounting rules, and even interest earned can skew the net earnings number, so examining gross earnings before these expenses are taken out allows measurement of a company's profitability at a gross level. Earnings are not generally tied to one particular investment or project, but are totaled across the entire company. As an evaluation tool, EBIT facilitates comparison of relative profitability and efficiency despite variation in the tax, financial, and accounting structures of different companies.

12.2.2 Government Accounting

Standards for financial reporting and accounting for government contracts and grants is a highly specialized subset of GAAP. There is a formally defined set of Federal Financial Accounting Standards approved by the Government Accounting Standards Board [GASB, 2010].

12.2.3 Effects of Program Lifetime

Program lifetime can be altered with sometimes profound effects on total budgets. A powerful method of dealing with the interactions between program lifetime and projected cost is discussed below.

Critical Path Analysis and Program Evaluation and Review Technique (PERT) are management methodologies that formalize decision making in complex projects. PERT was first used during the late 1950s in the development of the Polaris submarine launched ballistic missile [Durrwachter, 2010].

In Fig. 12-1, a simplified flowchart for a hypothetical launch vehicle project is shown. The initial step of providing preliminary funding from internal sources is followed by 3 independent steps that can be performed in parallel:

- Vehicle Characterization
- Market Analysis
- Study Regulatory Process

According to PERT rules and the structure of the flowchart, all 3 of these steps must be completed before the following step of Cost Analysis can be started. Assume that the 3 tasks will require 2 wks, 12 wks, and 4 wks to complete, respectively. The time-limiting step is Market Analysis at 12 wks. Vehicle Characterization consumes 2 wks and has a slack of up to 10 wks and Study Regulatory Process has a slack of up to 8 wks without delaying the project under this scenario. The critical path in any project is the longest possible continuous pathway from the beginning of a project to its completion. It defines the total calendar time required for the project. Time delays along the critical path will delay project completion. The critical path is the longest parallel path (Market Analysis) in this simplified illustration.

Table 12-1, Fig. 12-0, Eq. 12-2◀

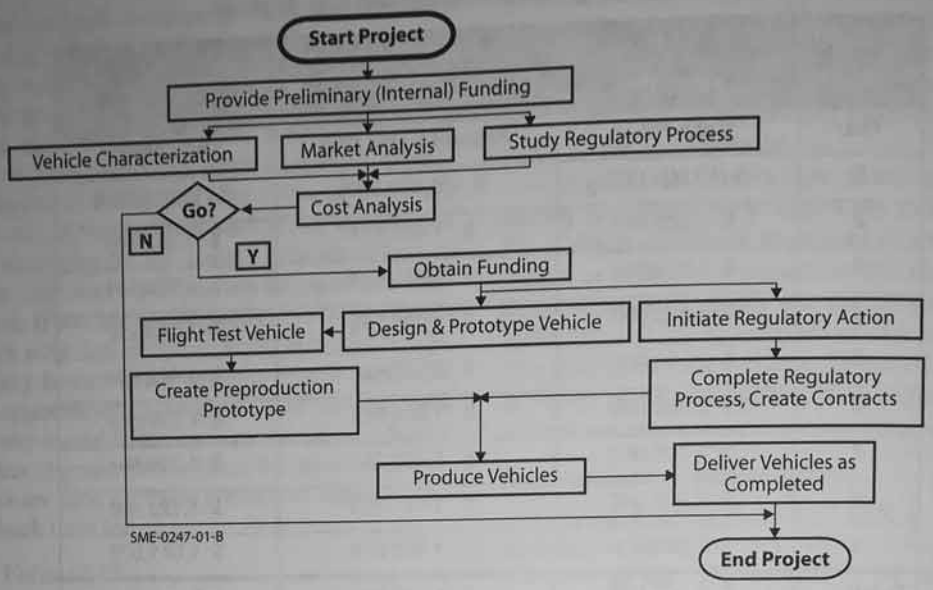


Fig. 12-1. Illustrative PERT Chart.

If resource limitations are responsible for the 12 wk requirement for the Market Analysis, managers could potentially divert resources from the two shorter tasks to reduce the time required for the time-critical Market Analysis and also consume some of the slack in the other 2 tasks. Of course, that assumes that resources can be interchanged freely and that adding additional resources reduces the time required to complete a task.

For each task or step, a minimum or optimistic, maximum or pessimistic, and most likely (best estimate) time and cost are specified for completion of that task or step. Frequently, 10th, 50th, and 90th percentile times and costs are estimated and used for each task or step. If the optimistic time is defined as the 10th percentile time, the probability of completing the project step within that time is 10%. If the pessimistic time is defined as the 90th percentile time, the probability of completing the project step within that time is 90%. The expected time (or cost) is given by the formula:

$$T_e = \left(T_o + 4 \times T_b + T_p\right) / 6 \qquad (12\text{-}3)$$

In this formula, T_e is the expected time, T_o is the optimistic time, T_b is the best estimate or most likely time, and T_p is the pessimistic time for a given task. Filler steps (slack) can be used as dummy tasks for clarity of the overall process. Slack is the time that a task can be delayed without causing a delay in overall project completion. Tasks or steps can be redefined with experience or as initially unanticipated problems arise. This is where the power of PERT becomes obvious. Software, such as Microsoft's Project, is available and is especially valuable for simplifying the analysis tasks or for dynamic reanalysis of changing conditions associated with the project. PERT is ideally suited for management of complex one time projects—especially if the project must be dynamic in that multiple changes are expected during the lifetime of the project. This property makes PERT useful for managing complex research and development projects. PERT can also monitor project compliance with schedules as part of the management or compliance monitoring process.

If a project has a "drop dead date," PERT can be used to examine the effects of changing tasks, changing their order, or running them in parallel on the final completion date. Dollars can be traded for time with some tasks. In other words, throwing money at a project by paying overtime or adding personnel can accelerate completion of some specific tasks. However, this is not always the case. Nine women cannot make a baby in one month. Delaying or stretching out a project can increase total project costs as well. Koelle [2007] developed an empirical relationship between relative cost and deviation (either acceleration or stretch-out) from the optimal time schedule in the aerospace industry. This is illustrated in Fig. 12-2.

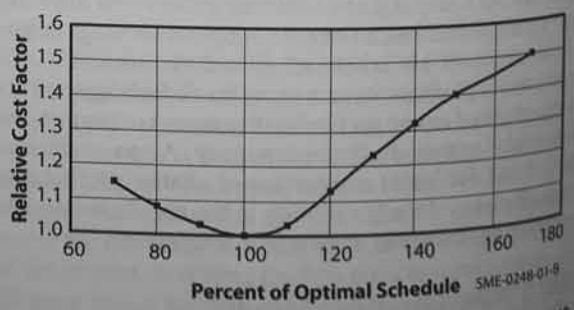


Fig. 12-2. US Aerospace Industry Cost Trends Versus Schedule Duration [Koelle, 2007].

PERT can be used to meet fixed project costs by iteratively refining or redefining tasks and their associated costs. With increasing experience, repetitive tasks, including

Table 12-1◀, Fig. 12-2, Eq. 12-3

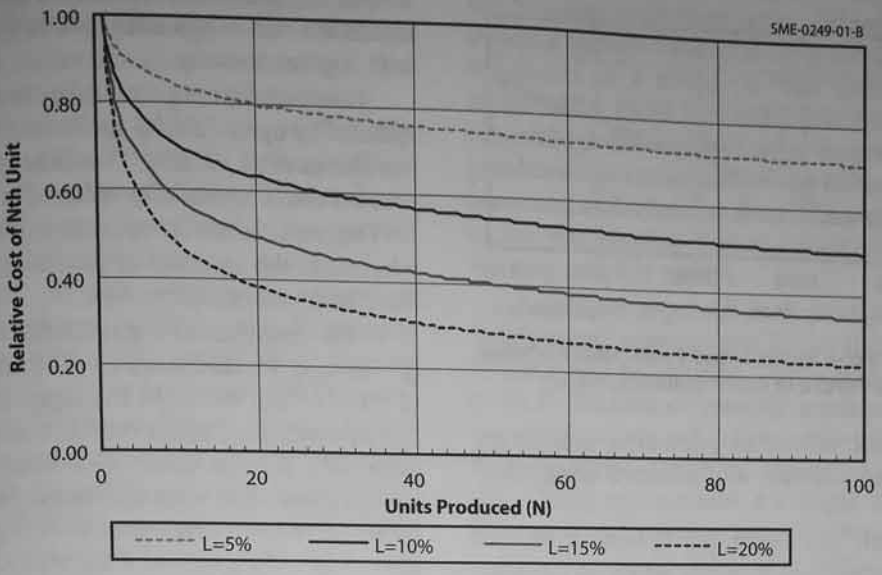


Fig. 12-3. Cost of Producing the nth Unit as a Function of Number of Units Produced and Learning Factor.

production, can gradually become more efficient. Empirical observations have shown that associated production efforts, as quantified by the surrogates of cost, man hours, or both, decrease by approximately constant percentages for each doubling of production. A learning or experience curve can frequently be applied to production costs of automobiles or other non space-related products, aircraft, or spacecraft. In addition, it can even be applied to such things as, for example, maintenance hours per flight hour of aircraft.

For aerospace production, the learning factor typically reduces production costs by 10%–15% for each doubling. If a learning factor of 12% is used, a production run of 6 vehicles should cost 4.93 times the cost of producing the 1st vehicle (TFU).

In simplistic terms, one of the several formulations of the learning or experience curve is described by the following power law equation:

$$C_n = C_1\left(1.0 - L/100\%\right)^{\log n/\log 2} \qquad (12\text{-}4)$$

Here, C_n is the cost or labor needed to produce the nth unit and C_1 is the cost of producing the first unit (usually described as TFU). L is the learning factor in percent.

Figure 12-3 illustrates the cost of units produced relative to that of the TFU for learning factors ranging from 5% to 20%. Underlying assumptions in the figure are that all units are identical and produced without major changes in production equipment or technology. These factors and major unit redesigns can produce discontinuities in the cost versus production number curves. Readers interested in more details should consult the relevant literature.

Six Sigma is another popular management tool that has been in use since the early 1980s. It is based on statistical tools and people specifically trained in the method to gain expertise in solving organizational problems. Companies interested in the Six Sigma approach have their staff trained and certified with sophisticated and progressive certification levels. Six Sigma projects follow specific paths with specific protocols.

Figure 12-4 illustrates the historical Consumer Price Index compiled by the US Bureau of Labor Statistics. Comparison of this graph with Fig 1-14 shows that aerospace industry labor costs inflated more than consumer prices in the US—particularly in the mid-1970s.

Interest accounts for 2 factors: The time cost of money as reflected by currency inflation and the risk associated with promises of future payment. NPV methodology has been discussed in terms of a discount rate or effective interest rate. That rate must compensate for inflation (decreased value of money with time) and also for perceived risk of a default in payment. Acceptable cash in hand is worth more than a promise of cash to be delivered in the future because events may transpire to prevent delivery on that promise.

Projects involving development of a product such as a launch vehicle, especially if funded by governmental sources, have a significant probability of being cancelled. Cancellation of such projects is usually supported by a contract clause allowing termination "for the convenience of the US government." That "convenience" may result from the vendor failing to meet project milestones or from changing procurement needs, budgetary changes imposed by Congress, or other non-technical factors. Financial planning for a specific project, whether involving space or not, should include compensation for the risk of cancellation. That risk may be addressed by building it into the pricing structure or it may be addressed explicitly by incorporation of cancellation terms into the project contract. Cancellation terms are especially important in government contracting because funding is subject to the whims of congressional appropriations—and the US Congress is sometimes perceived as being fickle. Cancellation can damage a contractor as it ripples into risks of dispersion of experienced design

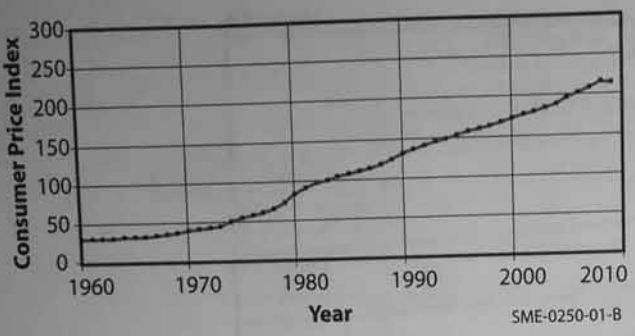


Fig. 12-4. Consumer Price Index Trend From 1960 Forward. Modified from CPI [US Bureau of Labor Statistics, 2010].

and fabrication teams, effects on plant size, specialized capital equipment acquisition, and assorted contractual agreements.

Projects may fail, and some perception of risk of failure should be considered in financial planning. Risk of project failure for technical reasons has been significant in an historic sense. Sometimes a project is initiated which is eventually seen as "a bridge too far."

12.2.4 Role of Insurance

Insurance is a way of protecting against relatively low or medium probability risks of unforeseen events with large economic consequences. In the space industry, insurance is needed by the launch provider in case of a catastrophic failure resulting in loss of payload, damage to uninvolved parties, or failure to properly place the payload. The payload provider needs insurance in case of a launch failure. The end customer also needs insurance against any failures to have a functional payload available for use. In the space industry, insurance is frequently offered as a percentage of the launch cost—often in the 8%–10% range.

Space launches involve three components of insurance. Those components are intended to provide protection against loss of payload, loss of vehicle, and liability coverage for potential damage to third parties from, for example, falling debris. Computation of insurance costs is extremely complex and dependent on specific circumstances. Regardless of how insurance is priced for the customer, costs are obtained by generally following an analysis similar to that described below. Specifics related to past events (actuarial data) are considered to be proprietary by insurance companies. The following discussion largely relates to liability coverage, but also discusses self insurance to cover potential loss of vehicle.

An insurance premium component typically consists of the product of three factors for each specified risk of loss:

- An underwriting multiplier to provide revenue to the insurer
- A risk factor, and
- The *Maximum Probable Economic Loss (MPL)* associated with the specified failure

These premium calculations are summed over each identified risk when there are multiple risks associated with a given activity.

The underwriting multiplier is typically 3. It can be reduced by up to 25% by auctioning to the lowest cost bidder [Jurist et al., 2005]. Therefore, an auction or bidding process could potentially reduce the multiplier to 2.25.

The risk factor is typically given by $(1+f)/(1+s)$, where f is the number of losses or failures and s is the number of successful events.

If 100 launches of a given vehicle results in 4 failures ($f=4$) and 96 successes ($s=96$), then the risk factor is $(1+4)/(1+96)$, or 0.0515. Larger programs with lower failure rates have lower risk factors. With increased program size, the risk factor approaches the ratio of failures to successes, and with decreased program size, the risk factor increases to reflect lack of experience. If the same statistical failure rate of 4 failures in 100 attempts holds for a program of 1,000 launches, the larger program will have 40 failures and 960 successes. In this case, the risk factor will be reduced from 0.0515 to 0.0427. As the program size increases without bound to even more launches, the risk factor will approach 0.0417. The way to minimize the risk factor is to design the launch program for maximum launch reliability and maximize the number of launches.

Assuming a total of 5 successful test and demonstration flights of a new launcher with zero losses as it enters commercial service, the risk factor is initially $(1+0)/(1+5)$, or 0.167. If there are two failures during the 5 test and demonstration flights, the risk factor for insurance purposes will be $(1+2)/(1+3)$, or 0.750. The value is greater than 1 if there are more failures than successes. The risk factor can be recalculated based on accumulated experience every time the insurance contract is renewed. If additional experience extends the string of successes, insurance premiums should be appropriately reduced.

The maximum probable third party damage or economic loss with a catastrophic failure is related to the energy dissipated in the loss event and to the value of damaged properties in the failure event debris field. This loss depends on vehicle mass, velocity, and location at the time of the failure. Examples of MPL will be given later.

One can minimize MPL by:

- Assuring that the vehicle trajectory does not overfly valuable properties (i.e., launch over uninhabited areas),
- Minimizing vehicle mass from launch to mission completion, and
- Minimizing potential velocity of the failed vehicle (or its fragments)

Minimizing potential velocity of vehicle fragments has been accomplished historically by command destruct mechanisms that blow the failing vehicle into small pieces so that air drag effectively decelerates the frag-

Table 12-1◀, Fig. 12-4, Eq. 12-4◀

ments and reduces their impact kinetic energy. Positive control of the vehicle and defaulting to an automatic abort mode if vehicle communications are lost will also reduce the potential damage to uninvolved parties.

Launch vehicles, including RLVs, can be initially tested with reduced propellant loads to reduce vehicle mass and maximum burnout velocities. This reduces maximum kinetic energy and potential flight range and also reduces the MPL factor for insurance purposes.

Incremental testing of each vehicle or spacecraft design during the development phase should provide meaningful failure statistics. For a hypothetical reusable vehicle development project, this occurs as vehicle kinetic energy is progressively increased. Finally, benign failure modes should be designed into each vehicle or spacecraft to reduce the probability of catastrophic loss. For vehicles, this will entail low vehicle empty weights and provision for dumping of environmentally benign propellant loads in the event of an abort or trajectory deviation.

The US government requires a minimum MPL of $3 million on liability coverage for commercial launches [Jurist, 2009]. For a single low mass, low fuel capacity vehicle such as those competing in the recent Northrop-Grumman Lunar Lander Challenge, the total insurance premium for a projected 2 day flight window was recently quoted as $22,000 [Breed, 2008]. Commercial orbital launcher cost factors use a liability MPL of about $1,000/kg of gross takeoff mass.

An underwriting multiplier of 2.25, an MPL of $3 million, and a risk factor of 0.167 for a string of 5 successful test flights with no failures yields a third party liability insurance premium of approximately $1,125,000 for the subsequent flight. Assume that the vehicle developer can establish a probability of loss of vehicle of 0.1% (1 in 1,000 flights)—either by analytical means or by direct flight experience. Then, the asymptotic risk factor would approach 0.001001 after many thousands of successful flights to establish the track record. This would eventually, after many thousands of successful flights, reduce the per flight insurance premium to $6,757.

A better strategy while accumulating risk data based on flight experience for a presumably reliable vehicle design would be to self-insure for liability. For example, one might place a security deposit of $6 million (twice the optimistic MPL) with an independent escrow agency. That amount can be retired over 10 years at an 18% annual interest rate with annual payments of $1,335,000 annually. This equates to $133,500 per flight for 10 flights/yr and $13,350 per flight for 100 flights/year assuming no payouts for third party losses.

However, the probability of escaping a payout over 10 years at 100 flights/year with a 99.9% reliable vehicle is only 37% ($0.999^{1,000} = 0.37$). The loss of two partly reusable Space Shuttles in about 126 flights gives a loss risk of about 0.016 compared to the postulated, and highly optimistic, loss rate of 0.001 used in this example.

Since the flight tests and demonstration flights need to be insured with no revenue stream, self-insurance for liability makes even more sense. Of course, the risk in self insuring against liability is the possibly catastrophic consequences of a string of "bad luck" events in which losses occur more frequently than the average predicted rate. A high deductible blanket insurance policy could be purchased to cover part of the MPL and the remainder escrowed and retired as described previously. Discussion of the optimization of such a business strategy is beyond the scope of this section.

A rational approach is to produce spare vehicles to "self insure" against vehicle loss if a probability of loss of 0.001 can be attained. This would add a minimum of 0.1% to the cost of vehicles produced and successfully launched and it would allow replacement of lost vehicles from the spare inventory. Insurance against vehicle loss for vehicle replacement is a single factor in the overall insurance cost and does not affect the costs for third party liability "insurance" based on MPL.

12.2.5 Price vs. Demand

Sale price has an effect on demand for many products. Gasoline might serve as a useful example. Cutting price at a given outlet may increase its sales at the expense of other nearby outlets as consumers become aware of the change and act upon it. In addition, sustained lower prices encourage more driving and more discretionary gasoline consumption. How this factor may come into play with space access and the NewSpace industry is presently unclear.

A market is *inelastic demand* when demand remains unchanged with changes in price. A market is described as *elastic demand* when demand increases as price decreases. The equilibrium elasticity, E, is defined as the percentage increase in demand divided by the corresponding percentage decrease in price for small changes. If $E > 1$, the total revenues increase as price decreases at equilibrium. If $E < 1$, total revenues decrease as price decreases. If $E = 0$, demand is unchanged with price changes and revenues decrease in direct proportion to any decrease in price. Equilibrium time is significant since markets take time to respond to changes in price structures.

This concept is illustrated in Figs. 12-5 and 12-6 for a hypothetical reusable suborbital sounding rocket [Jurist, 2009]. The figures show annual launch demand (in payload mass) and total annual revenues as a function of launch price and various hypothetical values of E. Both figures are based on current suborbital sounding rocket launch numbers. The threshold for elastic demand is assumed to be $250/kg with negligible market response time. In the region of inelastic demand ($250–$1,000/kg), total revenues decrease as price decreases to the elastic threshold, and then increase with further price decreases if $E > 1$.

If launch demand remains inelastic, the provider is motivated to keep prices as high as possible without going to the extreme of killing demand by making the launches unaffordable.

Standard economic practice is to put price on the vertical axis of the graph with revenue or demand on the

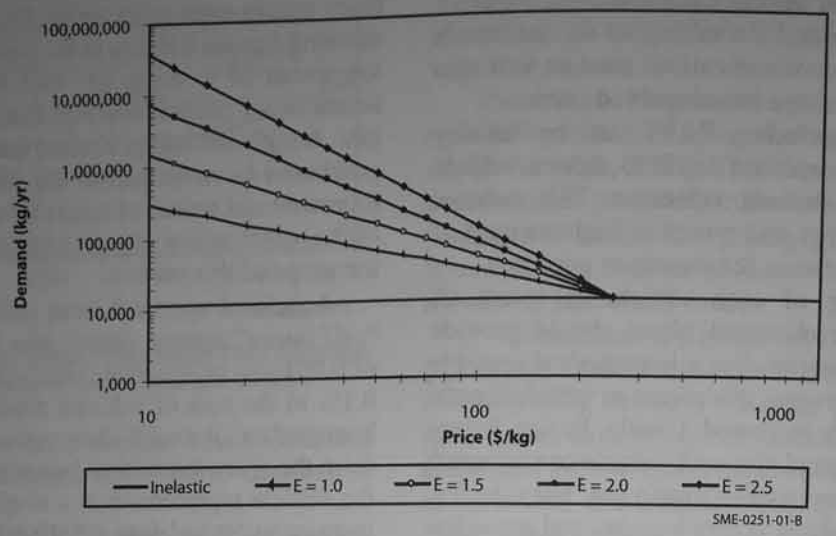


Fig. 12-5. Annual Launch Demand as a Function of Delivered Price.

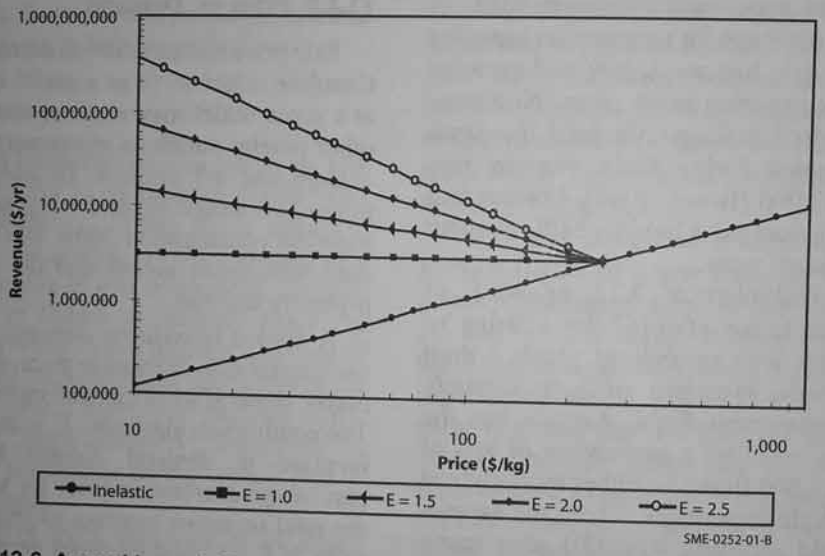


Fig. 12-6. Annual Launch Revenues as a Function of Delivered Price.

horizontal axis, but standard scientific and engineering practice is to put the independent variable (price in this case) on the horizontal axis. Engineering practice is used in these figures.

As mentioned previously, the response time to changes in price structure may be significant. With space operations, the entire system response time includes definition of the mission requirements, assembly of procurement documentation (such as requests for proposals or RFPs), negotiation of contracts, definition of contract compliance monitoring, mechanisms for dealing with delivery delays, R&D, vehicle or system production and testing. The entire response time is commonly measured in years or even decades.

The absence of solid documentation of market size for a potential vehicle or payload implies a "build it and they will come" mentality. Financial success of such an approach requires the reliable creation of demand for the product. This approach is frequently described as the "Field of Dreams Approach to Space Development."

Table 12-1◀, Fig. 12-6, Eq. 12-4◀

Determination of market size allows the definition of product lifespan, and production rates to optimize profitability or minimize costs. In the context of a launch vehicle, launch rates and program lifetime would determine the production rate, plant size, degree of automation, planning for learning curve influences on production economics, and other factors.

12.2.6 Project Life Cycle

The life cycle of a space project may contain many stages as well as reiterations of many of those stages. Adequate business planning needs to develop a realistic budget for each stage. Additionally, some estimate of the uncertainties in that stage's budget for contingencies that may arise must be made. Finally, the time for completion of each stage, and the overlap in time, if any, that impacts adjacent stages must be estimated. This is where PERT and Critical Path Analysis becomes very valuable in allowing dynamic modifications to be made to the plan. Some of the life cycle stages are listed in the boxed section:

Life Cycle Stages
• Concept or response to RFP
• Feasibility study
• Trade studies
• Detailed conceptualization
• Initial cost estimation
• Detailed planning (PERT or some other method)
• Detailed cost estimation
• Financing
• Assignment of assets to the project
• Acquisition of needed assets
• Performing the necessary R&D
• Testing at subsystem and system levels
• Fabrication of the first preproduction prototype
• Production planning
• Production asset and facility assignment
• Acquisition of additional assets
• Production
• Testing and quality assurance
• Shipping
• Testing at destination
• Evaluation of performance, and
• Consideration of the next generation

The rest of this textbook considers the stages outlined in the boxed section in much more detail.

12.3 Law and Policy Considerations

James D. Rendleman and Milton "Skip" Smith,
Sherman & Howard, L.L.C.

Civil and military, as well as national and multinational commercial space capabilities are vital enablers for successful 21st century militaries, economies, diplomatic communication and collaboration. Given their strategic importance, engineers should not be surprised to find that legal and policy concerns intrude on all elements of the space system mission design process. The complete span of international legal, policy, and diplomacy implications should, therefore, be fully considered when planning for and executing space activities.

Engineers crafting a space system must anticipate these issues in order to avoid potential "show stoppers"—adverse decisions that could be directed by government leaders, customers, suppliers, legislators or courts, and the international community. A perfect engineering solution is useless unless and until it can be implemented. Engineers also need to appreciate the time it may take to comply with applicable laws. A good example of this is Space Shuttle mission 51C to retrieve the Palapa and Westar VI satellites. These satellites remained in low-Earth orbit because the motors which should have transferred them to geostationary orbit failed to operate. Hughes and NASA engineers worked out the technical solutions in a little more than six weeks. The lawyers then spent more than six months resolving the legal questions concerning ownership, salvage rights, insurance coverage, and release of liability between the insurance carriers, Western Union Company and the Indonesian government. Hughes Aircraft Company and NASA had to agree on technical roles, liability, and compensation before the recovery effort could begin. The recovery efforts were successful, but legal fights related to the original launches continued for another 10 years.

International space law and domestic aspects of US space law and policy were forged within the crucible of the most dangerous parts of the Cold War. Comprehensive diplomatic, law and policy activities have since been executed to achieve important strategic objectives throughout the Space Age. In some ways, space activities provided a tremendous opportunity for competing powers to offer olive branches to each other on cooperative activities. Indeed, the International Space Station (ISS) was an important post-Cold War expression of the integration of Russia and the former Soviet Union into the Commonwealth of Nations. The ISS has been cited by many national representatives as a wonderful expression of the promise that such cooperation can provide.

President Eisenhower's administration developed the first comprehensive treatment of US space policy. Presidents since have followed suit, accompanied by a number of targeted policy pronouncements on more narrow space-related topics. Each of these national pronouncements was carefully considered, and reflects priorities of their time—to emphasize desires to fly national security and civil missions on the Space Shuttle, unhook national security missions from the Shuttle system after the Challenger explosion, energize a nascent commercial launch and remote sensing industry, encourage global use of the US Global Positioning System, and, affirmatively state US interests to protect its access to space capabilities following the September 11 attacks. A number of these considerations have also been incorporated into US law, and they affect how national security, civil, and commercial space programs are structured and funded.

This chapter reviews the major international agreements impacting space activities and then turns to US national laws impacting space activities. Thereafter, specific topics, such as satellite communication and space debris, that have particular relevance to space activities are addressed [International Institute of Space Law, 2011].

12.3.1 Foundations of International Space Law

US law and policy place great emphasis on diplomacy and international engagement; it is a centuries-old practice that has secured borders, enhanced commerce, and brokered and resolved disputes. Assuming space operators pay heed to customary and treaty-based provisions of international law and to best practices encouraged by the international space community, such efforts afford the space community a respectable measure of confidence they can all have assured access to space capabilities.

Table 12-1◀, Fig. 12-6◀, Eq. 12-4◀

So what are the foundations of international law that affect space activities? Treaties and other bilateral agreements to which sovereign states are signatories, and which govern issues of interest; and, second, multinational agreements among sovereigns. International agreements are governed, not by contract law, but by the *Vienna Convention on the Law of Treaties* [1969]. While the United States has not ratified the Vienna Convention, it accepts the bulk of its rules as customary international law, which is a third foundation of international law. The fourth formulation is general principles common to mature legal systems; and the fifth, subsidiary "municipal" determinations of law, see Burney [2008], (e.g., national laws and decisions such as those rendered by the US Supreme Court).

With relatively few treaty restrictions governing military and other space activities, some might think the US is faced with a dilemma—should it only abide by a permissive "letter of the law" standard or the "spirit of the law"? If only the letter of the law, what approach should it want to see adopted by current or fledging space nations? Actually, the choice is not between the letter and spirit of the law; on the whole, the United States, abides by both standards. Decades of senior policy-makers within the executive and congressional branches of the US government have recognized the importance of the space domain and the rule of law. They have assessed risks associated with not providing measured and calm global leadership to preserve access to the domain consistent with law, and have made innumerable decisions in accord with those assessments. While diplomatic engagement has been helpful, there is an element of risk in relying solely on it to assure access to space capabilities. Enforcement mechanisms for violating treaties and agreements relating to space are rather limited. There are no specific enforcement mechanisms in place to address violations of space related treaties, and this increases the risk of depending on such documents and handshakes to protect or assure access to space. Violations of treaties and other agreements should nominally be responded to through economic means and diplomatic consultation and, if necessary, other sanctions, assuming a nation or some part of the global community agree to them. In turn, the United States encourages thoughtful law and policy-making by other members of the global community.

Despite their minimalist nature, space-related treaties, conventions, and agreements help regularize space activities. They help protect capabilities of systems that have been or are about to be placed on orbit. Bi-lateral and multi-lateral arms control treaties also preserve some of the *sanctuary* aspects of space by prohibiting "interference" with "national technical means" (NTMs) such as missile warning and reconnaissance satellites used to verify treaty compliance. Confidence building procedures have been agreed to and these have improved opportunities for transparency between potential adversaries, perhaps improving dialogue to prevent any dispute from escalating into armed conflict or to a nuclear catastrophe. Other treaties and conventions such as those involving the International Telecommunications Union (ITU) address vexing spectrum management issues which have profound impacts on military, civil, and commercial space systems. The ITU presently attempts to equitably reconcile the explosion of information technologies and exponential user growth and needs, all within nature's limited useable bandwidth in the electromagnetic spectrum.

Of course, space missions are inherently an international activity. Satellites can overfly the sovereign territory of every nation on Earth. Mission designers must, therefore, consider the potential for objections as they develop their systems. Views of space-faring nations sometimes diverge from those of developing nations, and economic competitors or potential military adversaries may disagree on particular issues in order to gain near- and long-term advantages. Differences of opinions can arise from different cultures, economic status, political imperatives, and world views.

Each nation approaches international space policy and legal issues differently. For example, some nations argue for a legal definition of the demarcation between a country's air space and outer space should be defined as beginning at 100 kilometers above sea level. This is known as the Kármán Line, calculated by and named for Theodore von Kármán. This demarcation has been accepted by the Fédération Aéronautique Internationale (FAI) [2004]. The United States does not agree with this definition and rejects the need for it. It refuses to officially accept a specific "boundary" and, instead, employs a functional approach to assert space-related free passage and transit rights. In the US view, if boundaries for the definition of space are strictly defined by treaty or through customary international law, this could affect important space-related rights. The reader is reminded that the Soviet Union reserved the right to shoot or bring down aircraft in its "airspace", and did so with alarming and tragic deadly effect. The United States and its allies might be forced to employ deterrence strategies and prepare for conflict if a state wanting adoption of the Kármán Line also threatens spacecraft that cross below it above their territory. Given these considerations, the United States has not agreed to the definition.

Similarly, a number of the world's equatorial nations have argued (in vain) that their air and space sovereignty rights extend upwards to about 36,000 km (geostationary orbit). These nations seek to assert rights to control access and obtain compensation from space-faring nations who use the slots above their territory. No major space-faring nation agrees with this claimed right.

12.3.2 Outer Space Treaty

The *Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, Including the Moon and Other Celestial Bodies, 1967*, the "*Outer Space Treaty*" [1967], is the grandparent of international and US space law. The classic treatise on space policy [Christol, 1982]. The Treaty serves as an

Table 12-1◀, Fig. 12-6◀, Eq. 12-4◀

important foundation of the entire US military, civil and commercial space program. It forms the basis for most international space law, including its important legal principles and prohibitions, and incorporates important elements of what was emerging as customary international space law in the 1960s.

The Treaty was consummated at a time when US policy makers concluded space offered unique benefits for the military and political dimensions of the Cold War national security strategy. They hoped to fashion an agreement to preserve access to the domain and these motivations and the document have endured and continue to serve the US and its allies' national interests. Assuming the mantel of the world's leading space-faring nation, the US helped lead the way on discussions relating to the treaty's formation, crafting the treaty instruments and forging a global consensus to set a tone and worldview that space activities should be prosecuted for peace and the benefit of mankind. Other material is in the annual reports of the *Colloquium on the Law of Outer Space*, [AIAA, 2011].

Signatories of the *Outer Space Treaty*, have agreed to a document that supports freedom of access to space by all space venturing powers, with language that provides: "Outer space … shall be free for exploration and use by all States without discrimination of any kind…." *Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, including the Moon and Other Celestial Bodies* (Outer Space Treaty), Article I. The treaty states in pertinent part: Article I—The exploration and use of outer space, including the moon and other celestial bodies, shall be carried out for the benefit and in the interests of all countries, irrespective of their degree of economic or scientific development, and shall be the province of all mankind. Outer space, including the moon and other celestial bodies, shall be free for exploration and use by all States without discrimination of any kind, on a basis of equality and in accordance with international law, and there shall be free access to all areas of celestial bodies. There shall be freedom of scientific investigation in outer space, including the moon and other celestial bodies, and States shall facilitate and encourage international co-operation in such investigation. Article III—States Parties to the Treaty shall carry on activities in the exploration and use of outer space, including the moon and other celestial bodies, in accordance with international law, including the Charter of the United Nations, in the interest of maintaining international peace and security and promoting international co-operation and understanding. Article IV—States Parties to the Treaty undertake not to place in orbit around the earth any objects carrying nuclear weapons or any other kinds of weapons of mass destruction, install such weapons on celestial bodies, or station such weapons in outer space in any other manner. The treaty also declares nations should have "freedom of scientific investigation in outer space." Addressing topics that affect the potential for space conflict, the Outer Space Treaty provides that international law applies. "…Article III [of the Outer Space Treaty] incorporates the application of international law, and specifically the Charter of the United Nations, in outer space, making it a vital part of the *corpus juris spatialis*" [Blount, 2008]. This incorporation of *international law*, not just the United Nations Charter, is important.

Under the *Outer Space Treaty*, nations agree to share the global space commons, and every major space faring nation is now a signatory.

Rights (and obligations) of non-signatories can be found in international customary law. Customary international law "... consists of rules of law derived from the consistent conduct of States acting out of the belief that the law required them to act that way" [Rosenne, 1984]. *Outer Space Treaty* signatories can look to both treaty and customary law sources, as customary law may be applied whether or not a state is a treaty party. The vast majority of the world, including the United States, accepts in principle the existence of customary international law (although there are often differing opinions as to what rules are contained in it).

The *Outer Space Treaty* lays down widely accepted legal principles for outer space, codifying important, pre-existing customary international law principals and precedent relating to space. For example, the launch of the Sputnik satellite system over a half-century ago established the legal precedence and customary international law for free passage of space systems and over-flight rights over national territories while on-orbit. The Eisenhower Administration's objective to confirm universal acceptance of this concept of satellite free passage and over-flight rights was achieved years later when these principles were included as part of the *Outer Space Treaty*. In the meantime, customary law related to such issues was developed as statements of such principles were presented and discussed within various global community and United Nations' forums. They can be found in a number of disparate documents including the 1958 National Aeronautics and Space Act, and UN General Assembly resolutions.

The *Outer Space Treaty* provides:

- All nations may scientifically investigate space, with international cooperation encouraged
- No nation may claim sovereignty over outer space, including the Moon and other celestial bodies. As an illustration, when Neil Armstrong placed an American flag on the Moon, he did not do so to claim new territory, as Columbus and other explorers did when they arrived in the New World
- The laws relating to space activities will follow the established principles and rules of international law and the U.N. Charter
- No signatory will place nuclear weapons or any other weapons of mass destruction in orbit around the Earth, or on the Moon, or on other celestial bodies. Note, however, this restriction does not ap-

Table 12-1◀, Fig. 12-6◀, Eq. 12-4◀

ply to nuclear power sources; comparably, the 1963 *Treaty Banning Nuclear Weapon Tests in the Atmosphere, in Outer Space and Under Water*, also known as the *Limited Test Ban Treaty* (LTBT), prohibits "any nuclear weapon test explosion, or any other nuclear explosion" (emphasis added) in the atmosphere, underwater, or in outer space

- The *Outer Space Treaty* does not specifically prohibit testing weapons in outer space itself (as opposed to on celestial bodies); instead it proscribes the stationing of nuclear weapons on orbit. However, with the LTBT, testing and subsequent use of nuclear weapons in response to attacks on space systems would appear to be banned, unless employed in a possible narrow exception that allows such devices to be employed to preserve the "survival of a State." Employing nuclear weapon systems against conventional space systems probably could not be shown to support such a survival objective
- Nations must use the Moon and other celestial bodies exclusively for "peaceful purposes," but they may use military personnel in scientific research. The United States defines *peaceful* as "non-aggressive," thus permitting defensive measures
- Astronauts are *envoys of mankind* and must be returned to their home nation when rescued
- Nations bear international responsibility for their activities in outer space, whether done by governmental agencies or private citizens. Thus, the US must authorize and continuously supervise all space activities of its citizens. This requirement is unique to space activities, resulting from a compromise between the United States and the former Soviet Union. The Soviets had insisted that only governments should be permitted to go into space, whereas the United States insisted on permission for private entrepreneurs. The *Commercial Space Launch Act of 1984* was enacted and amended by the United States in part to carry out the its obligations under the *Outer Space Treaty* and *Liability Convention*. Even though the Treaty provides that States bear international responsibility for government and private space activities and must supervise and regulate national activities in space, some argue there is no effective enforcement mechanism in the Treaty to respond to a failure by a State to perform actions consistent with responsibilities established under the document.
- Launching nations are liable for direct damages to citizens of other nations caused by national and private launch activities
- A State Party to the Treaty retains jurisdiction, and control rights over their spacecraft, space objects and personnel. In addition, ownership rights of objects launched into outer space, and their component parts, is not affected by their presence in outer space. Unless these rights are relinquished, some argue that peacetime retrieval, alteration of orbit, or any other form of interference with foreign space objects would be unlawful without prior consent under treaty and customary international law, no matter how desirable the end result. If strictly interpreted, only a launching State can exercise *jurisdiction and control* of its space objects, even after satellites are defunct and long abandoned, and even if they threaten a particular orbital regime or another satellite with debris.* Any entity attempting to remove space junk could run into a claim from a State that the removal is a violation of international law†
- Nations must conduct space activities so as to avoid harming or adversely affecting space activities of other nations. The 1963 *Project West Ford* influenced this provision when the United States placed 480 million copper dipoles into orbit 2,000 miles above the Earth in an attempt to create an artificial ionosphere to enable radio messages to bounce from coast to coast. The in-house name for the project was *Needles*, which proved to be an unfortunate choice. Complaints came in from the astronomical community asserted by astronomers who were convinced the belt of copper needles would ruin their view of the universe. They also argued the experiment might lead to even worse experiments. The Soviets wryly observed and commented "What if one of the needles pierces the heart of a cosmonaut or puts out someone's eye when it reenters the atmosphere?"
- If one State's space activities pose a potential to cause harm or interference to space activities conducted by another State, consultations should be initiated before they proceed. Consistent with the point, military mission planners should review the 1977 *Convention on the Prohibition of Military or any Other Hostile Use of Environmental Modification Techniques*, or *Environmental Modification Convention* [1977], if they anticipate large-scale effects. The *Environmental Modification Convention* prohibits all military or hostile environmental modification techniques that might cause long-lasting, severe or widespread environmental changes in Earth's atmosphere or outer space. "Each State Party to this Convention undertakes not to engage in military or any other hostile use of

* As of 1989, jurisdiction and control of a State over its space objects was considered "permanent."

† "Ownership of space objects is also permanent, as Article VIII implies that the State of registration may not be divested of title to its space objects, (cit.om.) regardless of the use or condition of the space object. (cit.om.) Moreover, the rights of ownership include the rights of possession, use and disposal, thereby denying a right of encroachment without the consent of the State of registration.(cit.om.)" [Baker, 1989].

Table 12-1◀, Fig. 12-6◀, Eq. 12-4◀

environmental modification techniques having widespread, long-lasting or severe effects as the means of destruction, damage or injury to any other State Party." "Widespread" is defined as "encompassing an area on the scale of several hundred square kilometers"; "long-lasting" is defined as "lasting for a period of months, or approximately a season"; and "severe" is defined as "involving serious or significant disruption or harm to human life, natural and economic resources or other assets" [Understanding, Environmental Monitoring Convention, 1977]. The *Environmental Modification Treaty* focuses on proscribing employing military weapons, tactics and techniques that deliberately change natural processes. The 1992 version of the *Environmental Modification Convention* limited its purposes to military conflict*

- Stations on the Moon and other celestial bodies shall be open mutually to representatives of other nations after reasonable advance notice. The *Antarctic Treaty*, in contrast, allows unannounced inspection of stations, installations, and equipment; therefore we must not confuse these provisions with those of the *Outer Space Treaty*. The *Moon Treaty of 1979* has extensive inspection provisions but little force or effect, because no major space-faring nation has ratified it. The "common heritage of mankind" provisions, which envision sharing profits with all nations even though they have not contributed, is a major sticking point

12.3.3 Liability Convention

The 1972 *Convention on International Liability for Damage Caused by Space Objects* (the "Liability Convention") expands on a topic noted in the *Outer Space Treaty* that "launching states" are liable to other states for damage caused by space objects (including debris). States are liable only for direct damage caused by a space object (i.e., loss of life; personal injury or other impairment of health; or loss of or damage to property). If damage is caused to another space object in outer space, liability is based on fault. If damage is caused by a space object on earth or to an aircraft in flight, liability is absolute.

The *Liability Convention* was crafted to protect the interests of non-space-faring nations, so it held space-faring nations absolutely liable if they caused injury or damage on Earth. Absolute liability simply means that someone who is injured may claim compensation just by proving damage and who did it, without having to demonstrate negligence or fault. If damage occurs in space, however, the Convention provides that space-faring nations must prove fault in order to recover damages.

The procedure for settling claims calls first for negotiations by the nations involved. The only claimant under the *Liability Convention* is another nation, bringing a claim on its own behalf or on behalf of its citizens. If these negotiations fail, a three member claims commission is formed. Each nation involved appoints one member. The two appointed members then appoint a third, the chairman.

Notably, there can be more than one "launching State"—a launching state is any state that launches an object, procures the launch of an object, or from whose territory or facility an object is launched. If there is more than one launching State, joint and several liability rules would apply. States may make indemnification agreements and apportion liability among themselves. Since allied nations supporting space conflict activities could be construed as launching states, liability issues and allocation of liability issues should be resolved before engaging in such activities.

In the end, the *Liability Convention*'s real limitations on space activities arise out of its provision for liability associated with causing damage to third-parties. These liability issues must be evaluated, addressed and mitigated by law-abiding states before performing activities that could cause damage to third-parties. Planners must account for payment of damages or plan to limit such problems.

The first case involving a claim under the *Liability Convention* arose out of the reentry of Cosmos 954 over the Northwest Territory of Canada on January 24, 1978. The reentry scattered radioactive debris from the Cosmos' nuclear reactor over 124,000 km^2. The two nations did not submit their claims and defenses to a claims commission, because negotiations between Canada and the Soviet Union were successful. The search and recovery effort entitled *Operation Morning Light* cost Canada almost $14M Canadian dollars, but the claim was settled for $3M. Canada elected not to assert claims for personnel and equipment costs. The Soviets argued that Canada was too meticulous in the clean up operations and that if this event had occurred in Siberia, it would not have expended that much effort. Additionally, the Soviets asserted that it had offered to help clean up, but Canada rejected the offer. Thus, Canada had not mitigated its damages. The amicable settlement set a positive precedent for resolving claims in the future.

Table 12-1◀, Fig. 12-6◀, Eq. 12-4◀

* US attempts at environmental modification during the Southeast Asia conflicts were opposed by dissidents in Congress. As a result, in 1973 the US Senate passed a resolution urging President Nixon to initiate negotiations leading to a multilateral treaty. In 1974 US-USSR bilateral negotiations began. In 1975, the USSR and the US submitted separate but identical texts of a draft convention to the UN. The draft was then finalized in the Conference of the Committee of Disarmament (CCD) and adopted by the UN General Assembly on 10 December 1976. The Environmental Modification Convention opened for signature in May 1977 and entered into force 5 October 1978.
The US Delegation Statement provides: "The Environmental Modification Convention is not an Environmental Protection Treaty; it is not a treaty to prohibit damage to the environment resulting from armed conflict. Rather, the Environmental Modification Convention fills a special, but important niche reflecting the international community's consensus that the environment itself should not be used as an instrument of war."

Under the *Liability Convention*, the US must pay when her space activities injure citizens of other nations or their property. The Federal Tort Claims Act also provides that US citizens may also recover damages for personal injury and property loss caused by US procured government activities. Losses caused by US commercial space operators may be recovered in state and sometimes federal courts. In order to manage the potential liability arising out of space activities, and ensure that appropriate compensation is paid, the government through its FAA licensing activities requires insurance be obtained before authorizing a commercial launch. Government launches, both military and civil, do not have formal insurance coverage, as the government is a self-insurer. If the government suffers a loss, it only allocates more funds for replacement systems and mitigating damages from the public coffers.

12.3.4 Registration Convention and Rescue and Return Agreement

The 1975 *Convention on Registration of Objects Launched into Outer Space* (the "Registration Convention") requires a launching State to advise the U.N. Secretary General of the following: name of the launching nation, description or registration number, date and location of launch, basic orbital parameters (nodal period, inclination, apogee, and perigee) and general function of the space object, as soon as practicable. A nation need not update this information but may do so if it wishes. States often delay filing the requisite information.

The 1968 *Agreement on the Rescue of Astronauts, the Return of Astronauts and the Return of Objects Launched into Outer Space*, (the "*Rescue and Return Agreement*"), sets forth rights and obligations of states concerning the rescue of persons in space. It expands on the rescue provisions in the *Outer Space Treaty*. The Agreement requires that any state party that becomes aware that the personnel of a spacecraft are in distress must notify the launching authority and the Secretary General of the United Nations. In addition, any state party to the agreement must provide all possible assistance to rescue the personnel of a spacecraft who have landed within that state's territory, whether because of an accident, distress, emergency, or unintended landing. If the distress occurs in an area that is beyond the territory of any nation, then any state party that is in a position to do so shall, if necessary, extend assistance in the search and rescue operation.

The *Outer Space Treaty* states simply that *astronauts* are to be rendered assistance by parties to the treaty. It does not define the term "astronaut" and fails to clarify whether this applies to spacecraft personnel such as a space "tourist". The Rescue and Return Agreement expands on the term by referring to the "personnel of a spacecraft" rather than "astronauts."

Finally, the *Rescue and Return Agreement* provides that if a space object or its parts land in the territory of another state party, the state where the object lands is required (upon the request of the launching authority) to recover the space object and return it to the launching authority. The *Rescue and Return Agreement* provides that the launching state must compensate the state for the costs incurred in recovering and returning the space object. The Agreement does not describe the condition of the object or parts that are to be returned.

12.3.5 International Cooperation

Interest in international space cooperation is not new. International cooperation has been an important and long-standing hallmark of the US national space policies. The United States engages in a wide range of such activities.

Cooperation allows space-faring States to combine resources and reduce risk, improve efficiency, expand diplomatic engagement, enhance prestige, improve political sustainability of space projects, and provide workforce stability. Cooperative participants expect international projects will generate a calculus that one plus one will equal three—that diverse resources, skills, and technologies of the partners will provide synergy, adding up to more than the sum of their parts [Mukai, 2008]. Given the benefits sought, sentiments to support international cooperation and collaboration are growing and evolving, much more so than in past policy declarations. NASA, commercial, and European space activities have already achieved considerable success with their cooperative endeavors [Dale, 2006].

Fans of the new US National Space Policy may think that the number of times "international cooperation" or "cooperation" is mentioned is very important. The reality, however, often is that cooperative programs are expensive, difficult to resource, and hard to manage. Given the challenges, shrewd partners know that the direction to engage in cooperation is not worth much unless tied to money and programs. Some potential and even current international partners are taking a judicious *wait and see* attitude to the new direction. Changing business plans to respond to changing US international cooperation initiatives involves risks.

12.3.6 US National Space Policy

The US National Space Policy provides the President's statement of the "highest priorities for space, and reflects our principles and goals to be used in shaping the conduct of our space programs and activities" [Rose, 2010].* The policy has been updated a number of times. It sets forth a number of perspectives one would expect on such an important topic. A careful reading of the current document shows it continues the key principles of national

* Rose has been serving as point man for the Administration on the new policy. He has been traveling to a wide variety of venues to speak to and relate its key messages for the US and global community.

Table 12-1◀, Fig. 12-6◀, Eq. 12-4◀

space policy that were crafted long ago. The document is circulated within the US Government agencies to achieve consensus and moves the discussion forward incrementally. The decision to do this is wise. Revolutionary transmutations in policy and national interests should be driven by major shifts in the international scene, such as those foisted upon decision makers by wars, new countries, breakthroughs in technologies, or other significant events. No recent shift or significant event has occurred with regard to the space domain, so the continuing incremental evolution in policy is appropriate.*

Past versions of the US National Space Policy emphasized that the sustainability, stability, and free access to, and use of, space is vital to US national interests. The 2010 US National Space Policy endorses this view and contends that space systems are vital to secure access to and satisfy important communication, navigation and timing, weather, remote sensing, missile warning and defense needs. It also tips the hat to needs of the United States to secure its industrial base, improve the science, technical, engineering and math expertise of the population, and ensure the professionalism of its space community. Importantly, much of the new policy's foundational points are based on principles of international cooperation—*mitigate the dangers of space debris*, *improve space situational awareness (SSA)*, *achieve collision avoidance*, and *pursue pragmatic bilateral and multilateral transparency and confidence-building measures (TCBMs)* to mitigate the risk of mishaps, misperceptions, and mistrust (that could lead to space or terrestrial warfare.).

It is US policy to comply with the four basic international space treaties.† Each US space policy has supported the proposition that all nations have the right to explore and use space for peaceful purposes, and for the benefit of all humanity, in accordance with international law.

Importantly, the United States considers that its space systems possess the rights of passage through, and conduct of operations in, space without interference. It reserves the right to defend itself from threats and attacks against its space systems in each of the last three policies in one form or another. It has been longstanding US policy that purposeful interference with space systems, including supporting infrastructure, will be considered an infringement *on its rights*. Reflecting the expanding international flavor of the policy, the new US National Space Policy broadens this assertion by stating that purposeful interference with space systems will be considered an infringement of a *nation's rights* [National Space Policy, 2010]. Consistent with exercising its inherent right of self-defense or lawful collective self-defense, the policy asserts that the United States will "deter others from interference and attack, defend our space systems and contribute to the defense of allied space systems, and, if deterrence fails, defeat efforts to attack them" [National Space Policy, 2010].

The new policy also continues to emphasize the need to address the dangers posed by space debris. Warnings about space debris and mitigating its hazards have been part of US space policies for decades and understandably so. The 1980s saw the United States migrating to use only the Space Shuttle to support its national security and civil spacelift needs. In undertaking this responsibility, NASA concluded that the growing space debris threat endangered its astronauts and therefore commenced studies and analysis to better understand the risk. This concern was then embodied in early national space policies that emphasized the need to mitigate the generation of space debris. Consistent with this long-standing policy, the United States has been a strong proponent of activities conducted within the United Nation's *InterAgency Space Debris Coordination Committee (IADC)*. The IADC has been used as a vehicle to develop recommended space debris mitigation guidelines and obtain global space community compliance with the techniques space-faring nations should employ when launching, operating, and safely retiring and de-orbiting space objects. Unfortunately, compliance by the international community has been inconsistent to date.

12.3.7 Import and Export Controls and Technology Transfer Constraints

Unfortunately, achieving increasingly interoperable platforms, planning, and operations can be difficult and complicated because US law and policy imposes controls on the release of sensitive technologies and operations. Specific information on important technologies and information relating to some space operations and technologies may be determined by the US Government to be non-releasable to allies, close partners, and sometimes one or more members of a coalition. (Other nations also secure their technologies for diplomatic, military and economic reasons.)

Law and policy considerations run the gamut of international military space relationships: foreign military sales, direct commercial sales and programs, cooperative programs, officer exchanges and liaison officer relationships, visits including data releases and briefings, and

Table 12-1◀, Fig. 12-6◀, Eq. 12-4◀

* Space warfare zealots might argue the January 11, 2007 Chinese anti-satellite (ASAT) intercept was a watershed space security event. Perhaps so, but kinetic ASAT capabilities have been tested and even operational since the late 1960s.

† The four international treaties are: (1) the 1967 *Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, Including the Moon and Other Celestial Bodies* (the "Outer Space Treaty"); (2) the 1968 *Agreement on the Rescue of Astronauts, the Return of Astronauts and the Return of Objects Launched into Outer Space* (the "Rescue Agreement"); (3) the 1972 *Convention on International Liability for Damage Caused by Space Objects* (the "Liability Convention"); and (4) the 1975 *Convention on Registration of Objects Launched Into Outer Space* (the "Registration Convention"). For various reasons, the United States and the other major space-faring States have refused to ratify a fifth international treaty, the 1979 *Agreement Governing the Activities of States on the Moon and Other Celestial Bodies* (the "Moon Treaty").

operational activities including exercises and intelligence information. They must be continually assessed. Any proposed international space operation, cooperative effort, or technology transfer must be examined and assessed in its entirety. US government officials are charged to weigh the political and military advantages expected to accrue to the United States against the damage that could potentially result from the possible compromise of the information disclosed.

Important portions of releasability law and policy arise out of the *Arms Export Control Act (AECA)* [AECA, 1976], which provides the authority to control the export of defense articles and services. It governs the sale and export of defense articles and services and related technical data. The AECA is an important part of the statutory scheme to ensure US compliance with technology control regimes and agreements that seek to stop and slow the proliferation of missile technologies and the technologies that can be used to deliver weapons of mass destruction. Under the AECA, spacecraft, space related articles and services are designated to be subject to export control. The designated articles comprise part of the *US Munitions List (USML)*, which is contained within the *International Traffic in Arms Regulations (ITAR)*. The Munitions List specifically includes rockets, spacecraft, space electronics, and guidance equipment. *Defense services* are also included, defined as furnishing help to foreigners "in the design, engineering, development, production, processing, manufacturing, use, operations, overhaul, repair, maintenance, modification or reconstruction of articles." Included is the furnishing to foreigners of any technical data-whether in the United States or abroad. Finally, no one may provide foreigners "technical data", meaning information classified or related to defense articles, without a license or approval issued by the Department of State. Under these regulations, an export does not have to cross a border. For example, an export occurs when an individual discloses technical data concerning a spacecraft or rocket to foreigners, even if the disclosure is part of a potential sale and occurs within the US.

The AECA requires that exports of space articles, services and related technical data meet US national security interests. It also requires the proposed recipients offer assurances to the United States before transfers are made: First, the recipient nation must agree to not transfer the articles, services, or data to third parties without prior US government consent; second, the recipient must agree to use the articles, services, or data only for purpose for which they were furnished; third, the recipient must agree to maintain the security of the defense article and services and provide substantially the same degree of security as the US Government.

There is a substantial paperwork component associated with each of these releasability compliance regimes. The rules are driving some small suppliers out of the export marketplace as they usually lack the economies of scale to properly respond to export legal requirements. International partners are also wary of the legal rules and procedures. Nearly all members of the space community, foreign and domestic, consider the rules to be burdensome and onerous. The requirement for assurances, and the potential for US criminal liability arising out of them, is generally agreed to have cost the US space industry billions of dollars in sales in the international marketplace. The US communications satellite industry has lost market share to international competitors who claim their systems, products and services are ITAR-free.

Of course, other US laws, regulations, and policies apply to exports of space data, hardware, and services, such as the release of classified information.* For example, another control involves patents containing military technology. When the Departments of Defense and Energy review an export application, either department may classify the information to keep it secret, thus preventing export without additional security controls. In addition, dual use military/civilian use commodities and technical data on the *Commodities Control List (CCL)* fall under the licensing requirements of the Export Administrative Act of 1979. The Commerce Department runs this program, has published a series of detailed regulations, and licenses exports under them.

Unless a project has federal funding, the restriction on imports into the United States is not a substantial issue unless the import requires sharing of data proscribed by ITAR or other export constraints. If federal funding is involved, the *Buy American Act* and domestic preference provisions in various other acts and regulations could affect purchases. The *Buy American Act* restricts foreign involvement in US Government satellite launches. In general, the *Buy American Act* prohibits buying foreign goods but does not apply to services. NASA has held that a supplier of space services may not use foreign goods to provide the service. Under the Act, a product is not *foreign* if more than 50% of the cost of all components comes from domestic sources. The DoD may buy from NATO countries, and there are other exceptions. Given the global economy and on-going specialization in various technologies, the Act's provisions and rules can give American mission planners headaches, especially if waivers to the law and relief from policy constraints are needed. Waivers have been granted, however, to enable launches of important civil space missions. (See also Federation Acquisition Regulation (FAR) Part 25.)

* The *Export Administration Act of 1979* (EAA) governs the export of most dual-use unclassified articles and services (having both civilian and military uses) not covered by the AECA. The EAA controls exports on the basis of their impact on national security, foreign policy, or supply availability. With the expiration of EAA in 1994, the President declared a national emergency and exercised authority under the International Emergency Economic Powers Act (P.L. 95-223; 50 U.S.C. 1701 et seq.) to continue the EAA export control regulations then in effect by issuing EO 12924 on August 19, 1994.

12.3.8 Commercial Space Launch Act of 1984

Mission planners must consider the *Commercial Space Launch Act of 1984*. As amended, it requires the Department of Transportation's (DOT) Office of Commercial Space Transportation (a branch of the Federal Aviation Administration, generally referred to as FAA/AST, or AST) to approve and license all commercial expendable vehicles if launched from US territory or by a US citizen or corporation outside the US. AST approves any commercial rocket launch operations—that is, any launches that are not classified as model, amateur, or "by and for the government." The Act proclaimed two objectives: 1) to encourage, facilitate, and promote commercial space launches by the private sector; and 2) to foster the development of licensing requirements in consultation with other US Government agencies.

The creation of a new single regulatory agency within the FAA to handle this was hoped to speed development of the then-nascent US commercial launch industry. AST licenses private space vehicles and spaceports within the United States. In contrast, NASA, a research and development agency, neither operates nor regulates the commercial space transportation industry. It only launches satellites and spacecraft on vehicles developed by private companies.

Before issuing a launch license the AST requires both a mission review and a safety review. The first review focuses on the payload itself and the flight plan to ensure that the mission is in accord with US policy and meets US responsibilities under international treaties. Regulations require AST to review how missions affect national interests. This authority could be used to block missions, and the *Commercial Space Launch Act* gives AST considerable power over commercial activity. Still, Congress has declared privately conducted commercial launches to be consistent with the national security and foreign policy interests of the United States. Thus, US policy seeks to encourage commercial space activities.

The AST also reviews commercial operator launch safety efforts, including:

- Procedures for safety controls for launch sites and flight corridors
- Range safety expertise
- Procedures for ground and flight safety
- Range tracking and instrumentation
- Vehicle safety systems
- Proposed vehicle design

Given the damage that could occur, and US international obligations to conduct space activities safely and responsibly, mission designers have found that these reviews have been conducted conservatively. Missions using proven government launchers and national launching facilities usually receive quick approval. A mission proposing to use a new and untested vehicle or launch site will encounter close scrutiny.

The AST also establishes liability insurance requirements for commercial launch activities taking into account the requirements of international law and US obligations under it. In general, insurance is required to cover the Maximum Probable Loss (MPL) amount, which is not to exceed $500M. Above that, the US Government provides indemnification for third party loses up to $1.5B plus the MPL. Also the US Government covers losses of its own property above the MPL.

The AST must coordinate commercial launches with all other interested agencies within the Federal Government, particularly the Department of Defense and the Department of State.

The present US Commercial Space Policy requires maximum utilization of commercial launch capability by the federal government. This includes planning missions in such a way that they match the space transportation capabilities of US commercial launch providers. The main exceptions to this requirement are national security or international collaborative efforts relating to science and technology, or space exploration. As part of its efforts to support development of the private launch industry, Congress provided in the *Commercial Space Launch Act* that excess or unused Government launch property should be made available to private launch companies through lease or sale agreements. Specifically, the Secretary of Transportation was directed to "take such actions as may be necessary to facilitate and encourage the acquisition (by lease, sale, transaction in lieu of sale, or otherwise) by the private sector of launch property of the United States which is excess or is otherwise not needed for public use" [USC, 1984].

12.3.9 Satellite Communication Law

The provision of communication services is the most commercially profitable use of space, with important applications in telephone, telegraph, television, and data transmissions. As a consequence, considerable attention is paid to law and policy issues that pose a potential to affect the business activities of communication satellite systems. Importantly, the *International Telecommunication Union (ITU)* is chartered by treaty to regulate the use of satellite orbits and frequencies. The ITU has extensive Radio Regulations that govern the use of orbits and frequencies.

Once a satellite is registered with the ITU, no subsequent system may interfere with its signal transmission. Generally, a similar replacement satellite will retain the rights of a registered satellite.

Since 1973, orbital positions have been associated with particular frequencies and technical characteristics. The unique properties of the stationary orbit over the equator (the "geostationary-satellite orbit" or "GSO") permits "stationary" positioning of a satellite.

The GSO is a limited resource because satellites need separation between them to limit radio frequency interference. The required separation depends on many factors including the frequencies used, geographic area served, and technical and operating characteristics.

Table 12-1◀, Fig. 12-6◀, Eq. 12-4◀

There have been a number of concerns over the reservation of orbital slots, and frequencies based on the "first-come, first-served" regulatory regime. In particular, developing nations have been worried that no slots will be left for them. An ITU compromise was crafted that left the conventional part of C and K_u radio frequency bands allocated to the Fixed Satellite Service on a "first-come, first-served" basis. At the same time, an allotment scheme provided all nations with orbital slots and associated K_a band frequencies, *World Radiocommunication Conferences (WRC)**, organized by the ITU, review, and, as necessary, revise the Radio Regulations governing the use of the radio-frequency spectrum and geostationary and non-geostationary satellite orbits. The ITU is required to perform these activities "in order to avoid harmful interference between the radio communication services of different countries" [ITU, 2011]. The nations also occasionally hold multilateral planning meetings to resolve area conflicts between themselves.

According to the ITU, "Equitable access to spectrum and orbital resources is of special concern, given the uneven needs of developed and developing countries. As a consequence, the principle of *a priori* planning of spectrum and orbit resources is considered in conjunction with a series of plans established by radio communication conferences" [ITU, 2011]. In exercising its registration function, the ITU's Radio Regulation Board and Radiocommunication Bureau do not evaluate the particular nation's reasons for making a particular assignment. They only seek to ensure that the national assignment conforms to identified services for the assigned frequency and that transmissions do not interfere with other broadcasts. Nations then determine how they will parcel out the allocated frequencies.

If a proposed satellite could cause interference, the sponsoring nation resolves issues under negotiation procedures established by the ITU. The ITU conducts much of its technical activities through the Radio Regulation Board and Radiocommunication Bureau. In the United States, the FCC allocates orbital slots and frequencies upon formal application. The approval process may take 3 to 5 years for a new communications service or up to a year for an existing satellite.

Enforcement mechanisms for violating the ITU treaties and agreements are rather limited. The ITU has been described as a "gentlemen's club" depending on the "goodwill of its members. There is no mechanism for forcing an administration into compliance with the rules" [de Selding, 2010].†

* Prior to 1993 these were called the World Administrative Radio Conference (WARC).

† When commenting on Iranian jamming of European communication satellites: "Rancy, a veteran international-frequency regulator who chaired the ITU's World Radiocommunication Conference in late 2007, said that while he hoped ITU pressure would affect Iran's behavior, he was not counting on an immediate stop to the practice."

Table 12-1◀, Fig. 12-6◀, Eq. 12-4◀

12.3.10 Intellectual Property Rights

Outer space activities utilize sophisticated technologies about which intellectual property and patent law play an important role to protect rights to the benefits of inventions, scientific discoveries, industrial designs, and protection against unfair competition. Space activities have been shifting from state-owned activities to private and commercial activities and this has heightened the emphasis on protecting property rights, whether in tangible or intangible form. Private financing has to be motivated by expectations that research and development and other investments can be recovered and profited on.

Patent law and intellectual property rights in the United States stem from the first national patent law enacted in 1790. It was based upon the US Constitution (1787), article 1, section 8, paragraph 8 that states, "Congress shall have power to... promote the progress of science and useful arts by securing for limited times to authors and inventors the exclusive right to their respective writings and discoveries." That limited period in the US is usually 20 years from the date of filing for a patent with the US Patent and Trademark Office.

According to the World Intellectual Property Organization, "A patent is an exclusive right granted for an invention, which can be a product or a process that provides a new and inventive way of doing something, or offers a new and inventive solution to a problem. A patent owner has exclusive rights to prevent third parties not having the owner's consent from the acts of making, using, offering for sale, selling, or importing for these purposes the patented invention... The patent owner may give permission (license) to other parties to use the invention on mutually agreed terms. The owner may also sell the right to the invention to a third party, who will then become the new owner of the patent. Once a patent expires, the protection ends, and the invention enters the public domain. Patent owners are obliged, in return for patent protection, to publicly disclose information on their invention in order to enrich the total body of technical knowledge in the world. This body of public knowledge promotes further creativity and innovation by others. In this way, patents provide not only protection for the owner, but valuable information and inspiration for future generations of researchers and inventors" [IBWIPO, 2004].

The United States has enacted law relating to protecting invention for space activities; the 1990 law, found at 35 USC § 105, provides in pertinent part:

Any invention made, used, or sold in outer space on a space object or component thereof under the jurisdiction or control of the United States shall be considered to be made, used or sold within the United States for the purposes of this title, except with respect to any space object or component thereof that is specifically identified and otherwise provided for by an international agreement to which the United States is a party, or with respect to any space object or component thereof that is carried on the registry of a foreign state in accordance with the Convention on Registration of Objects Launched into Outer Space.

Any invention made, used or sold in outer space on a space object or component thereof that is carried out on the registry of a foreign state in accordance with the Convention on Registration of Objects Launched into Outer Space, shall be considered to be made, used or sold within the United States for the purposes of this title if specifically so agreed in an international agreement between the United States and the state of registry.

Mission designers should be aware of two patent cases. Hughes Aircraft Company/Williams in 1973 obtained a patent relating to the spin stabilization of satellites to assure obtaining and maintaining satellite attitude on orbit. Hughes Aircraft Company sued the US Government for infringement of its patent, because the government used the concept without paying for it, and transferred it to third parties. It was not until 1983 that a multi-billion-dollar decision against the Government was handed down after the United States and others had launched 108 satellites, which were held to have infringed the patent rights of Hughes Aircraft Company. The lesson is clear; patent infringement may have very significant economic consequences for mission designers and systems engineers.

The second case stems from a matter where TRW obtained a "patent" for a particular system orbital architecture. It claimed rights to an "orbital shell" above the Earth from 5,600 to 10,000 nautical miles, to be reserved for TRW exclusively, for all communications applications to mobile handsets. Soon after the patent was granted TRW sued ICO Global Communications Ltd. to prevent them from launching satellites into the proscribed altitudes. The case was dismissed because there was no present infringement, and TRW and ICO separately resolved their issues. Nevertheless, this is a troublesome patent, because it flies directly in the face of the Outer Space Treaty provisions on non-appropriation of space by any nation including its private entities. The patent could be fatally flawed, especially if it is interpreted to give any property rights to a spatial location.

12.3.11 Insurance

The *Outer Space Treaty* requires a nation to *authorize* and *continually supervise* all of its space activity, including those of its citizens acting privately, to assure they conform to treaties and international law obligations. The treaty also imposed liability for space activity, and this provision was amplified by the *Liability Convention*. In order to satisfy these mandates, commercial space activities are required under US law to procure insurance to address the potential for damages. There are a number of different types of insurance that apply to space activities:

- ***Pre-Launch.*** Pre-launch coverage insures against the risks from damage during shipment of the launcher and the satellite, as well as during integration and movement to the launch pad
- ***Government Equipment.*** The government requires launches on commercial expendable vehicles to provide insurance for loss to government launch equipment and facilities
- ***Third-Party Liability.*** Launch vehicle and space operations may cause injury to others on launch, on-orbit, and through the reentry. The risk on reentry is small because these objects generally burn up in the atmosphere during reentry. If they do not, they would typically strike the ocean or one of the uninhabited areas that make up most of the Earth's surface. Because of this low risk, coverage is less expensive though still required by AST for launch vehicles and satellites and by NASA for commercial satellites. There are comparable risks for other launch and on-orbit activities, which are described in greater detail below
- When NASA was the only means of access to space for commercial satellites, their regulations required an insurance policy of $500M, with the government assuming the risk above that amount. This initial limit was a practical one because the insurance community could not insure a greater amount. Now that commercial launch vehicles are becoming the prime means of transportation into space, Congress has established and extended a similar cap of $500M, until the world's insurance community can provide greater coverage. The purpose of this provision of this amendment to the *Commercial Space Launch Act* was to attempt to level the commercial space launch playing field. This amount was based upon the prior NASA policy of requiring $500M coverage, adjusted for inflation
- Private operators of launch vehicles must also concern themselves with liability claims from citizens of the US and around the world. The AST regulations require coverage against this risk before issuing a launch license
- ***Launch Failure.*** The most expensive insurance coverage insures the satellite value from the moment of lift-off to on-orbit checkout. The prices for this coverage range from 5% to 30% of the satellite value depending on the perceived or actual risk, and availability of indemnification monies in the commercial space marketplace. Man-rated vehicle such as the Space Shuttle have enjoyed some discounts because the insurance community had higher confidence in their reliability. However, the insurance marketplace can change based on problems in the space community. Premiums rose dramatically with the loss of the Shuttle Challenger in 1986 and several significant failures of expendable launch vehicles shortly thereafter. In fact, during that timeframe, the space insurance community paid out 3 times more in loss claims than they took in through premiums. Once the United States and

Table 12-1◀, Fig. 12-6◀, Eq. 12-4◀

Ariane launch vehicle programs were back on track, premiums moderated. Mission designers may want to choose launch vehicles and satellites with a history of proven success. Otherwise, insurance could be prohibitively priced

- ***Loss to Others on the Flight.*** Under provisions of NASA launch contracts, the risk of damage to a payload by another payload or by NASA's actions is addressed and mitigated by use of a *hold harmless clause*. This contractual clause requires that all parties to the launch agree not to sue each other regardless of what happens. This practical remedy came about because it is very difficult to calculate the risk of such problems and determine the premium. Given their size and perception that they are a deep pocket for any liability issues, large spacecraft and launch vehicle integrators monitor these agreements and insurance requirements for the system

- ***On-Orbit Performance.*** Insurance can also be obtained to address concerns on whether a satellite will continue to perform its mission activities for its planned lifetime. Launch failure insurance terminates after on-orbit checkout. Satellite owners obtain on-orbit performance coverage if they do not have sufficient resources to "self insure" the risks. Banks that finance communications systems usually demand the coverage. An alternative risk mitigation strategy may involve placing satellites on orbit to back-up the satellite losses

Unfortunately, space launch campaigns and on-orbit operations are relatively new activities. There have not been enough total launches to derive actuarially correct figures. Therefore, the price of insurance tends to depend on "feel" and various intangibles. Failures drive prices up, and successes decrease them. Interestingly, these variations occur whether or not a failure was insured.

12.3.12 Remote Sensing

The use of commercial remote sensing satellites is part of a trend toward global transparency. High-resolution satellite imagery of the globe is available now and provides the space industry an opportunity to make profits from a wide variety of consumers. US Governmental policies permit and encourage such systems. US national policy is to "advance and protect US national security and foreign policy interests by maintaining the nation's leadership in remote sensing space activities, and by sustaining and enhancing the US remote sensing industry. Doing so will also foster economic growth, contribute to environmental stewardship, and enable scientific and technological excellence" [Commercial Remote Sensing Policy, 2003]. To do this, the US Government is supposed to "Rely to the maximum practical extent on US commercial remote sensing space capabilities for filling imagery and geospatial needs for military, intelligence, foreign policy, homeland security, and civil users" [Commercial Remote Sensing Policy, 2003].

Of course, not all uses of remote sensing information are benign, and the provision of information shifts power from the former holders of secrets to the newly informed, which has implications for national sovereignty, for the ability of corporations to keep proprietary information secret, and for the balance of power between state and non-state actors. Sovereignty is directly challenged by new satellite systems, and this has important policy implications. If permitted to image territories anywhere in the world and sell such imagery to anyone, governments lose a significant degree of control over information about their territory. As noted in the literature,

Under international law, countries have no grounds for objecting to being imaged from outer space. The existing standards, the result largely of long-standing US efforts to render legitimate both military reconnaissance and civilian imaging from space, are codified in two UN documents. The Outer Space Treaty of 1967 establishes the notion that outer space cannot be claimed as national territory, legitimating satellite travel in space over any country's territory. And despite years of efforts by then-Soviet bloc and developing countries to claim a right of prior consent in order to review and possibly withhold any data about their territory, in 1986 the UN General Assembly adopted a set of legal principles on civilian remote sensing that make no mention of any right of prior consent. Instead, the principles merely require that "As soon as the primary data and the processed data concerning the territory under its jurisdiction are produced, the sensed State shall have access to them on a nondiscriminatory basis and on reasonable cost terms" [Florini and Dehqanzada, 1999].

Publically available imagery has improved from ten, to five, to one, and to 0.5 meter resolution. Higher resolution imagery is being demanded. If a country being imaged asks for a copy, it is entitled to buy one at the going rate. Military installations could be imaged. Industrial concerns may find themselves being observed by competitors. This can be considered a form of corporate espionage, but satellite imaging is legal.

Pursuant to the *1992 Land Remote Sensing Policy Act*, responsibilities have been delegated from the Secretary of Commerce to the Assistant Administrator for NOAA Satellite and Information Services (NOAA/NESDIS) for the licensing of the operations of private space-based remote sensing systems [USC, 5601]. In accordance with the Act, the *15 CFR Part 960* regulations concerning the licensing of private remote sensing space systems have been issued. Given national security and diplomatic sensitivities and pursuant to regulation, it is thus unlawful for any person who is subject to the jurisdiction or control of the United States, directly or through any subsidiary or affiliate, to operate a private remote sensing space system without possession of a valid license issued under the Act and the regulations.

Table 12-1◀, Fig. 12-6◀, Eq. 12-4◀

The ability to image the globe is not unfettered and can be limited by the terms of the license and in accord with *15 CFR Part 960*. Consistent with the requirement that licensees operate their systems in a manner that protects national security concerns, foreign policy and international obligations, the Kyl-Bingaman Amendment requires that "[a] department or agency of the United States may issue a license for the collection or dissemination by a non-Federal entity of satellite imagery with respect to Israel only if such imagery is no more detailed or precise than satellite imagery of Israel that is available from commercial sources" [National Defense Authorization Act, 1997]. So the Department of Commerce must make findings as to the level of detail or precision of satellite imagery of Israel available from non-US commercial sources. NOAA requires an applicant to submit a plan explaining how its proposed system will be able to restrict the collection and/or dissemination of imagery of Israeli territory at a level of resolution determined by the Commerce Department. As an additional licensing requirement, licensees are required, "upon termination of operations under the license, to make disposition of any satellites in space in a manner satisfactory to the President," in accordance with Section 202(b)(4) of the *Land Remote Sensing Policy Act*. Applicants are also required to provide at the time of application a plan for post-mission disposition of remote sensing satellites.

12.3.13 Space Debris

As discussed in Sec. 7.5, the global expansion of space technologies, and especially the availability of diverse small spacecraft technologies and providers, is leading to a rapid expansion in the population of active spacecraft, and associated space debris. As of 2011, the DoD tracks over 22,000 objects in orbit. Of these objects, 928 are operating satellites with 437 owned by the United States, Russia 95 and China 58. Of the operating satellites, 449 are in low earth orbit (LEO), 59 in medium earth orbit, 381 in geosynchronous orbit (GEO) and 39 in elliptical orbits.

Space Situational Awareness (SSA) is crucial to monitoring space, and is absolutely essential to mission success and protection of space assets [Space Foundation, 2008]. It enables a space power to "detect, identify, assess, and track space objects and events to support space operations. SSA is also critical to space support operations, such as placing satellites in orbit," or performing anomaly or recovery operations [del Monte, 2007].* A basic and fundamental component of establishing SSA is space surveillance. Space surveillance is the "observation of space and of the activities occurring in space." Space surveillance tasks can be accomplished through a variety of ground and space-based radar and electro-optical sensors. Space-faring nations can use their awareness to avoid space hazards, or advise other nations on those dangers and perhaps on the means and ways to avoid producing them. Surveillance enables deterrence and defense activities against man-made and environmental threats, and promotes stability in the interactions among space operators.

A perfect surveillance system would provide "continual awareness of orbiting objects; real-time search and targeting-quality information; threat detection, identification, and location; predictive intelligence analysis of foreign space capability and intent in a geopolitical context; and a global reporting capability for friendly space systems" [Space Foundation, 2008]. If all operators had contemporaneous and exquisite awareness of their satellites, and threats to them, they could then react and maneuver their spacecraft out of harm's way if such a collision was imminent. Unfortunately, that perfect level of surveillance and awareness is nothing but science fiction. The current US and other space surveillance systems cannot observe everything in orbit continuously, nor can they provide any real-time services. The limited number of sensors, their locations, their fidelity, and plain fundamental physics do not enable that level of capability. The system only tracks satellites when they pass over or near one of its radar or optical sensors.

Owner/operators should expect to possess positional information on their satellites that are often an order of magnitude more accurate and timely than that gathered by US and other space surveillance systems. And that knowledge is even more significant when one takes into account their knowledge of maneuver plans. Despite this advantage, no single operator knows the location of all space objects with the best fidelity; and few have insight into what other non-US surveillance sensor networks observe and report. Even the organization with access to the most comprehensive catalog of SSA information, US Strategic Command (USSTRATCOM), which publishes

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* The attributes of the SSA definition have been examined in a number of venues. For example, an ESA User Expert Group performing an SSA requirement study preliminarily defined Space Situational Awareness (SSA) "as a comprehensive knowledge of the population of space objects, of existing threats/risks, and of the space environment. According to del Monte, the study concluded that SSA is the understanding and maintained awareness of (a) the Earth orbital population, (b) the space environment, and (c) possible threats. "(a) Earth orbital population. Detection and/or tracking of man-made objects (e.g., US SSN Catalog, comprising spacecraft, rocket bodies, mission-related objects, and fragments); identification and characterization of detected objects (e.g., launching nation, owner, operator, object status, and function); determination of orbit state and covariance information; identification of spacecraft maneuvers; determination of spacecraft attitude; determination of antenna/instrument pointing. "(b) Space environment—Detection and/or tracking natural objects (e.g., near-Earth objects [NEO]); detect and understand man-made, induced environment; detect and understand interferences; forecast space weather and its effects; predict the natural particulate environment and its effects (meteoroids). "(c) Possible threats—Predict and assess the risk to humans and property on ground and in air space due to re-entries; detect and assess adversary use or preparations for adversary use of or upon space systems (e.g., ground- or space-based); detect on-orbit explosions and release events (accidental or intentional); predict and/or detect on-orbit collisions (accidental or intentional); predict and detect permanent or temporary disruption of mission and service capabilities."

Table 12-1◀, Fig. 12-6◀, Eq. 12-4◀

the information about these objects through its SSA Sharing Program, doesn't know where all orbiting objects are, all the time. Even though it has more global sensor coverage and an ability to track smaller objects, the USSTRATCOM's SSA system would profit by obtaining owner/operator information and maneuver plans.

To be sure, the 22,000 orbiting objects described above are only those that can be tracked within the capability of USSTRATCOM's sensors, which have difficulty tracking objects smaller than the size of a grapefruit or 10 cm. Compounding the challenges, there may be millions more objects on orbit and up to 330 million objects of 1 mm to 1 cm size and 560,000 objects of 1 cm to 10 cm (see Sec. 7.5). The probability of collision by satellites with one or more of these small objects is almost a certainty given any amount of time on orbit. And small objects can be very destructive.

Another issue confronting satellite operators is that even if they know where all the threatening objects are and their ephemerides, they may not have sufficient time, fuel or maneuvering capability to avoid them. One must also consider the variety of collision scenarios. For example, two live, maneuverable satellites could both perform maneuvers to avoid a possible collision. Unless these maneuvers are coordinated, they could end up making the situation worse rather than better and wasting valuable fuel. Of course, this is not a new problem or one that operators are unaware of, but the recent growth in on-orbit debris has made solving or mitigating the problem more urgent.

In addition, the control of satellites is already a challenging task, given the space environment the satellites operate in and the distances involved in the communications chain, 500 to 35,000 km. Satellite operations are based on the ability to communicate with and control satellites, hopefully without hindrance from man-made objects or actions. Indeed, some satellites, such as Galaxy 15, have stopped responding to commands from ground operators. Galaxy 15, dubbed "Zombiesat" by the media, began to drift, while its receiver and transmitter equipment continued to function. The combination of uncontrolled drifting, with an active communications payload, meant that Galaxy 15 posed serious radio interference concerns to other satellite operations in the geosynchronous orbit (GEO) belt. Since its communications payloads receive and transmit functions were still working, the satellite could pickup and re-broadcast C-band signals intended for other active C-Band satellites. So as it drifted through the GEO belt near other satellites, Galaxy 15 could pickup and re-broadcast C-band emissions, potentially causing multi-path interference for other satellite operations. Rigorous and sophisticated operations, data sharing and coordination among several satellite system operators were necessary to mitigate the satellite's threat to other operating satellites, before it could be recovered by the operator.

Satellites fail in the active GEO belt, on an average about one per year, [Choc and Jehn, 2010], as documented in an annual report published by the European Space Agency. There are a variety of causes and scenarios for these failures. These failures leave satellites in active orbits and they then present danger to other systems.

Under the recently adopted United Nations Space Debris Mitigation Guidelines, which are based on the more extensive United Nations Interagency Space Debris Coordinating Committee (IADC) Guidelines, spacecraft operators are supposed to perform an end-of-life disposal maneuver to remove their satellite from the GEO region. This usually involves boosting it at least 250 km (155 miles) above the GEO belt. Unfortunately, the guidelines don't resolve problems associated with spacecraft left on orbit during the early years of the space age. Compliance with the IADC guidelines for new spacecraft is still spotty at best. Of the 21 GEO spacecraft that reached end-of-life in 2009, only 11 were disposed of properly. Several were moved, but not into an orbit high enough to ensure that they do not cause problems in the near future. Three spacecraft were abandoned in the active belt and are now librating about the 75 deg East libration point. Four rocket bodies, used to place payloads in geosynchronous orbit, also orbit within the protected zone [Weeden, 2010].

The continuing and expanding numbers of space debris and orbiting systems is heightening safety of flight concerns among space-faring states. Some argue we are nearing the point where whole sectors of the space domain are saturated and thus lost to safe and secure operations. Experts forecast that the debris may eventually and continually collide with other debris and active space objects; these collisions will then multiply and create even greater numbers of objects that could damage other spacecraft (see Sec. 7.5).

In addition to being what some would argue is one of the most physically congested regions, with significant space debris and satellite collision threats; the GEO belt is also a region in space that suffers from significant electromagnetic radio frequency inference (RFI) between satellites. This interference threat exists in part because many satellites in GEO are all relatively close and most broadcast and receive in the same frequency bands. The Galaxy 15 incident described above serves as a representative example of the threat radio frequency interference poses to on-orbit systems. Inadvertent and intentional jamming and RFI have caused problems for satellites for many years.

Space debris is a significant environmental issue. Mission designers must plan missions to prevent or abate space debris as much as possible. They may also want to design satellites to survive the risk of collision with debris. Finally designers should consider designing systems that support an effective SSA system to ascertain and attribute on-orbit events so that appropriate responses can be developed and implemented by affected space-operators and, if necessary, interested States.

12.3.14 Environmental Concerns

Space systems introduce pollution contamination in various degrees into different parts of the space environment. A spacecraft and its launch system produces

exhaust gases, and a mix of cooling water, sand and dust when launching. Chemical releases from spacecraft on orbit could have a depleting effect on the ozone layer, and reduce the density of electrons in the ionosphere, potentially distorting radio communications. Reentry involves the production of metal vapors into the upper atmosphere. Space activities could introduce biological pollutants into the space environment.

The *Partial Test Ban Treaty* prohibits nuclear explosions in the atmosphere and in space. The *Convention on the Prohibition of Military and Other Hostile Use of Environmental Modification Techniques* also helps protect the environment.

The United States still has obligations under Article IX of the *Outer Space Treaty,* which provides for international protection of the outer-space environment. Thus, it must comply with either the letter of the Act's provisions or their essence. If a proposed mission "has been done before," environmental questions may quickly be resolved. But if the mission includes something new, then approval may take some time, depending on how strongly the mission affects outer space or, especially, any celestial body.

The *National Environmental Policy Act* (NEPA) and its implementing regulations established US policy promoting the enhancement of the environment. It sets up procedural requirements for all US Government agencies to prepare Environmental Assessments (EAs) and Environmental Impact Statements (EISs) describing the environmental effects of proposed actions. NEPA's requirements apply to all Federal agencies procuring and operating space systems. These agencies must integrate environmental values into their decision-making processes by considering the environmental impacts of their proposed actions and the reasonable alternatives to those actions. NASA follows the NEPA regulations promulgated by the Council on Environmental Quality (CEQ) and has developed agency-specific NEPA policies and regulations to ensure compliance with the NEPA statute, implementing regulations, and related Executive Orders.

The NEPA is an "umbrella" legislation and its processes often involves coordinating compliance with the following laws:

- ***Clean Air Act***—The Clean Air Act, incorporated as Title 42-Chapter 85 in US law, provides regulations to protect and improve the US's air quality as well as the ozone layer
- ***Clean Water Act***—The Clean Water Act regulates the restoration and maintenance of the integrity of the nation's waters and wetlands, and provides assistance for the improvement of wastewater treatment
- ***National Historic Preservation Act***—The National Historic Preservation Act promotes Federal leadership for the preservation of historically significant, cultural resources and to "foster conditions under which our modern society and our prehistoric resources can exist in productive harmony"
- ***Endangered Species Act***—The Endangered Species Act regulates the conservation of threatened or endangered animals, plants and habitats

Under NEPA, the responsible agency must analyze the potential environmental effects. If more than one agency is involved, a lead agency supervises the preparation of the environmental documentation. For a private or commercial launch, AST would require an environmental impact assessment under its mandate from the *Commercial Space Launch Act.*

An EA helps determine the need for an impact statement on a particular program. It must include enough information to allow one to determine whether the proposed action is major and whether it significantly harms the environment of the "global commons." It must assess the environmental effects and the need for the proposed action plus available alternatives. Unless security restrictions intervene, the assessment is usually made available to the public upon request. There is no need to obtain public comment. An EA is less formal and rigorous than an EIS. Typically, the EA applies to:

- Spacecraft development projects in space science and in space and terrestrial applications
- Specific experimental projects in space and energy technology
- Development and operations of new space transportation systems
- Advanced development of new space transportation and spacecraft systems

In contrast, there is a need to file an EIS when an action is "expected to have a significant impact on the quality of the human environment." The draft statement should be thorough enough to permit analysis and comment. If it is complete enough after agency review, it goes to the public for comment. At this time, the Department of State, the Council on Environmental Quality, and other federal agencies have a chance to comment. For the proposed action and reasonable alternatives, the EIS: (1) considers purpose and need; (2) provides a detailed description; (3) analyzes the environmental effects; (4) briefly describes the affected environment of the space "global commons"; and (5) compares the alternatives' effects on the space "global commons."

If something is not known because it is unavailable or scientifically uncertain, the EIS must say so. Public hearings are not required, but may be appropriate, depending on circumstances. After agencies and the public have commented, the EIS is revised as necessary, further analyzing any issues they have raised. It is then published in the *Federal Register* before forwarding it to the US Government decision maker for final action.

Normally, an EIS is required for research and development activities associated with developing and operating new launch vehicles, space vehicles likely to release large amounts of foreign materials into the

Table 12-1◀, Fig. 12-6◀, Eq. 12-4◀

Earth's atmosphere or into space, and certain systems for nuclear propulsion and generating power.

In some cases, space activities may affect the environment of a foreign nation or a resource designated as one of global importance. If so, slightly different procedures would be required. First, an environmental survey or review would need to be prepared. An environmental survey is a cooperative action and may be bilateral or multilateral. Whether or not a review is conducted depends on consultations which determine if the proposed action would do significant harm. The content is flexible, but generally includes (1) a review of the affected environment, (2) the predicted environmental effects, and (3) significant known actions that governmental entities are taking to protect or improve the environment against the proposed action.

The US Government prepares an environmental review unilaterally. In effect, it is an internal action, by which one or more governmental agencies surveys the important environmental issues associated with the proposed action. It contains essentially the same information as an environmental survey.

12.3.15 Lasers, Particle Beams, and Nuclear Power

Lasers and particle beams illuminating into, through, and from space have occurred since the beginnings of laser technology. Use of laser systems is contemplated and used for a wide variety of needs: communications, calibration, ranging, active and passive tracking and space surveillance, power transfer, targeting, and negation purposes. Laser communications present great opportunities to increase data through-puts and security. Unfortunately, while not all lasing is malicious, the use of laser capabilities poses the potential to damage systems and payloads on orbit, spacecraft personnel, and ground-based populations, depending on the design and employment of the system. It is alleged that some countries have secretly fired powerful laser weapons to disable imagery satellites by "blinding" their payloads [Harris, 2006].

Laser beam propagation to and from the ground to space has executed the cognizance of various government agencies, namely: the Occupational Safety and Health Administration (OSHA) responsible for protecting workforce personnel; the Federal Aviation Administration (FAA) responsible for protecting pilots and aircraft; and the Laser Clearinghouse of US Strategic Command (USSTRATCOM) responsible for protecting space assets.

The Laser Clearinghouse (LCH) and laser owner/operators (LO/Os) are required to work together to provide safe and responsible laser activities, as required by Department of Defense Instruction (DODI) 3100.11. US Government LO/Os with non-waived lasers require Joint Functional Component Command (JFCC) —Space authorization to conduct laser activities according to US Strategic Command Instructions. DoD LO/Os must obtain predictive avoidance open firing windows from the LCH or the Joint Space Operations Center (JSpOC) prior to conducting laser activities [Cheyenne Mountain Operations Center, 2008].

The use of nuclear power in space has enabled the extension of space activities into regions that would not have been possible with non-nuclear power sources. The cold, hostile, and poorly lit environments of the outer planets require a rugged, solar-independent power source that can survive and function for long periods of time. Missions closer to the Sun have environments that can be too harsh for conventional power sources. As the need for nuclear power increases, it will not be banned from space, but its use will be closely regulated.

The main design requirement on US space nuclear power systems has been safety. Extensive safety testing and analyses have been conducted beginning with the very first nuclear power systems (SNAP-3B). Formal safety analysis and independent safety reviews are required before each launch. With the enactment of NEPA, an independent EA is also conducted. The primary safety requirement for radioisotope power sources is to contain or immobilize the radioisotope fuel so that there is no interaction with humans and the environment. Designs are assessed through a wide combination of safety tests. The primary safety requirement is to maintain the reactor in a subcritical condition.

Design for safety must include both system and mission design. The methods to reduce risks from nuclear materials include containing them within radioisotope thermoelectric generators, diluting and dispersing them with reactors, delaying their effects by boosting them into a higher orbit, and possibly retrieving them using a vehicle like the Space Shuttle.

The US safety program includes an Interagency Nuclear Safety Review Panel composed of three coordinators appointed by the Secretary of Defense, the Administrator of NASA, and the Secretary of Energy. The Nuclear Regulatory Agency, the Environmental Protection Agency, and the National Oceanic and Atmospheric Administration also participate in these reviews. The safety review ascertains whether the benefits of using nuclear power are worth the risks. The policy of the United States in using radioisotope thermoelectric generators following an aborted SNAP-A mission in April 1964 was to design the container so that all nuclear material would survive intact, regardless of the nature of the accident. The policy makers specifically envisioned reentry and impact on earth. This effort had important consequences. On May 18, 1968, the range safety officer aborted the launch of NIMBUS-81 at an altitude of 30 km over the Santa Barbara Channel. The generator capsules were recovered without incident. Those who launched SNAP 10A with a nuclear reactor in 1965 launched the reactor in a subcritical mode, designed it to remain subcritical at or after impact should it reenter the atmosphere before start-up, and delayed its start-up until it had reached orbit. It is in an almost circular polar orbit, which has a decay life of 4,000 years. Additionally, this reactor package was designed to come apart on reentry.

Table 12-1◀, Fig. 12-6◀, Eq. 12-4◀

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13 Reducing Space Mission Cost and Schedule

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13.1 The Need to Reinvent Space
The Need for Low-Cost, Responsive SmallSat Missions; The Need for Mission Diversity

13.2 It's Possible, But It Isn't Easy

13.3 Counterproductive Approaches to Reducing Cost

13.4 Cost vs. Reliability: Focusing on the Mission Objectives—*Mike Hurley* and *Bill Purdy*
Understanding What Reliability Is and Is Not; On-Orbit Reliability—History and Examples; Linking Reliability to Schedule; Considerations for Small and Large Satellites and Systems; Considerations for Achieving On-Orbit Availability; Conclusion

13.5 Principal Methods for Reducing Cost and Schedule
Attitude; Personnel; Programmatic, Government/Customer; Systems Engineering; Mission; Launch; Spacecraft Technology; Operations; Summary

13.6 Avoiding Cost and Schedule Overruns
Traditional Causes and Solutions; An Alternative View of Causes and Solutions

This chapter focuses on perhaps the most fundamental problem facing space missions today—they cost too much and take too long. (See Sec. 1.1.) Obviously no one sets out to design a space mission that costs too much and takes too long, yet fixing this problem has proven remarkably elusive. A number of major and important activities in space, such as the human exploration of Mars and the Wide Field Infrared Survey Telescope (WFIRST) have been delayed for a decade or more due to budget pressure. On the other hand, there are a great many missions flying, largely using small spacecraft*, that are very low cost and also have high reliability. What we would like to find is what these small, low-cost missions can teach us about reducing the cost and schedule on all space missions.

There is a substantial bibliography on reducing space mission cost on the website. Several of the books are somewhat older, but remain the standard reference works on this topic. Wertz and Larson [1996] provide an overview of the entire subject, with some updates in Wertz and Dawson [1996] and Wertz, et al. [2011]. Sarsfield [1998] gives the results of a RAND study on low-cost science missions and London [1994] provides an overview of both the reasons for high launch costs and methods to overcome them. AMSAT has created a number of documents providing detailed recipes for building low-cost communications satellites [Davidoff, 1998; Ford, 2010]. Fleeter [2000] gives a detailed description of the use of smallsats to reduce cost and schedule. While not directly related to reducing cost, Harland and Lorenz [2005], provide an excellent discussion of space system failures, which is, of course, a key element of mission design that may, or may not, be related to attempts to reduce cost. Also important are the two principal annual conferences on this topic listed in the bibliography—the Utah State University SmallSat Conference in Logan, Utah, in August and the Reinventing Space Conference in Los Angeles in April or May. There are also several international SmallSat conferences.

Most of the specific missions cited in this section are SmallSat missions simply because that is where most of the experience base resides in methods for reducing cost and schedule. (See Sec. 25.3 for a discussion of SmallSats and 25.4 for CubeSats.) What we would like to do is take this experience base and make it applicable to more traditional larger missions (including science, military, interplanetary, commercial, and human spaceflight) by using any or all of the following:

- Applying some of the processes and methods first developed and tested on small satellite programs to larger, more traditional programs
- Using small satellites to test new technology in space much faster and at lower cost than would otherwise be possible
- Redefine the missions themselves so that they can be done, at least in part, by one or multiple small satellites

I teach a course in "Reinventing Space—The Design of Low Cost Space Missions" biannually at the University of Southern California where the question of applicability to

* What constitutes a "small" spacecraft is intentionally a bit nebulous here. In the context of this chapter, "small" represents more a state of mind than physical size. A small spacecraft is one that has a limited budget and schedule, constrained objectives, one, or at most a few, payloads, and a willingness to be less risk averse than a traditional large spacecraft. As discussed in more detail in Sec. 2.1.8, these spacecraft are typically less than 500 kg, but it is more a matter of how they are built than the actual physical size.

large missions arises often. At the end of the course, I apply the methods developed primarily on smallsats to the largest space program that seems practical in the near-term, a 1000-person permanent colony on the Moon. I am convinced that such a "Luniversity" can be created for substantially less than the cost of the ISS. [Wertz, 1999]. The point here is not to begin a push for large-scale lunar colonies, but simply to point out that the goal of this chapter is to use methods, processes, and technology developed largely for smallsat missions to dramatically reduce the cost of the space mission enterprise as a whole.

13.1 The Need to Reinvent Space

What is "Reinventing Space"? In my view, this means combining the current dramatic advances in microelectronics, software, and material science with the goal-oriented, much more driven spirit of the Apollo, Explorer, or Corona missions to create smaller, lower-cost, more responsive systems capable of doing more and doing it more quickly and at lower cost. In short,

> ***Reinventing Space* is using modern technology and old-fashioned drive, determination, and some willingness to accept risk to do much more, much faster, with fewer resources.**

In practice, this means creating what I call *Mission Diversity*, i.e., a mix of large, typically very expensive programs and much smaller, quicker, more flexible space systems—the PT boats of space. In some respects, reinventing space is similar to the call by NASA Administrator Dan Goldin in the late 1990's for "faster, better, cheaper" space missions, with two critical differences:

- The problem today is substantially more acute
  - For example, in 2007 the top 10 DoD space programs were overrun by over $32 billion and it continues to get worse because of the Space Spiral shown in Fig. 1-2 in Sec. 1.1.
  - The economic decline prior to 2010 has created an unprecedented worldwide demand to reduce cost
  - To be relevant today, systems must be responsive in hours or days, not months, years, or decades
- The technology is available to make it happen
  - Microelectronics are greatly increasing the capability of small, low-cost systems
  - Computer technology and software are allowing major advances in on-board processing
  - Advanced materials technology is creating a new realm of lighter, stronger spacecraft and launch vehicles

In addition, it is likely that failing to aggressively attack this fundamental problem will mean that the United States will lose its dominant position in space. It is already the case that most commercial space launches are now done from Europe, Russia, India, or China. The US is rapidly losing its primary position in human spaceflight and this is likely to continue. Other nations have long-standing, very active programs to build both small launch systems and increasing capable small satellites, taking full advantage of modern technology to create increasingly competent systems that will be capable of being launched quickly and in large numbers. If the downward Space Spiral of Fig. 1-2 continues, the American space program has the potential of significantly falling behind other global space programs, while remaining the most expensive in the world.

Much of traditional mission assurance for space systems focuses on parts reliability and the capability of a system to work in space for a decade or more. Unfortunately, as discussed in Sec. 13.4, parts reliability does not prevent the most likely causes of mission failures or lack of data, such as:

- Launch failures (OCO, Challenger)—on the average, approximately 10% of launches fail
- System failures (USA 193)
- Collisions (Iridium 33/Cosmos 2251) which typically leave thousands of untrackable debris particles
- Operator errors (Huygens Titan probe wind experiment, Viking 1 Mars lander)
- Assembly errors (Genesis)
- Data unavailable due to cost or schedule overruns
  - Extended delays (Chandra, JWST, NPOESS)
  - Program cancellation due to cost (T-Sat, Constellation, Human mission to Mars)

For most traditional large satellite programs, the consequences of any of the above is that the asset isn't available for any current need and will take years or decades and a great deal of money to replace. In addition, 15 years after the successful launch of a large, perfectly assembled satellite with no orbit failures, no collisions, and no fatal operator errors, we have:

- A satellite built with 25-year old technology to meet 25-year old mission needs and trying to cover the entire world, and
- No production line and no one who knows how to build a replacement.

13.1.1 The Need for Low-Cost, Responsive SmallSat Missions*

The arguments for creating mission diversity by adding substantially more smallsats and responsive, smallsat launch capability to the broad mix of satellite systems

* For a more detailed discussion of this topic see Wertz [2010]

fall primarily into the following categories, each of which is discussed below:

- They create the potential to reduce long-term space mission cost by billions
- They reduce mission risk and fragility
- They can be put into operational orbits that aren't feasible with more traditional missions and effectively mitigate the growing problem of orbital debris
- They increase efficiency in the use of both human and financial resources
- They enable missions that aren't otherwise affordable or feasible
- They make newer, better technology available to users
- They can be far more responsive to world events and the needs of the end user
- They create a more economically sustainable business model for the space industry

The Potential to Reduce Long-Term Space Mission Cost by Billions. As discussed in Sec. 2.1.8 and in more detail in Sec. 25.3, there are many smallsat missions with a high level of utility and high reliability that are flying, being built, and being planned. Unlike more traditional satellites, smallsats are evolving rapidly. A smallsat that uses technology more than a few years old is effectively a dinosaur waiting to be replaced by systems much faster, much cheaper, and much more capable.

Nonetheless, this very substantial technological progress would appear to have very little impact on long-term space mission cost for several reasons. Some communities, such as human spaceflight, don't see the relevance of smallsats unless we are able to microminiaturize people, which is still a ways off. Also, the cost of the major space programs in DoD, NASA, the intelligence community, and the large international programs are dominated by the cost of the "programs of record" that are typically very large spacecraft built and flown over decades. Until we find a way to reduce the cost of large programs, it appears that any resulting savings will be small. But changing how the large programs are built and flown is very difficult because of the Space Spiral that we have discussed. The cost of these programs and of the associated launch vehicles continues to escalate with little potential for being reduced. By the time the programs get to the level of the NASA Director, the Secretary of the Air Force, or Congress, the only alternative to accepting high cost is to cancel the program and this has now occurred on a number of very expensive, high profile systems.

This issue is perhaps best addressed by an airplane analogy. Assume that an air fleet is dominated by 747s with very expensive equipment on board and we are trying to find a way to reduce cost. We can't change the 747 or any of the equipment without risking the entire mission and we can't afford to do that. The right answer is to begin off-loading some of the 747's tasks to small, cheaper, quicker-to-build airplanes and to begin testing on the small planes the technology that can ultimately reduce cost on the larger ones. This also reduces mission risk by having smaller systems available that can be put in place quickly, should there be any problems with the 747 fleet. Eventually, we arrive at a balanced fleet that can do far more with a combination of newer and better technology in small systems and a smaller number of large, expensive systems that are still cheaper than previously because they have less to do, can use more modern technology to get it done, and have back-ups available should problems arise. Returning to our space example, low-cost small spacecraft can't deliver astronauts to the Moon or the Space Station, but they can bring water, peanut butter sandwiches, medicine, and a left-handed torque wrench when it's needed. If we can do some entire missions, and parts of other missions, far quicker and at much lower cost, we can expand what we can do in space, be more responsive to the needs of the end user, and do more with fewer resources. Specifically, we can reasonably expect cost savings (or equivalently, increases in what can be done) of billions of dollars by reducing launch costs, reducing the "time to market," offloading some onto more economical, smaller systems, bringing newer technology to bear, and using small systems to test the processes and technologies needed to reduce cost on more traditional spacecraft [Wertz, 2010].

A key issue for space systems is that launch is a driver of mission cost. It is typically not the most expensive element for large missions, which has led some observers to conclude that launch cost is relatively unimportant in terms of reducing overall mission cost. But that isn't the case. It doesn't make sense to launch a $5 million satellite on a $25 million launch vehicle. So long as a dedicated launch costs $25 million or more, the mission (payload, spacecraft bus, launch, and initial operations) will typically cost $75 million to $100 million or more, because it doesn't make economic sense for them to be much cheaper.

Similarly, mission responsiveness is driven by both the cost and responsiveness of launch systems. The Soviets/Russians have had systems in inventory and launch-on-demand (within hours) for over 30 years. The Soviets launched 29 payloads in 69 days to support and monitor the 1982 Falklands War that was sufficiently far south that there was relatively little coverage from traditional observation satellites [Cooper, 1992]. But for most US missions, $50 million to $100 million satellites are too expensive to build to inventory, and this is typically not done.

Weather, particularly winds aloft, is one of the most common causes of launch delays. This further increases cost by requiring a "standing army" of launch operations personnel. Again, the Soviets/Russians have had all weather launch capability for decades, such that this is not a major concern for them.

From a purely economic perspective, large satellites need nearly all of the money spent up front before there is a return on investment. This is both economically adverse and tends to enforce the approach of being very risk-averse, because there is no way to fix a problem once the system has been launched. For SmallSat systems, money is spent and results are generated more-or-less continuously over the life of the program as systems are continually being built and launched. This is both cheaper (money spent later costs less, due to the cost of money) and more amenable to a build, test, fly approach in which corrections can be introduced during the program, and new technology can be rapidly inserted.

Satellites with high utility costing $5 million to $10 million can have a strong, positive, near-term impact on the ability of space to support the modern warfighter, respond to natural or man-made disasters, recover from on-orbit failures, or test space systems rapidly and at low cost. But **operational** systems in this price range cannot come about unless there is a dedicated, responsive launch at comparable cost. While launch itself is not the highest cost element for most missions, the high cost and long timelines of current launch systems prevent us from creating low-cost, truly responsive missions.

Reducing Mission Risk and Fragility. As defined in Sec. 24.3, mission risk is the product of reliability and the cost of failure. Lower cost missions are inherently lower risk because fewer resources are at stake. Most current large space systems are inherently fragile because any failure can lead to major consequences and the loss of a single spacecraft can lead to a lost capability that can be replaced only after many years and at very high cost. Alternatively, if one of many small spacecraft is destroyed by a collision with invisible debris, for example, there is a loss of capability, but that loss is mitigated by having multiple spacecraft on orbit and an inventory of others that can be launched in a short time. Given the rapid evolution of small spacecraft, it is likely that the replacement on the ground waiting to be launched will have a greater capability than the on-orbit asset that was lost. In a fleet of 20 small spacecraft, we can plan for launch and on-orbit failures and take that into account in our program by building 10% or 20% more spacecraft than are needed. (This is done in virtually all constellations.) If most or all of a capability is in a small number of large spacecraft, the risk is higher and far more difficult to mitigate.

As pointed out with specific examples in the introduction to Sec. 13.1 above, there is no failure-free system. **Reduced risk and high mission assurance come instead from system robustness—i.e., having sufficient back-ups and options available to ensure mission success even when failures do occur.** In addition, current space systems are designed primarily to have a high parts reliability, but parts failures aren't typically why missions fail. By driving up cost and schedule, we are making the system less reliable from the point of view of the end user. As discussed in Sec. 13.4, many current space systems are effectively being designed to have a "reliability of zero," because they are not launched on time—not a good scenario from the perspective of the end user.

There is often a presumption that what you "buy" with high cost space systems is a greater level of reliability, at least for the spacecraft itself, but even this may not be true. There is no statistical data available on the reliability of low-cost vs. high-cost spacecraft. However, anecdotal evidence and historical experience suggest that small, low-cost spacecraft may be equally as reliable as larger, more complex spacecraft.

For SmallSats, we would expect a decrease in reliability because of lower quality parts, fewer, lower-cost procedures, lack of redundancy, and more willingness to accept risk. On the other hand, we would expect an increase in reliability for SmallSats vs. traditional larger satellites because there are fewer parts, simpler designs, larger margins, less emphasis on an optimized design, and a higher level of personal responsibility [Wertz, 2010]. For example, on a SmallSat there may be only one person responsible for building the power system. If that system fails, everyone knows whose fault it was. This means that the power system engineer who designed, built, integrated, and tested the entire power system for that spacecraft has as their very first goal that the power system will not fail. They feel that personal responsibility and work very hard to ensure success.

The potential for better operational orbits that aren't feasible with more traditional missions and which can also mitigate the growing problem with orbital debris. Orbit design for low-cost, responsive, smallsat is fundamentally different than that of more traditional missions because the basic mission objectives are different. Specifically, traditional space missions are very expensive and have a long on-orbit lifetime. Therefore, the orbits are designed for worldwide coverage because it isn't known at the time of launch what area of the world may need coverage, and the area needing coverage is likely to change many times over the life of the mission. Consequently, traditional mission orbit objectives are:

- Global coverage
- Long life
- Use whatever launch vehicle it takes to get there

In contrast, operational SmallSat mission orbit objectives are significantly different:

- Coverage of a specific region, event, or set of events—i.e., localized in both time and space
- Willing to give up long life to get good performance at low cost
- Want low-cost launch commensurate with the satellite cost
- Want to minimize creation and accumulation of orbital debris

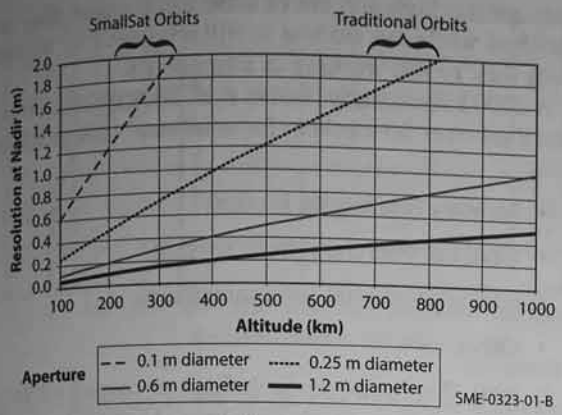


Fig. 13-1. Resolution vs. Altitude for Low Altitude SmallSat Orbits and Higher Altitude Orbits Used for More Traditional Systems [Wertz, 2010].

Most operational SmallSat mission orbits fall into one of 2 broad categories—observations and communications. For SmallSat observation missions, a low altitude, prograde *Repeat Coverage Orbit* (Sec. 10.4.2.6) provides better coverage and better resolution with lower collision probability and rapid debris decay such that orbital debris cannot accumulate. For SmallSat communications missions, circular orbits at 15,000 to 20,000 km altitude and 0 or 90 deg inclination provides both continuous coverage and one-hop communications to CONUS. These orbits provide better coverage, lower vulnerability to attack, and fewer collision opportunities than highly elliptical orbits (HEOs), which have often been considered the best choice for SmallSat communications systems [Wertz, 2007]. See Sec. 10.4.2.7 and Wertz [2005, 2007, 2010] for a discussion of the advantages and disadvantages of these types of orbits.

Because traditional observation satellites are expensive, they are normally placed into high-altitude orbits typically in the range of 700 to 900 km in order to provide both a wide swath and long mission life. In contrast, for SmallSat missions we would like to fly in the 200 to 400 km range with sufficient delta V to allow a 1 to 3 year life and substantial maneuvering. This approach provides comparable resolution using a much smaller telescope, as shown in Fig. 13-1 [Wertz, 2010]. For example, Hubble Space Telescope (2.4 m aperture) initially cost $2.5 billion in FY10$ and, if looking at the Earth, would provide 0.22 m resolution at 800 km altitude. Essentially the same resolution can be achieved with a 1 m instrument at 300 km or an 0.5 m instrument at 150 to 200 km, but at a cost of millions, rather than billions of dollars per mission.

While low altitude can be important for observation missions, it is even more important for missions with active sensors, such as a wind lidar or space-based radar. Resolution falls off as the square of the distance, but the power requirement for an active sensor falls as the distance to the 4th power. (See Chap. 15.) This means that going from 600 km to 300 km means a power reduction by a factor 16 and going to 200 km reduces the power by a factor 81. Flying at low altitudes can have a dramatic impact on cost, although it will also reduce coverage.

Low altitude works well for systems intended to provide focused coverage over a defined region, rather than needing to provide coverage of the entire world. In addition, *agile spacecraft,* i.e., ones with substantial on-board delta V, can adjust the altitude and coverage parameters as needed to optimize resolution, coverage, lifetime, and maneuvering to confuse the enemy. This high level of agility is more easily achieved on a lightweight SmallSat than with a large spacecraft. NanoEye, for example, has a 20 kg dry mass with 50 kg of propellant and is intended to provide approximately 2 km/sec of delta V, a much higher level of agility than would be possible with a traditional large and heavy system. [Van Allen et al., 2011].

It has been suggested that the orbital debris problem, made more visible by the Iridium 33/Comos 2251 collision in 2009, is an argument for large, multi-purpose space assets, rather than smaller, shorter-lived satellites or constellations. The essence of this argument is that fewer satellites leads to fewer collisions which leads to less debris. However, this is an incorrect argument for several reasons.

First, active satellites are only a very small part of the debris material in low Earth orbit. In LEO there are roughly 900 active satellites vs. approximately 500,000 debris particles larger than 1 cm, which is large enough to destroy virtually any spacecraft, as discussed in Sec. 7.5, and is about 10 times smaller than the smallest particle that can be tracked. Significantly increasing the number of active satellites doesn't change the amount of "dangerous stuff" in orbit by much at all.

Second, SmallSats have a smaller collision cross section and, therefore, are less likely to collide with other satellites. Traditional LEO satellites have a cross section of 20 to 200 m^2, largely due to the deployable solar arrays. On the other hand, SmallSats are typically 0.5 to 5 m^2 in cross section and CubeSats are about 0.01 m^2. Because of the much larger surface area to volume ratio, small satellites often do not have deployable arrays. Because of the small cross section, SmallSats have a lower probability of hitting another satellite.

Most important, operational SmallSats are typically at low altitudes where the debris cloud decays and re-enters over months or a very few years. As shown in Fig. 13-2, single large satellites are typically at high altitudes where the debris cloud decays over 100's or 1000's of years. Of course, SmallSats deployed with large satellites and into the same orbit could begin to create a debris problem, which is a possible argument against using rideshare for SmallSat deployment.

In terms of space debris, altitude is far more important than the number of spacecraft. Even large numbers of SmallSats at low altitude will **not** create a long-term debris hazard, simply because any debris will decay and re-enter in a short time. Low altitude is essentially "self-cleaning," even without an active deorbit system, which adds another failure mode for large, high-altitude spacecraft.

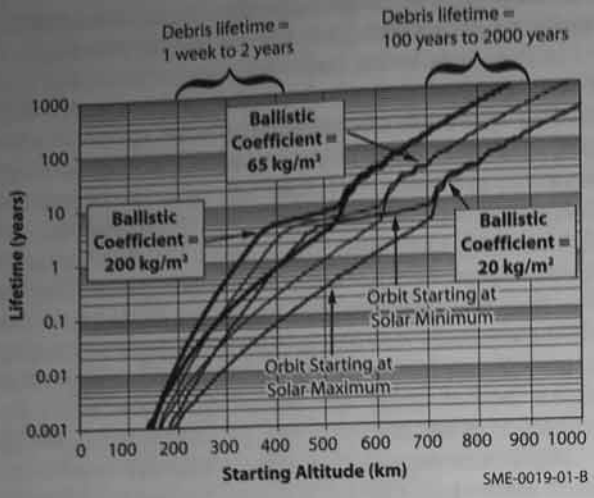


Fig. 13-2. Operational SmallSats Will Typically Inhabit a Regime Where Orbital Debris Cannot Accumulate. (Underlying chart is Fig. 9-22 from Sec. 9.4.4.)

Increased Efficiency in the Use of Scarce Resources. Smallsats can be more responsive to current needs than more traditional missions, simply because they are quicker and lower cost to build and can be built to inventory if needed. However, they are difficult to get funded because they don't have the political clout of larger missions which create large numbers of jobs. On the other hand, once funded they are less likely to be canceled, because of both their lower cost and quicker schedules. In most countries, funding decisions on both new and ongoing programs are made on an annual basis. In addition, in the United States, a major review of space goals, objectives, and programs is likely every 4 or 8 years when administrations change. This means that major programs which may well stretch out over several changes of administration, are subject to several major programmatic reviews in their lifetime, any of which can result in the program getting cancelled or stretched out. (It rarely occurs that a new administration comes in and puts emphasis on speeding up ongoing programs or gives them extra money to finish sooner or do a better job.) This means that a program that can be completed in a few years has a distinct advantage of being less likely to be cancelled. While canceling some programs may be appropriate as perceived needs change over time, it is certainly a waste of both human and economic resources. Smaller programs let us make progress in small steps with much less risk of large backward steps that occur when major programs have schedule overruns, or worse, are cancelled after extensive resources have been spent.

Enabling Missions that Aren't Otherwise Affordable or Feasible. There are many missions that are relevant and important, but which won't occur or will be greatly stretched out in time if they are too expensive. An obvious example is student and training missions. These are extremely important if we are to give younger engineers a sense of what real missions and real mission problems are like. But they simply won't occur if the costs are too high and are of little value unless they are launched while the student is still a student, or at least while they're still working in astronautics.

A 2006 Futron study [Foust, et al. 2008; Wertz, 2010], identified over 30 markets for smallsats* in 4 principal areas:

- Military (the largest market)
- Civil/commercial remote sensing
- Civil/commercial communications
- Other

In total, the Futron study defined an overall addressable market of 39 to 76 satellites/year. At an average total cost of $7.5 million per spacecraft, this represents an overall market of $290 million to $570 million per year. Raising the cost to $8 million/spacecraft, to account for inflation since the time of the study, provides an estimated current market value (payload, spacecraft, and launch, excluding operations) of $312 million to $608 million per year. It is clear from the Futron study that, while specific numbers and missions are uncertain, the smallsat market is very robust and could begin to reinvigorate the space business enterprise, **if** there is dedicated, responsive launch at a cost commensurate with the spacecraft cost.

In another example, the DoD Space Experiment Review Board (SERB) list currently contains 62 approved and validated payloads or spacecraft needing a ride to orbit. 59 of these provide sufficient information to estimate the Equivalent Mass to the Reference Orbit (EMRO). In order to estimate the launch requirements, payloads on the SERB list were first converted to an estimated spacecraft mass, then the spacecraft mass was converted to an EMRO by multiplying by the Orbit Cost Function (OCF), defined in Sec. 10.3.2. Complete spacecraft on the SERB list were simply multiplied by the OCF. The overall mass distribution is shown in Fig. 13-3.

The results of the analysis are as follows:

- 36 of the entries could be launched by a Very Small Launch Vehicle (VSLV = 100 kg to a due east orbit)
- 14 additional entries could be launched by a Small Launch Vehicle (SLV = 450 kg to a due east orbit).
- 9 entries are too large (or in too energetic an orbit) to use a small launch vehicle and would require a larger Light Launch Vehicle (LLV = 1800 kg to a due east orbit).

Some of the SERB missions could be launched as piggyback payloads with another mission. Most, however,

* For the Futron study, a small satellite was defined as less than 100 to 200 kg total mass with a payload mass fraction greater than 50%; 200 W orbit average power, 500 W peak; 1–2 year mission life; and total spacecraft cost of $5 million to $10 million with low-cost launch options available. It is clear that there are a great many additional smallsat missions that don't meet all of these criteria.

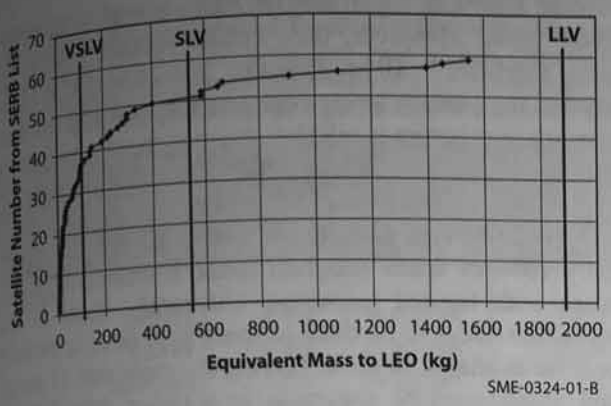


Fig. 13-3. Satellites in the SERB List Sorted by Equivalent Mass to LEO. (Data compiled by Nicola Sarzi-Amade, Microcosm.)

have specific orbit requirements that would require a dedicated launch. This indicates that at any given time, there is a substantial backlog of payloads and systems to be put in orbit from even a single space community, if there were an opportunity to do so.

Making Newer, Better Technology Available to Users. As we have previously discussed, smallsat technology is evolving rapidly and will continue to do so for some time as changes in the electronics marketplace and materials technology begin to make their way into space technology.* The result of this is a continuing influx of new technology and a greater willingness to use this technology because the risk (i.e., the resources and time at stake) is far less. If we are launching 5 to 10 low cost observation satellites per year, then there is relatively little risk associated with introducing a new sensor technology into the satellite stream. If, for any reason, it proves unsuccessful, there are still a number of on-orbit satellites using the prior sensor technology that are still providing data. In addition, a new spacecraft which either reverts to the older technology or finds a way to fix the new technology can be available and launched in a short time. It is a very different process and mindset relative to what occurs for traditional space systems.

An example of this process is the computer technology flown by Surrey Satellite Technology Limited (SSTL). Surrey typically launches a few satellites per year and flies a new computer with nearly every mission flown. At the same time, they also fly the computer flown and validated on the last mission such that they are continuously flying the latest in computer technology but still mitigating risk by flying one that is known to work. This means that Surrey is flying low-cost smallsats with current computer technology vs. large traditional satellites that may well be flying computer technology that is 2 decades or more old. The difference in performance between the new, small low-cost system, and the old, large, very expensive system may not be as great as one would otherwise expect.

Creating Missions that are Far More Responsive to World Events and the Needs of the End User. There is a continuing demand for more responsive solutions to modern issues—in part because of the nearly instant global communications that make problems apparent to the world virtually as soon as they happen. Traditional space systems can respond only if they are already on orbit, have on board the relevant instrumentation, and are in an orbit that provides the appropriate coverage. In contrast, the Chinese are developing launch-on-demand systems [Office of the Secretary of Defense, 2007] and the Soviets/Russians have had systems in inventory and launch-on-demand (within hours) for over 30 years [Cooper, 1992].

At the present time, if critical US military assets are destroyed by a system failure, collision with invisible debris, or enemy attack, they cannot be replaced in a time that would be applicable to any current needs. Because of their irreplaceable character and high cost, they also become inviting targets for a potential enemy attack. However, if some subset of their capability can be replaced within hours or days by inexpensive smallsats, they become a less valuable target and, because of low cost and rapid replacement, the back-up smallsats are also uninviting targets.

Perhaps most important, low-cost smallsats and launch-on-demand allow us the capacity to respond to world events and either natural or man-made disasters in time to help solve problems and potentially save lives and reduce suffering. If we can launch satellites quickly in direct response to a hurricane, tsunami, earthquake, or major terrorist attack, we might be able to find and rescue people who are stranded, washed out to sea, cut off by rubble, or in the path of a moving forest fire or rising flood. For these systems, our goal is not to evaluate what happened, but to find ways to help save lives, protect property, and reduce hardship while there is still a chance to help.

Creating an Economically Sustainable Business Model. One of the problems that arises with the traditional approach of high cost, long lived systems is that they don't provide an economic basis for a sustained industrial base. The Space Shuttle was intended to meet the needs for human spaceflight and large payloads for the indefinite future. But, once the Shuttle was designed, built, and delivered, there was no way to maintain the skill base required to design and build it. There was no continuous production process that would provide for changes, upgrades, maintenance, and incorporation of new technology. The same is true of other large systems that are intended to live for a decade or more on orbit and for which only a relatively small number are needed. Similarly, the cancellation or completion of a single large program can have a major impact on even large aerospace companies.

* Moore's Law suggests that the speed of modern electronics doubles every 2 years. This means that a program using 10 year old computer technology is operating roughly 32 times slower than one using more modern technology. A program that takes 10 years to design and build and then lives on-orbit for 15 years is operating at roughly 1/4000th (0.00025) of the speed and throughput currently available.

In contrast, a mix of spacecraft sizes provides a much more robust economic basis to build on. Building a larger number of satellites with shorter lifetimes not only reduces the overall cost, but leads to a more sustainable manufacturing process as satellites can be built more-or-less continuously with some speed up and slow down to meet the changing demand for any particular type of satellite. New technology can be incorporated as it becomes available, the people who understand how the satellite is built are still around, and the component manufacturers can produce a small, but continuing stream of components, rather than one set every 5 or 10 years, which is often the case currently.

Unfortunately, the scientists, engineers, and technicians who form the underlying space systems knowledge base are not commodities that can be easily "stored" for the next time they are needed. SmallSats and SmallSat launches can provide a continuous flow of business that can allow SmallSat manufacturers to continue to grow over an extended period of time.

13.1.2 The Need for Mission Diversity

Although there are significant cost advantages, SmallSats should not be viewed as a replacement for traditional, larger satellite systems because of their inherent limitations. Of these, perhaps the most fundamental is the limitation due to physical scale. Nearly all observation spacecraft use diffraction-limited optics, such that the only way to improve angular resolution is to make them larger. (Linear resolution can be improved by flying lower as discussed in Sec. 13.1.1, and it is possible that better resolution can be achieved by using multiple small satellites in formation, although this creates a challenging control problem.) Similarly, high data rates require a large power-aperture, and high power implies the need for large solar arrays and a large satellite. Physically large satellites and launch systems will continue to be required for human spaceflight. Of course, this doesn't mean that small responsive satellites cannot play an important role in human spaceflight or other large satellite missions by offloading tasks that could otherwise increase mission cost.

In addition, because SmallSats are inherently near-term and low cost, it doesn't make sense to spend billions in R&D for their development. If a particular problem requires billions of dollars in research investment, then it is likely that it will be implemented in a more traditional, unique, large spacecraft system. SmallSats built to inventory with launch-on-demand have significant economic and technological advantages over traditional large satellites for some applications and can help larger systems significantly reduce cost by offloading tasks, providing a level of back-up in case of system failures, and serving as a testbed to validate both technologies and processes needed to significantly reduce cost of the larger systems.

It is clear that smallsat systems can be lower cost, more flexible, and more responsive to both changing technology and changing user needs than traditional satellites. They can significantly reduce overall space program costs. However, they cannot replace traditional large satellites in all applications. It is clear that what is needed for a much lower cost, much robust and responsive space program is *mission diversity* in which we have a mix of larger traditional satellite programs (but with newer, lower-cost technology and a reduced work load) and lower cost, newer smallsats. This not only dramatically reduces space program costs, it allows us to do more with limited resources. Ultimately, there is far more that needs to be done in space than there is money or time available to do it. Having the right mix of large and small satellite programs is a major step toward becoming both more efficient and more capable.

13.2 It's Possible, But It Isn't Easy

It has been clear for some time that is very important to the future of the space program to substantially reduce space mission cost and schedule and that many competent people have worked hard to reduce both cost and schedule and constrain cost and schedule overruns. It has proven very difficult to do, which makes it clear from experience that it isn't an easy task, but is it possible?

Essentially all space programs are run so as to try to create the lowest cost system consistent with the mission objectives, requirements, and constraints. As mentioned at the opening of the chapter, no one sets out to create a program that costs too much and takes too long. In addition, it should be clear that if we build the same system that we built last time and build it the same way, it will cost the same to within minor variations. This means that genuinely reducing cost is not a matter of buying the same spacecraft that we bought last time for a lot less money. Dramatically reducing cost requires that the system be fundamentally "different" in some aspects. So the real question is not "can we buy the same spacecraft for a lot less money?" (for which the answer is "probably not"), but, rather "can we achieve these broad mission objectives for a lot less money than we would expect to pay for a traditional space mission?"

There are three different lines of evidence that suggest that dramatic cost reduction is indeed possible. First, because the USCM8 cost model presented in Chap. 11 and previously in SMAD continually overestimated the cost of small spacecraft, the Aerospace Corp. created the Small Satellite Cost Model (SSCM), also presented in Chap. 11, specifically for the purpose of more correctly estimating smallsat costs. As discussed in Sec. 2.1.8 (Fig. 2-15), the behavior of the SSCM and the introduction of CubeSats (Sec. 25.4) suggests that there may be a need for an even newer Very Small Satellite Cost Model that continues trying to model the trend of ever lower cost, smaller, more competent systems. Second, there is direct evidence from the cost data presented in the *Reducing Space Mission Cost* (RSMC) book [Wertz and Larson, 1996]. Table 13-1 shows the actual cost of each of the 10 case study missions as a percentage of the projected cost from the USCM8/SMAD cost model,

Table 13-1. Actual Cost for the 10 Case Study Missions in RSMC as a Percentage of the Projected Cost from the USCM8/SMAD Cost Model for Traditional Space Systems as Given in SMAD III [Wertz and Larson, 1999]. The actual dollar cost, inflated to $FY10, is shown in the right-most column. Data from RSMC [Wertz and Larson, 1996], which also provides an explanation of the cost data and extended discussion of each of the missions with the actual data cost data and explanations of the data. The other columns show the actual cost as a percentage of the model's projected cost.

	Bus	Payload	Space Seg.	Launch	Grnd Seg	Ops + Main	Total Prog	Total Program Cost (FY10$M)
AMSAT								
AO-13	2.8%	in spcraft	1.8%	0.4%	N/A	N/A	1.0%	1.87M
AO-16	0.9%	in spcraft	0.5%	0.2%	N/A	N/A	0.3%	0.30M
Average	1.8%	in spcraft	1.1%	0.3%	N/A	N/A	0.7%	
Other LEO								
Ørsted	31.8%	27.1%	29.8%	12.1%	4.2%	33.9%	19.6%	27.8M
Freja	28.2%	18.5%	24.1%	7.2%	0.7%	4.1%	9.0%	37.0M
SAMPEX	82.2%	41.1%	64.7%	93.9%	14.2%	109.2%	51.1%	109.6M
HETE	16.3%	59.4%	35.1%	42.2%	1.4%	in payload	14.4%	45.5M
RADCAL	12.3%	in spcraft	8.2%	73.3%	in spcraft	N/A	14.7%	25.1M
ORBCOMM	42.7%	in spcraft	24.9%	18.1%	0.2%	6.1%	4.2%	23.7M
PoSAT-1	4.4%	1.9%	3.3%	1.6%	0.7%	4.2%	2.2%	3.2M
Average	31.1%	28.9%	26.9%	35.5%	3.6%	31.5%	16.5%	
Interplanetary								
Clementine	88.5%	24.0%	72.2%	32.1%	1.9%	54.1%	36.3%	128.4M
Pluto Express	19.8%	18.0%	19.7%	7.6%	in spcraft	14.0%	15.8%	429.0M
Average	54.2%	21.0%	46.0%	19.9%	1.9%	34.0%	26.1%	
Avg - All Exc. AO	**36.3%**	**27.1%**	**31.3%**	**32.0%**	**3.3%**	**32.2%**	**18.6%**	
Avg - All Missions	30.0%	27.1%	25.8%	26.2%	3.3%	32.2%	15.3%	

adjusted for inflation. From this table, it is clear that there are large cost reductions for all of the elements of the space mission—i.e., the spacecraft bus, payload, launch, ground segment, and operations. This must be true in order to have large cost reductions for the entire program. If it were not the case, the elements where cost savings weren't possible would become the dominant cost and keep the full mission cost from coming down by a significant factor. Finally, there is the anecdotal evidence from individual missions other than those in RSMC. These include missions such as BremSat [Wertz and Dawson, 1996, 1997], TacSat-3 and Low-Cost Wind Lidar [Wertz, 2010], NanoEye [Van Allen, et al. 2010], NASA's Small Explorer program, and the various SmallSat and CubeSat missions described in Sec. 25.3 and 25.4.

Because we will likely never build both a low-cost smallsat and a traditional satellite with the identical mission objectives, there is no way to prove conclusively that cost reduction is real. Perhaps the most that we can conclude is that the "preponderance of the evidence" suggests that it's possible to dramatically reduce cost (i.e., by a factor of 2 to 10 or more) without significantly impacting reliability. However, it does require changing strongly the way we do business in space.

Nearly everyone would agree that it's desirable to reduce cost and increase responsiveness by a large fraction, and it appears to be possible, so why is it so hard? For one, inertia and culture can be two of the strongest forces in any organization. Dramatically reducing cost and schedule requires not just a different type of spacecraft in some way, but it also requires "doing business differently" to allow that spacecraft to be built quickly and at low cost. However doing business differently is hard. The culture of how we do business in space has come about over decades in response to specific successes and, more often, specific problems that have occurred in prior programs. For example, we may do a specific set of tests because in past experience failures occurred when those tests weren't done. The space culture is a result, in part, of our space experience.

We talked in the opening paragraph of the book about NASA Administrator Dan Goldin's attempt to change the culture by instituting a policy of "faster, better, cheaper." Surely, if anyone can cause the culture to change, the Administrator of NASA would seem to have a good chance. But, in the end, the old culture won. ("Faster, better, cheaper—pick any two.") Culture can be among the hardest forces to overcome—it will take great delight in pointing out every failure of a low-cost program and will point out that a low-cost program that fails is not as valuable as a high-cost one that works successfully on-orbit for many years. No matter how much one points out that occasional failures can be acceptable and taken into account if the program is sufficiently low cost, there is still an underlying belief that "failure is not an option," and that every program must be built such that

the failure probability is driven to as near zero as possible, irrespective of the cost consequences. This is reinforced by both the funding hierarchy and Congress that notice, and then quickly forget, major cost reductions in programs, but remember system failures for a much longer time when deciding whether to fund the next phase of a program or a new program intended to be done more quickly and at much lower cost. It is further exacerbated by the response of the media and the public, i.e., all of us who ultimately pay for the public space program. Failures, especially very expensive ones that involve spectacular explosions, become front page photos on major newspapers and headlines on the evening news. On the other hand, it is typically quite a long time between front-page stories in the *Washington Post* about small, low-cost, high reliability spacecraft built by any small company. Small successes are not as newsworthy as expensive failures.

The cultural problem is reinforced by ideas that most of us share. A cheap product is expected to fall apart quickly, whereas an expensive product is expected to be one of high quality that lasts a long time. Of course, in reality that may depend on the application. A Ferrari is much more expensive than a Ford and is typically thought of as a "better" car. At winning races, the Ferrari is likely to be the better car. Every extra ounce is removed to provide higher performance. But, if our objective is to take the kids safely to school every morning and make trips to the grocery store with high reliability, it's possible that the Ford can do that job better, with higher reliability. Cost is not necessarily a good measure of long-term reliability and what system is "better" depends, at least in part, on your objectives.

All of this implies that to make major changes in the cost and responsiveness of space systems will require that we change, at least in part, both what we buy and how we buy it. Low-cost, quick and responsive programs are not consistent with the traditional space culture and will require a lot of work to make them happen. Unless there is a well-formulated "plan of attack," and a process for implementing it with management approval and support, the inertia and culture within the space community will largely prevent it from happening. Reversing the space spiral of Fig. 1-2 is possible, but hard.

In my opinion, the first lesson in creating a new way of doing business in space is that to successfully reduce cost, we have to talk about real cost and make cost data known. One way to ensure competitive pricing, is to tell the community what the government paid for something last time. If we paid $1 million each for reaction wheels of a specific size and make that data known, the next time we ask for reaction wheels of that size it is unlikely that we will get any bids over $1 million. While this seems obvious, there is a strong tendency to not want to do it. Cost and price data is sensitive and much of it is proprietary. In addition, cost data is hard to compare across programs and components. For example, our $1 million price may or may not include testing, spare parts, shipping, and a special, all-composite, light-weight housing. It may or may not include non-recurring development cost. In addition, cost data is frequently hidden to make costs appear lower. For example, it is virtually unknown what a Space Shuttle launch actually costs. Revealing costs may make it harder to get a program funded. Nonetheless,

Reducing space mission cost is hard if we know what the costs are, and virtually impossible if we don't.

As discussed in Sec. 13.1.1, small spacecraft have a disproportionate advantage for reducing cost. Unlike many other areas, small organizations also appear to have an advantage. In automobile production, low cost cars are built by large companies and the expensive, unique cars are typically built by smaller organizations. In space, the reverse is the case. The most expensive spacecraft are built by large companies and the small, low-cost spacecraft are typically built by small companies. Why is this? The major advantages that large organizations traditionally use to drive down cost—such as large factories with the associated economies of scale, large research and support groups, and extensive (and expensive) facilities and operations—are typically not applicable to reducing spacecraft costs and may, in some cases, be handicaps. It would be challenging for an organization that is used to multi-billion dollar contracts to survive on a modest number of multi-million dollar contracts. On the other hand, small organizations could thrive and prosper on these and, in addition, have a number of characteristics that are advantageous to reducing cost. They tend to be innovative in their approach. Rapid decision-making is relatively easy. They live in a world where cost is always critical and they don't have to support a large and hungry infrastructure.

This dichotomy also suggests one of the biggest hurdles for government organizations that want to drive down cost. Governments are used to dealing with and depending on large organizations and spreading the decision-making process over years or decades and a great many organizations and people. But small organizations that are capable of dramatically driving down cost, simply can't survive that process. It can often take the government nearly as long to make a decision on spending a few million dollars as it does to spend a few hundred million dollars. If a small satellite program is ongoing and the government decides to take a funding break for 6 months or a year, it not only drives up the cost of the program, it's possible that nobody will be there at the small company to answer the phone when someone calls back to get things going again. Creating multiple small spacecraft creates a much more sustainable economic model for businesses, but only if the government supports the start of that process with moderately level or growing funding. (Once there is an ongoing small production line, it's likely spacecraft would be bought by multiple organizations and the ups and downs of the government procurement process would likely be much less of a problem.)

Table 13-1◀, Fig. 13-3◀, Eq. 13-0

13.3 Counterproductive Approaches to Reducing Cost

One of the key issues in reducing mission cost is being creative and inventive in terms of new technology, new processes, new ways of doing business, and an understanding of how these will work in practice. This leads to the fundamental question of what it is that creates creativity and inventiveness—money or ideas? Where do new inventions, new ideas, and new ways of applying approaches and technology come from? Large organizations, such as those in the pharmaceutical industry and often the government, would argue that it comes from spending large amounts of money on R&D. Others would argue that it comes largely from small companies and individuals who are creative and can think differently about a problem—such as the movie dramatizing the invention of the intermittent windshield wiper blade for cars by an individual inventor working in his garage. Historically, it seems clear that both processes can work and, to an extent, we need an element of both. It is likely that no amount of research will yield truly breakthrough results without someone in the organization with a creative streak that can look at a problem from a different perspective. Similarly, small organizations can come up with great ideas and inventions, but typically need some level of funding to get them started and nourish ideas until they can blossom into real and useful inventions. This is a part of the reasoning behind the creation of the Small Business Innovative Research (SBIR) program that has helped many small businesses develop new ideas in space, astronautics, and a great many other fields.

All of the above suggests that if we really want to reduce cost, we should ensure that those organizations that have been successful at doing so or are likely to be able to help, indeed have the opportunity to continue. Therefore,

> **A litmus test for any new approach, regulation, or policy intended to reduce space mission cost or schedule should be: Would it encourage or discourage participation by small businesses, individuals, or other groups that have been successful in the past?**

If a proposed approach tends to discourage any participation by those groups or individuals who really understand how to reduce cost and schedule, have ideas on how to do it, or who have done it in the past, then it is unlikely to be successful and is more likely to drive up costs and schedule by inserting unproductive rules, regulations, and processes into the way we do business. If the heads of the major government space agencies convene a meeting with the CEOs of the 3 or 4 major aerospace prime contractors, it is unlikely to result in creating new ways of doing business that are faster, better, cheaper, and less comfortable for the major aerospace companies.

An example of a potentially counterproductive approach is cost-sharing (or risk-sharing) as a means of reducing cost. Creative, low-cost organizations rarely have excess internal funds to carry additional burden or risk and this implies that they likely cannot share in program cost. Organizations that are willing to cost share typically plan to recover that cost plus any prior cost-sharing activity not ultimately funded plus additional return due to cost-sharing risk. This suggests that demanding cost-sharing, particularly in the up-front development, on something for which the principal user is likely to be the government will probably make the ultimate item more expensive rather than less expensive and, in addition, damage those organizations who may have good ideas or worked on low cost approaches. On the other hand, cost-sharing may be appropriate and helpful for space technology if the end product involves technology that has many applications in the commercial world, such as batteries.

Using this approach implies that we can construct a list of potentially counterproductive "cost-cutting" approaches that have a high potential of increasing, rather than decreasing, cost. Some of these are shown in Table 13-2. Although all of these have the potential to reduce costs in some respects, it is more likely that they will increase costs by tending to eliminate or marginalize those individuals or organizations that are most likely to be helpful to the problem.

There is a good analogy here with software development. Software is extremely expensive and many attempts have been made to reduce its cost. However, lowering software development cost by reducing the up-front planning and design has almost always proven counterproductive. Beginning to write code in the first day or month of a software project may be psychologically satisfying, but in the end it is likely to substantially increase costs for development and maintenance as people try to fix software that wasn't well designed in the first place. (See Secs. 3.1 and 20.2.)

In most programs, additional resources should also be allocated to operations planning. This should be done in two distinct stages. First, operations planning should be a major part of the mission engineering process. How are we going to realistically operate this mission to achieve our end objectives? Not taking operational issues into account can significantly increase the life-cycle cost. Second, detailed planning and ground operations systems development needs to begin well in advance of the launch of the first satellite. (See Sec. 29.1.) In many programs, there is a strong tendency to put off problems. Costs are constrained at essentially all stages and it is frequently seen as more productive to simply proceed and resolve problems at a later stage. Ultimately, these become operations problems that tend to dramatically drive up the cost of mission operations.

Another area which typically receives inadequate support and funding is exploring options. Too often, in the name of reducing both cost and schedule, we curtail looking at options as "serious" engineering gets underway. However, options for reducing mission cost should be explored with substantial care during mission definition and continued throughout the mission life. The prag-

Table 13-2. Potentially Counterproductive Approaches to Reducing Space Mission Cost. Each of these approaches has the potential to significantly increase cost rather than reduce it.

Approach	Intended Benefit	Potential Negative Impact
Consolidated, Centralized Acquisition for Multiple Programs	Provides greater accountability; gives appearance of reducing waste	Restricts competition; likely to force out small organizations and innovative approaches; likely to institutionalize the high cost approaches
Contractor Sharing of Development Costs	Reduces cost to the government if the major customer base is the private sector (e.g., personal computer development)	Forces out the small player; acquisition will be driven by investment economics—will require very large ROI, since the government is a high-risk customer
Contractor Cost Sharing in Up-Front Studies	Reduces study cost	Forces out the small, innovative contractor; contractor costs will be recovered in higher indirect rates
Cost Guarantee on R&D Programs	Limits government cost commitment; good for achieving accountability, but **not** cost reduction	Forces out the small contractor and doesn't permit taking risks that could dramatically reduce cost
Doing Work In House	Eliminates subcontracting cost; can be effective if the group has experience with low-cost efficient production; provides good training for government engineers	May drive up costs due to lack of efficiency and knowledge; may be largely an excuse to maintain a large infrastructure; on-the-job training can be expensive
Reducing the Amount of Up-Front Systems Engineering	Shortens program schedule and avoids over-engineering a strawman design	Ignorance is rarely of value in reducing cost or improving performance; principal problem is not adequately addressing alternatives
Delaying Operations Planning	Postpones up-front costs when budgets are small and tight	Can significantly increase operations cost and risk and drive up life-cycle cost
Not Selling the Program to the Funding Community and End Users	Reduces both money and effort devoted to non-engineering activities	Program may get canceled as people and priorities change. A canceled program is rarely cost effective
Not Spending on Knowledge Capture at Mission End	Cuts off expenditures when all (or more) of the money has been spent	A very small investment can be extremely valuable in reducing cost for future programs. Not doing that loses this opportunity

matic approach is clear. Given the current state-of-the-program, is it cheaper to proceed with the current plan or choose a lower-cost option, taking into account the time and cost of changing to a new baseline? In initial mission design, the cost of changing is very low. As we go further downstream in a program, re-engineering becomes more and more expensive. The cost of looking at options, however, does not change significantly and, if costs can be reduced, changes may be worthwhile, even late in the program development cycle. Even if the cost of implementing changes is judged too expensive or too time-consuming, it may still be appropriate to evaluate options such that follow-on programs can take advantage of the knowledge gained.

What we have called by the fancy title of *mission utility analysis* (Sec. 5.4) is really a matter of selling the program. This is an extremely important process, which does not necessarily reduce cost directly, but does so indirectly in a number of ways. First, it keeps the effort focused on the mission objectives. Is the program achieving what the customer intended and what the end user needs? Second, keeping the program firmly sold helps reduce the start-stop syndrome of many programs, which is a major source of cost and schedule overruns. Third, keeping the program sold keeps the program alive so that it can achieve the original objectives, assuming that they are still worth doing. Programs can be cut or eliminated at any stage, and cancelled programs are rarely cost effective.

In the government arena, selling a program can have a negative connotation of convincing the government or the public to do something that isn't in their best interest. However, analyzing mission utility is really about establishing, as quantitatively as possible, the benefits of the program and how it can meet the objectives of whoever is providing the funding. This is a positive and critical step for any new mission or at any time mission funding decisions are being reviewed.

Finally, we should provide adequate funding at the end of mission life. At this time, basically all of the money has been spent and there is no further potential for cost savings or schedule reduction. Still, spending some money on assessing and recording "lessons learned," while they're still fresh, can cut costs and improve performance of subsequent programs.

13.4 Cost vs. Reliability—Focusing on the Mission Objectives

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The space industry's understanding of, and approach to, reliability can be one of the most important drivers of a program's cost and schedule. This section discusses what reliability is and what it is not, while highlighting common misunderstandings that often mislead designers

and managers. Misunderstanding and misapplication of reliability can lead a program to set aside real mission objectives and effectively design and manage programs to a "reliability of zero." For instance, if extreme reliability requirements result in schedule delays and cost overruns leading to program cancellation, the achieved reliability is zero.

The reliability analysis presented in Chap. 24 is a quantitative analysis of the likelihood of part failures; for example, if a specific relay fails, then a given capability is lost. This section presents a complementary mission-oriented view of reliability with an emphasis on system availability to users. We examine historical space systems and on-orbit data to confirm key points. Ultimately, this section strives to advance the industry-wide understanding that is necessary to better achieve reliable space systems, available to users when needed.

The concept of reliability is heavily influenced by perspective. Personnel in space-system program offices and associated system developers almost always consider reliability to be a set of requirements. Those requirements specify a number that the system's reliability analysis must meet along with related processes and paperwork to assure the product properly made. These reliability requirements are met at design reviews or other development milestones; they will be met (or not) completely independent of cost, schedule, and actual on-orbit performance. On the other hand, from an end-user perspective, anything that makes the system unavailable when it is expected is considered unreliable, regardless of whether the reliability requirements were satisfied in a design or acquisition phase. Similarly, if a program delivers late, then the effective reliability is zero for every day (usually every year) it is late. If a program's cost doubles, those additional funds are unavailable for a second space system. These examples collectively give a sense of the important concept of availability: assuring a capability is available to users when required. Availability is further discussed in Sec. 13.4.4.

Ironically, efforts to achieve high reliability by analysis and processes can prove counterproductive. If a parts-reliability number is used as a mission design requirement, the reliability number can drive program decisions harder than real mission objectives. We have witnessed many programs implement hard (inflexible) reliability requirements, with little consideration for cost or schedule, as if meeting these requirements were a mission objective unto themselves. Although budget overruns and schedule delays are common, often resulting in an effective reliability of zero for many years (or permanently if canceled), cost and schedule are rarely considered to be related to a system's reliability. So it is essential to have a crisp understanding of reliability and to keep one's focus on achieving the mission objectives.

13.4.1 Understanding What Reliability Is and Is Not

True reliability, in our case on-orbit reliability, is a measure of how well a system performs in its operational environment. Regardless of the number of parts, design reviews, quality inspections, or acquisition processes, used to develop the system, in the end the on-orbit system may prove highly reliable or terribly unreliable. According to the Reliability Analysis Center, reliability is "the probability that an item can perform its intended function for a specified interval under stated conditions without failure." Unfortunately, nearly all space-system development only has estimated numbers to deal with and a perceived sense of what the system's reliability will be. In contrast, the aircraft industry has 10,000,000 commercial flights per year enabling them to measure operational reliability and iteratively improve their designs and processes.

What is the Reliability Prediction?

The *reliability prediction* is a calculated likelihood of avoiding part failures that induce loss of a spacecraft or mission. In other words, one minus the reliability number equals the probability of mission loss or degradation due to a part failure. See Chap. 24 of this book for a full description and calculations. The reliability analysis pulls upon mean time between failure data for standardized parts and electrical connections. In the US, MIL-HDBK-217 [DoD, 1991] is the guideline for performing such analyses. The analysis considers the failure of electronic parts, connections, and moving mechanical assemblies in the system. **It is especially important to understand the contributors to on-orbit reliability that are and are not considered in this reliability prediction, as shown in Table 13-3.**

Table 13-3. Failure Modes Included, and Not Included, in Reliability Predictions.

Failure Modes Considered in the Reliability Prediction	Failure Modes NOT Considered in the Reliability Prediction
• Electronic part failure	• Design failure
• Solder joint failure	• Software failure
• Connector or pin failure	• Operator error
• Moving mechanical elements such as bearing failure	• Improper build, assembly & workmanship
	• Late launch or launch failure
	• Program cancellation due to cost overruns

Limitations of Reliability Prediction

MIL-HDBK-217 revision F, section 3 [DoD, 1991] provides some of the best explanations of limitations on predicting reliability.

The direct extractions are shown in italics.

3.3 Limitations of Reliability Predictions – This handbook provides a common basis for reliability predictions, based on analysis of the best possible data at the time of issue. It is intended to make reliability prediction as good a tool as possible. However, like any tool, reliability prediction must be used intelligently, with due consideration of its limitations. The first limitation is that the failure rate models are point estimates which are based on available data.

Even when used in similar environments, the differences between system applications can be significant. ***Predicted and achieved reliability have always been closer for ground electronic systems than for avionic systems****, because the environmental stresses vary less from system to system on the ground and hence the field conditions are, in general, closer to the environment under which the data was collected for the prediction model. However, failure rates are also impacted by operational scenarios, operator characteristics, maintenance practices, measurement techniques and differences in definition of failure.* ***Hence, a reliability prediction should never be assumed to represent the expected field reliability as measured by the user … note that none of the applications discussed above requires the predicted reliability to match the field measurement****.*

So the reliability handbook itself points out that it was never intended to be an accurate predictor of operational reliability or the probability of success. On-orbit data strongly confirm this point, as discussed in Sec. 13.4.2. It should be noted that the MIL-HDBK-217F, while still the recognized industry standard, is officially obsolete. It is no longer maintained or updated by the US government; its last update was in 1991. The handbook's analytical process is still sound but much of its parts data, upon which the quantified calculations are based, are badly outdated. The lack of ongoing support for the military handbook, coupled with the rapidly evolving world of electronics, creates a serious problem for developers to obtain statistically valid data sets from relevant environments for quantitative calculations.

One should also be aware of how politically charged a single reliability number can become for a program. For example, a single reliability number is often presented to outside organizations in an attempt to provide a simplified understanding of how likely the system is to work. At a System Requirements and Design Review (SRDR), we witnessed a program office order that the reliability analysis be completed by the Preliminary Design Review (PDR) and at the same time announce that the reliability for the space system including launch will be 90%! While this 90% number may have been useful to create perceived on-orbit reliability for sponsors or others necessary to politically support the program, clearly this pre-determined "analysis" did not add value to the design or final operations. Such political emphasis and simplified understanding can be major obstacles to properly applying reliability analysis and balanced processes to space systems.

Value of Reliability Prediction

MIL-HDBK-217 revision F, section 3 [DoD, 1991] also provides some of the best explanations of the value added by predictions of reliability.

*3.2 The Role of Reliability Prediction—****Reliability prediction provides the quantitative baseline needed to assess progress in reliability engineering****. A prediction made of a proposed design may be used in several ways. Once a design is selected, the reliability prediction may be used as a guide to improvement by showing the highest contributors to failure. If the part stress analysis method is used, it may also reveal other fruitful areas for change (e.g., over stressed parts). The impact of proposed design changes on reliability can be determined only by comparing the reliability predictions of the existing and proposed designs. The ability of the design to maintain an acceptable reliability level under environmental extremes may be assessed through reliability predictions. The predictions may be used to evaluate the need for environmental control systems.*

So, reliability prediction analysis is a quantitative approach for comparing and improving designs. This prediction analysis, along with associated reliability analyses such as the failure modes and effects analysis (FMEA) and parts stress analysis over temperature are excellent for identifying weak links in a design and making improvements. Design improvements include electronics parts substitution, implementation of select redundancy, changes in thermal design requirements, and simplification to reduce complexity.

13.4.2 On-Orbit Reliability—History and Examples

Tafazoli [2008] studied more than 4000 spacecraft launched in the past 25 years and identified 156 on-orbit failures that occurred on 129 different spacecraft from 1980 to 2005. In this study he observed that 41% of all failures happen within 1 year of on-orbit activities. See Table 13-4. He also observed that the data suggest insufficient testing and inadequate modeling of the spacecraft and its environment. The study concludes that the reliability lessons learned are that adequate testing, redundancy, and flexibility are the keys to producing a reliable spacecraft. An example of flexibility would be the ability to upload new software to run new attitude control modes to work around a failed sun sensor. Notice

Table 13-4. Spacecraft Failures Over Time [Tafazoli, 2008].

Failure Distribution Grouped by Years On-Orbit				
0–1	1–3	3–5	5–8	>8
41%	17%	20%	16%	6%

that only 1 of these 3 characteristics, redundancy, is even included in reliability prediction. For students or those new to this field, refer to Chap. 24 to better understand the reliability prediction and what is (and is not) included in its calculation. Although testing and flexibility are not included in reliability predictions, programs must consider them carefully to realize a reliable on-orbit space system.

A second assessment of on-orbit spacecraft failure history is provided in a study by Robertson and Stoneking [2003], which investigated 750 spacecraft from 1990 to 2002 reporting 63 failures. See Fig. 13-4. Like Tafazoli's study, the data also show the majority of failures (48%) occur in the first 10% of design life. Failures then drop to a much lower rate and continue to drop slowly for the remainder of the planned life.

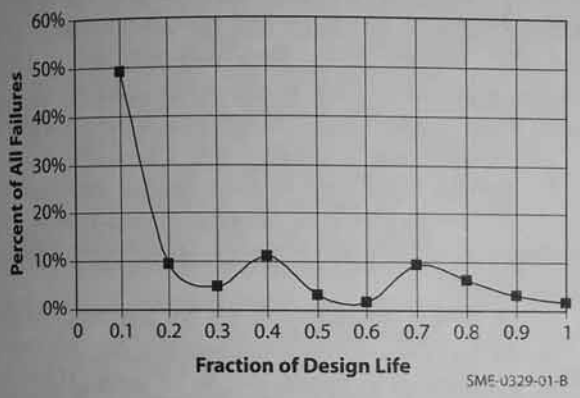


Fig. 13-4. Spacecraft Failure by Fraction of Design Life. (Graphic courtesy of Robertson and Stoneking [2003].)

The commonly used reliability Eq. (13-1) is often called P_S, for Probability of Success, with a value of 1.0 representing no failures, equivalent to 100% reliability. Examples of P_S calculations over time are shown in Table 13-5 in Sec. 13.4.3.

$$P_s = e^{-\lambda t} \neq On\text{-}Orbit\ P_s \qquad (13\text{-}1)$$

P_S = Probability of Success (used interchangeably with Reliability)
λ = Failures In Time (FIT) per billion hours
1 FIT denotes 1 failure per billion hours
t = time in hours (1 billion hours = 114,100 years)

The historical on-orbit data do not match the reliability equation's predictions. This equation fundamentally misses the fact that most failures occur early in the life, especially in the first year (or first 10%) of life as shown by the on-orbit failure data above. In contrast, the reliability equation predicts a near constant rate of failures every year because a constant failure rate (λ) over time is typically used. This inconsistency highlights that failures most frequent in early operations, such as design failures, are not considered in the reliability equation (recall Table 13-3). Tafozili's and Robertson's results indicate that infant mortality holds true for spacecraft. The belief that the preponderance of failures in the first year is driven by design or workmanship faults that expose themselves early in a mission is corroborated in a variety of reports. [Tafazoli, 2008; Robertson and Stoneking, 2003; Sarsfield, 1998; Bloomquist, 1984]

Also once functioning, spacecraft life often far exceeds predictions as we will see later in Sec. 13.4.4. This is largely because, in the absence of a measured failure rate, a constant failure rate is often used in reliability calculations even though on-orbit data show failure rates decrease for spacecraft over time. Sarsfield's [1998] work in this area provides an excellent example of failure rates decreasing with time for the Voyager spacecraft. This is a case where measured on-orbit data do exist for a given system and therefore, the reliability calculations can be used effectively to match the on-orbit failures. When such data exist for and comparable applications and environments, which is sometimes the case when multiple, similar geosynchronous communications satellites are built and launched serially, then reliability analysis can be used effectively to predict the on-orbit failures beyond the first year.

Launch Vehicle Contribution to On-Orbit Reliability of Space Systems

A related and key factor in all space systems is the launch vehicle. While it is not the intent of this section to discuss launch vehicle reliability in depth, we want to highlight that the measured reliability of proven launch vehicles is only 90–95%. Research shows, "Of the 4,378 space launches conducted worldwide between 1957 and 1999, 390 launches failed (the success rate was 91.1%), with an associated loss or significant reduction of service life of 455 satellites (some launches included multiple payloads)" [Chang, 2001]. Regardless of the attempted reliability for all other aspects of the spacecraft, the space-system reliability cannot exceed the launch vehicle's reliability, and a launch failure results in $R = 0$. Users, of course, do not care about the source of mission loss; all the user knows is that the capability they were depending on is not available.

One must consider the impacts of a single launch failure on overall system capability. The bigger the spacecraft, the bigger the impacts to the cost, the schedule, and the users. Consider the implications of a single launch failure on a constellation of 2 large and highly capable spacecraft versus the impact of a single launch failure on a constellation of 4 less capable spacecraft. In this theoretical case, a single launch vehicle loss creates far more loss of availability to the constellation of 2 large satellites. Furthermore, the true reliability of the rocket that the mission cares about is not well known to the mission designer unless that rocket has a long track record of similar missions from which meaningful assessment of reliability can be calculated. The bottom line is that launch vehicle reliability is a huge component of mission availability, which must be recognized by the mission designer. The mission designer should consider things like the constellation design (one or multiple), spacecraft

Table 13-5. Reliability Prediction Analysis Using Eq. (13-1) Compares the Expected Probability of Success as a Function of Reliability Level and Launch Delay.

Case	Predicted Reliability	Delivery Date; Year Started	Probability of Success at End of Year					Comment
			1	2	3	4	5	
1	90%	0	98%	96%	94%	92%	90%	High Reliability, deliver on time
2	70%	0	93%	87%	81%	76%	70%	Medium Reliability, deliver on time
3	17%	0	70%	50%	35%	25%	17%	Low Reliability, deliver on time
4	90%	4	0%	0%	0%	98%	96%	High Reliability, deliver late

sparing, and reasonably pairing the resources put toward the spacecraft with the cost and reliability class of the launch vehicle.

13.4.3 Linking Reliability to Schedule

A reliability prediction and on-orbit reliability are inextricably linked to the variable of time. As discussed earlier, the user's perspective on reliability cares not if the satellite will work eventually in the distant future, but only that it is available when promised.

A reliability requirement is typically defined as a probability of success, P_s, at some number of years after launch. This equates to R_1 at year 1, R_2 at year 2, and so on. Let's examine the probability of success for a single spacecraft from a user's perspective. Our user has been promised a space-based capability to start in year 1 and continue through year 5. Table 13-5 was created using the predicted reliability of Eq. (13-1) for four different bounding cases: High, medium, and low-reliability systems delivered on schedule and a fourth case of a high-reliability system delivered three years late in year 4. Case four shows the problematic situation of the user being delivered a wonderfully reliable spacecraft, but 3 years late. For the full first three years the user has no capability; a reliability of zero has been achieved. This is no different from a launch failure followed by the building of a second spacecraft over the following 3 years. The users are the ultimate arbiter of value, and the authors suspect that most users would prefer cases one, two, or even three over case four. Although not many sponsors would agree to the case-three spacecraft requirement, "The spacecraft shall have a predicted reliability of 17% at the end of 5 years."

13.4.4 Considerations for Small and Large Satellites and Systems

Early in the design, or even proposal process, a mission designer must make decisions on a mission-fulfillment approach using small satellites, large satellites or a mixture of both. This decision has far reaching implications on mission cost, schedule, and availability, which play out much differently for small versus large satellites with advantages and disadvantages for both. Neither approach is universally better; the trade-off must be made for each specific mission.

Considerations for Small Satellites

Small satellites and systems have some inherent benefits toward addressing reliability both mathematically and in real terms. At a macro level, the quantity of small satellites tends to be larger and the cost, smaller. Missions that require more than one satellite typically degrade gracefully; and single small-satellite failures tend to have no appreciable effect on a National scale, as the investment and capability are inherently modest. Small satellites also tend to have shorter development schedules. The shorter time between design and on-orbit operations inherently allow newer, often better technologies to be used.

The predicted reliability levels for generally single-string, small satellites would indicate that they are highly unlikely to last more than a couple of years. However, many single string satellites that survive the 1st year last for 5 to 10 years. NASA's Earth Observation-1 (EO-I) satellite is a single-string system that launched in November 2000 and is still operating in 2010, which is statistically unlikely based upon conventional reliability analysis misused as an on-orbit reliability prediction. The EO-1 spacecraft bus had a predicted reliability, or probability of success (P_s), of 75% for 1 year, which equates to 6% for 10 years. Note, this is only the bus reliability—the predicted payload reliability reduces these numbers much further. The WindSat/Coriolos mission is an example of a moderate (but not high) reliability mission. The WindSat payload is a 22 channel radiometer designed mostly single-string. It was launched in 2003 with a reliability prediction that gave it only a 3% chance to still be working in 2010, 7 years after launch. Yet WindSat is still operating 24/7 in 2011. The Coriolis spacecraft upon which the WindSat payload resides is fully redundant; yet in 8 years it has not experienced a failure on its primary side, thus functioning as if it were single string and lasting 8 years.

An interesting case study of an alternative approach to reliability comes from small-satellite builder, Surrey Satellite Technology LTD (SSTL). Company data on 20 satellites from 1981 to 2003 show an average Mean Time To Failure (MTTF) for their satellites of 6.4 years, yet the average design life was only 2.1 years.* In the last 10 years, with the incorporation of better battery technology, their satellites' MTTF has improved to 7 years. In

* Data courtesy of SSTL.

Table 13-5, Fig. 13-4◀, Eq. 13-1◀

Table 13-6. Advantages of Small and Large Satellites.

Small Satellite Advantages	Large Satellite Advantages
• Larger quantities affordable	• Enable capabilities limited by physics in small satellites, e.g., large aperture for high signal gain
• Single string reliability historically higher than reliability analysis indicates	• Greater design oversight affordable
• Reduced cost and capability impact of a single loss	• More extensive quality assurance affordable
• More rapid design cycles allow implementation of current, typically better technologies	• Can perform technology R&D within program to support future capability growth
	• Redundancy and large margins often easier accommodate within mass and cost budgets

contrast to typical space industry practices which emphasize high reliability parts, SSTL uses commercial parts extensively and avoids quantified reliability analysis. In their words they prefer to "Concentrate efforts on improving reliability, not quantifying it" and do this largely by starting with mission objectives, performing Failure Modes and Effects Analysis (FMEA) to uncover modes which may prevent these objectives from being met, and then focusing resources on preventing these potential failures. Bottom line, in addition to the inherent benefits of small satellites such as larger numbers (for a given cost), shorter schedules, and more current technology, their on-orbit performance often substantially exceeds their reliability-analysis prediction. This is especially true if they were well tested and designed with high margins and flexibility.

Considerations for Large Satellites and Systems

Larger systems allow for enabling capabilities, such as larger aperture and higher power, simply not physically possible on smaller systems due to physics. Large satellites and systems also have some inherent benefits toward addressing reliability. In particular, larger systems and programs can afford and justify more thorough quality assurance, testing (such as parts radiation testing), and independent reviewers. The extensive use of redundancy and large margins can be more affordable within mass budgets and financial budgets on large platforms because they take a relatively small percentage of overall program resources. Large margins and extensive redundancy can provide the confidence necessary for mission users to plan for very long satellite lifetimes. Large communications satellites built, following these principals, are often designed for 15 year missions.

Technology can help by making the job much easier or by fundamentally enabling the mission. However, new technology is often viewed negatively by acquisition programs due to the risk of undiscovered flaws. The Technology Readiness Level (TRL) concept provides a sound process to mature a new technology into a flight qualified item; however, this concept can be misused by programs to eliminate valuable technologies by requiring that "TRL-7 technologies or greater." This type of blanket requirement eliminates new technologies entirely instead of creating a sound plan to develop and qualify the technology on a time frame consistent with the program's schedule. A blanket TRL requirement can also pressure space programs to pretend that needed technologies are at a higher TRL than they are to get a mission go-ahead. This situation causes an artificial feeling of comfort, often resulting in a lack of focus and lack of resources needed early in the program to ensure that development does not affect the schedule. We have watched programs worry extensively over a decision to use a new "high risk" technology while at the same time deleting the test units to save cost and schedule, although the latter created a much higher risk.

Large programs, with extensively designed and calculated reliability numbers need to remain alert and not allow themselves to become fooled by this number. Even the highest reliability commercial communication spacecraft with a P_s >95% for five years and >90% for 10 years regularly have their mission lives substantially reduced. The "Satellites & Launches Trend Down" article by the Teal Group [Caceres, 2004] "estimates that during 2001–2003, an average of about a dozen satellites failed or malfunctioned each year." The rate of a dozen per year was similar from 1991–2000, but there were more satellites on-orbit in the 1990s. The study details over 12 high-reliability geosynchronous spacecraft which experienced failures that reduced their mission life substantially or ended the mission within the first year. These failures covered a broad range of companies and spacecraft types. So programs with high-reliability calculations need to be vigilant in all practices, such as thorough testing and designing in flexibility to assure on-orbit reliability is realized.

Summary of Considerations for Large and Small Satellites

Large satellite programs are well suited to large-aperture, high-power and very long-life missions. Small satellites are well suited for quicker-response missions and shorter development schedules inherently make use of current, usually better technology. Small satellites in high-quantity constellations can be quite robust against individual launch or satellite failures in contrast to small-quantity constellations of large satellites being very sensitive to individual losses. A mix of both small and large space systems can best address the wide range of space missions, users, and reliability needs. (See Table 13-6.)

13.4.5 Considerations for Achieving On-Orbit Availability

It has been shown that reliability analysis is great for improving a design but is fundamentally misapplied as a predictor of spacecraft success on orbit. So let's consider availability and how best to achieve it with limited resources.

There are many availability definitions and analytical methodologies in the literature [NASA, 1998] that bring the aspect of time into the reliability assessment as well as other considerations such as service interruptions and the time to restore service. The simplest representation for **availability** is the ratio of the expected amount of the up time of a system (the time the system is working properly and available to users) divided by the total time on-orbit.

Focusing on the simple up-time versus down-time equation reflects the user's perspective of **'did I get the capability I required at the time I required it?'** which is in fact the true purpose of the mission. This availability-based viewpoint allows the mission designer to consider other mission impacts including schedule delays, launch failure, design problems, and software failures. Clearly, the likelihood of faults such as design failures is incalculable; yet history shows many on-orbit problems that were in fact design problems rather than hardware failures. Optimizing mission availability requires the mission designer to consider many qualitative factors to best fulfill the mission's objectives and satisfy users' needs.

Availability depends upon many factors: meeting mission performance, surviving the environments, avoiding parts failures, proper manufacture such that the spacecraft is built as designed, staying on budget (or at least close enough that the project is not cancelled), meeting the schedule promised to users, proper operations, and robust software.

These factors are addressed by space programs through a variety of practices. Every program must select, prioritize and pay for a combination of these availability practices that best support their mission. These practices include good design, thorough testing, large margins and design/operational flexibility, redundancy, use of mass-produced components (although this is rarely available in the low production world of spacecraft), reliability analysis, manufacturing and process controls (including quality assurance and acquisition processes), preparation of mission simulators, budgeting for operations training, and constellation design. The best availability depends on the program having budget and schedule for every practice selected but the question mission designers must answer is "*how much can the program afford for each practice?*"

We have provided our opinions on the relative contributions of each of these practices to each of the elements of availability in Table 13-7. As this is a matter of judgment, one can argue with any of the individual positions. However, the value of this analysis is in the observation of the broad trends between practice and result. **An important take away from this table is that there are at least 9 important practices programs can use to address failure modes and improve on-orbit reliability and that each practice has resource and schedule impacts as well.** To be used effectively, an honest assessment of the relative merits for each practice must be considered in light of the unique circumstances of each mission.

Let's study each practice and its impact on on-orbit reliability in the table above.

Good Design: Good design is essential to meeting mission performance and surviving the environment. The simplicity of a good design has important availability benefits through schedule control because of the efficiency with which a good design can be built and tested and because of the reduced quantity of time-consuming test failures. As a design element, software derives great reliability benefits from good design. Because nothing is free, the effort required for good design requires time and funding, increasing schedule and cost. Still, the broad trend is that good design pays many reliability dividends, particularly reducing the likelihood of infant mortality, indicating that a program gets much value for the application of resources towards good design.

Thorough Testing: Testing is crucial in finding problems on the ground before they become reliability problems on orbit. Good testing pays large dividends, including exposing problems in meeting thermal and launch environments, and in finding flaws in manufacturing and software. Additionally, when one follows the "test like you fly" adage, flight operation's reliability improves from using and refining the operational ground software and procedures. Testing carries the broad trend of large reliability benefits with low negative impacts. Programs are often tempted to cut testing towards the end of the development cycle because at this point in a program even modest cost and schedule problems tend to have accumulated into larger problems. This often proves to be a false savings as shown by the preponderance of infant mortality failures. Ensuing failure investigations frequently determine these failures could have been caught with the proper ground test. Sometimes one has to ask, "Can we afford not to spend this money?"

Flexibility and Margins: A spacecraft with inherent flexibility in its operation—either through forms of redundancy or a design that can perform core functions multiple ways—allows continued operations in the face of component failures or operations errors. Similarly, large margins allow a spacecraft to continue operations in the face of specific types of failures. Let's examine how the power subsystem design affects the spacecraft:

- Case 1 is a spacecraft with modest power margins and a single array on a single gimbal. A gimbal failure will disable this spacecraft and end its mission.

Table 13-7. Author's Opinions—Effects of Development Practices on True Spacecraft Reliability.

	Failure Modes							
Practice to Address Failure Mode	Meets Mission Performance	Survives Environments —Stress & Thermal	Avoidance of Parts Failure, Radiation, & Wear Out	Built as Designed	Meets Budget	Meets Schedule	Operator Error	Software Failure
Good Design	++ strong benefit	++ strong benefit	+ weak benefit via simplicity and robustness	NA	– moderately higher cost	++ strong benefit via simplicity	+ weak benefit via simplicity	++ strong benefit
Good Testing	++ strong benefit	++ strong benefit	+ catch infant mortality	++ strong benefit	– moderately higher cost	– moderately longer schedule	++ if test like you fly	++ strong benefit
Flexibility & Margins	NA	++ ability to survive after component failures	+ margins enable work around for some part failures	NA	– moderately higher cost	+ weak benefit	++ more likely can recover from op errors	NA
Redundancy	NA	++ ability to survive after component failures	++ strong benefit	NA	–– high cost of parts & complexity	– increased build and test schedule	NA	NA
Use of Mass Production Components if Available	NA	+ part capabilities known in advance	++ measured reliability data exists & learning curve complete	+ weak benefit	++ production efficiency	++ production efficiency or truly off the shelf	+ ops of component often well understood	depends on specific component type
Reliability Analysis	+ circuit improvements	+ parts thermal stress analysis	++ strong benefit	NA	– or – – because of cost of Hi-REL parts if chosen	– or – – because of lead time of Hi-REL parts if chosen	NA	NA
Rigorous QA Controls	NA	NA	++ strong benefit	++ strong benefit	– or – – pending level chosen	– moderately longer schedule	+ QA & config control of ops procedures	++ strong benefit through software QA
Mission Simulation & Training	++ "flying" scenarios before launch, increases on-orbit availability	NA	NA	+ find undesired features	– cost for mission simulator & training	+ often enables parallel testing	++ strong benefit	++ wring out errors & inefficiencies in both ground & flight SW
Constellation Design (multiple S/C) or launch on demand replacement	NA	NA	NA	NA	– or – – cost pending specifics of the mission	++ strong benefit	++ learning curve ops benefits if multiple spacecraft	+ problems on first spacecraft fixed for others

Key: ++ Strong Benefit, + Weak Benefit, NA not applicable, – Weak Negative, – – Strong Negative

- Case 2 is the same spacecraft with large power margins and a single array on a single gimbal. A gimbal failure will limit the capability of this spacecraft, but the high margins may create enough power without gimballing the array to continue operations with modest degradation.
- Case 3 is the same spacecraft with large power margins but, instead of a single large array on a single gimbal, it has two half size arrays on two gimbals. Should a gimbal failure occur, the high margins, coupled with the second gimbaled array, are likely to give ground operators the flexibility needed to maintain most, if not all, of the mission capability.

These three cases illustrate the benefits and costs of margins and flexibility. More margins increase cost and mass for the solar array, as does a second gimbaled array.

Broadly speaking, the benefits and costs of flexibility and margins will be unique for each mission and each subsystem. A program's reliability will benefit from a continuous search for opportunities to provide flexibility and margins wherever available within a tolerable cost.

Redundancy: Redundancy can significantly increase reliability by providing insurance against bad parts or environmental failure. It is also one of the most expensive options in terms of financial cost, weight, modest increases in production and test schedules, and software complexity. Redundancy will not protect against fundamental design flaws; the redundant item will have the same flaws as the primary item. We have frequently witnessed spacecraft redundancy extend operational mission life by many years. We have also seen several cases where entire missions were lost, or greatly degraded, due to the blind application of redundancy as a hard requirement. The latter case has been especially prevalent for mechanisms. Often mechanisms become extremely complex when driven to redundancy such that they can be made more reliable by rigorously testing a simple non-redundant design. The most important point is to use redundancy smartly, not as a rigid requirement, to maximize its benefit-to-cost ratio.

***Use of Mass-Production Components*:** Aircraft and automotive reliability gains tremendous improvement from the fact that mass-production components are available for key functions. Mass-produced components have typically completed their learning curve, so that most or all reliability weaknesses have been removed. Mass production provides great benefits in manufacturing by consistently producing the intended design and driving down the cost and schedule. Unfortunately, very few spacecraft components are mass-produced, other than electronics piece parts and similar items.

Reliability Analysis: Reliability prediction analysis, along with associated reliability analyses such as the failure modes and effects analysis (FMEA) and parts-stress analysis over temperature, are excellent for identifying weak links in a design and making improvements. If misused as a predictor of true spacecraft reliability, the analytical predictions often drive programs towards very expensive and long lead, "class 1" maximum-reliability parts. Proper reliability analysis can be one of the most economical practices for improving spacecraft reliability. However, its misuse as a predictor of on-orbit spacecraft reliability can lead to great cost and schedule expense with little-to-no true reliability increase; and it can potentially even lead to major reliability decreases such as program cancellation.

Rigorous Manufacturing, QA, and Processes: Manufacturing a spacecraft using rigorous production-process control provides strong reliability by avoiding problems resulting from bad parts slipping into the build. It also ensures that the built spacecraft meets the engineered design, and implementation through software-quality assurance. These practices are always necessary at some level, as they can prevent costly errors at the system level and catch some items that testing is simply unable to screen. The price one pays for strict process control is cost and schedule increases that can be surprisingly large; for example in major acquisition programs large teams of people are dedicated to providing extensive process documentation. The degree of process control is best decided on the basis of careful consideration of the costs and benefits relative to the unique circumstances of a program's reliability needs.

Mission Simulation and Training: Sound flight operations carry large increases in on-orbit reliability through increased spacecraft availability and avoidance of mission-ending operations failures. The preparation of a realistic mission simulator, supported by sufficient training, provides these benefits at relatively low cost. The decision to design and build the simulator is often a substantial program decision made relatively early in the program. We have consistently seen programs that chose not to develop a good simulator pay much greater costs downstream due to technically inaccurate testing, schedule delays to support operations testing, and poor spacecraft availability during the first year on orbit—because operators are still learning how to fly the spacecraft instead of having learned using a simulator prior to launch. Schedule delays often result from programs being forced to test ground software and mission operations using the flight satellite for lack of a mission simulator alternative. For almost all spacecraft programs, we recommend producing a simulator that, at a minimum, enables commands and telemetry to be sent, and represents attitude determination and control events (i.e., slews, and ground contacts). A simulator and associated training costs are almost always worth the operational availability benefits.

Constellation Design and Launch-On-Demand: The mission designer has many options that drive availability to consider in a constellation design. Simplistically, one can (1) build only the number of satellites needed, initiating the build of additional spacecraft only as dictated by circumstances; (2) build and launch on-orbit spares; or (3) build spares and store them for launch on demand. The pre-planned building of spares provides benefits to availability issues should any spacecraft be lost; however, the cost is significant. On the other hand the lack of spares saves cost; however, the time to build a replacement once a failure occurs has significant impacts on availability. A strong point to consider in this analysis is that the cost of the last unit built of a limited production run is generally below the cost of the initial units and can be a relatively small percentage of overall program cost. Spare satellites can be viewed as insurance for availability and more spares are better if a program can afford them. Additionally, an advantage for constellations containing multiple similar satellites is that on-orbit lessons learned on one satellite can be applied to the others provided that satellites are designed with sufficient flexibility. For instance, if you find out that an attitude control device does not have sufficient life capability on one satellite, software patches for contingency attitude control modes can be created and installed throughout

Table 13-7◀, Fig. 13-4◀, Eq. 13-1◀

the constellation, making all the other satellites robust against this now known defect.

13.4.6 Conclusion

- Focusing on meeting mission objectives and maintaining a user perspective are essential to success. Remember that schedule delays and large cost overruns, both notorious in space systems, provide an effective reliability of zero whenever the system is not available to users as expected.
- Reliability analysis can be one of the most economical practices for improving true spacecraft reliability through improved design. However, this analysis is fundamentally misapplied as a predictor of spacecraft success on orbit. Both MIL-STD-217F and on-orbit data confirm the use of these calculations to predict on-orbit reliability is inappropriate and inaccurate. Misuse of reliability analysis as a predictor of on-orbit spacecraft performance can lead to great cost and schedule expense with little-to-no true reliability increase—and potentially even major reliability decreases such as program cancellation. Examples include implementing full redundancy as a hard requirement and mandating all class 1 electronics parts.
- The concept of system availability to users allows broader consideration of program practices, such as design with high margins and flexibility, which enable programs to better address on-orbit reliability. Foundationally this starts by focusing on mission objectives and eliminating failure modes that could prevent these objectives from being met.
- We recommend each reliability practice receive an honest assessment of the question, "Is the increase in on-orbit reliability provided by this specific practice worth the inherent cost and schedule increase?" Each space-mission developer should create a program availability plan that considers its unique program circumstances and resource limitations. This plan must carefully consider all failure modes, discussed in Sec. 13.4.5, and evaluate all 9 practices to address these failure modes.

13.5 Principal Methods for Reducing Cost and Schedule

As shown in Fig. 13-5, our goal is to find ways to reverse the Space Spiral, defined in Sec. 1.1, of ever increasing cost, fewer missions, longer schedules, and demand for higher reliability. Essentially any process that breaks that cycle and begins to reverse it—i.e., that lowers cost, shortens schedules, increases the number of missions, or reduces the demand for higher reliability—will help the process. However, it's clear that breaking through the barriers of inertia and culture discussed in Sec. 13.2 requires that we take moderately large steps that make a significant and visible change. Also as discussed in Sec. 13.4, the "reliability" of Fig. 13-5 is largely parts reliability and numerical reliability computations. Actual ***mission reliability***, i.e., whether the end user will have the data they need when they need, is increased by lower cost, shorter schedules, and more missions because the system becomes much less sensitive to individual failures, delays, or program cancellations.

The following sections discuss some of the major steps that can be taken to drive down cost and schedule. However, two broad caveats apply to most of these items. First, because reducing cost and schedule is hard, it is unlikely that any single step will have a large enough effect to make a major change. While even small steps help, really turning the Space Spiral around and beginning to make large reductions in cost and schedule will require a concerted effort and combining multiple elements to have an effect large enough to be truly helpful to a program or to the space community more broadly. Second, it is clear that almost any step can have either positive or negative effects (and possibly both) depending on how it is implemented. For example, we have

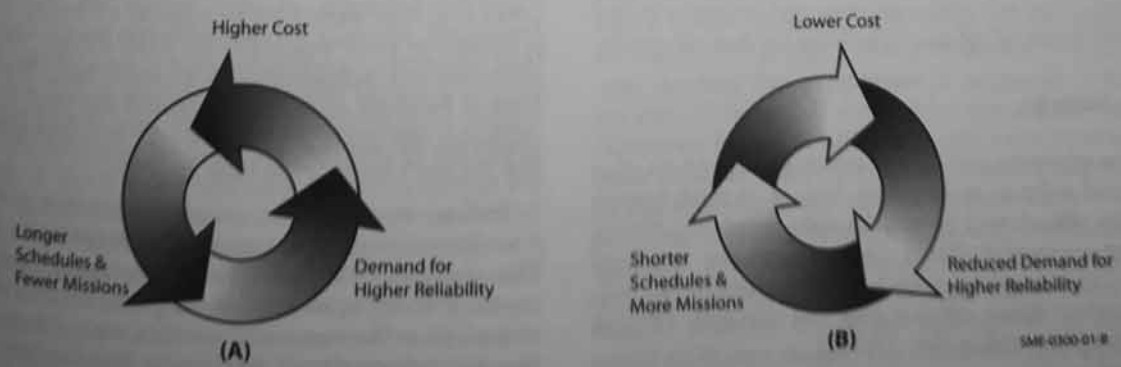


Fig. 13-5. Our Objective is to Find Ways to Reverse the Space Spiral in (A). As shown in (B) any actions that reduce cost, shorten schedules, increase the number of missions or reduce the demand for higher reliability will do this.

Table 13-7◀, Fig. 13-5, Eq. 13-1◀

Table 13-8. Summary of Attitude-Related Approaches to Reducing Mission Cost and Schedule. The right attitude and personnel (Sec. 13.5.2) are truly critical for any of the other approaches to succeed. In this and the subsequent tables on reducing cost and schedule, the most important or productive approaches are shaded. In this table, all of them are important.

Reducing Space Mission Cost—Attitude		© 2010, Microcosm, Inc.
Technique or Action	**Mechanism**	**Comment**
ATTITUDE		
A1. Recognize that it's Possible and Achievable	Eliminates major impediment of being unwilling to attempt to reduce cost	Biggest single impediment to reducing cost is the view expressed by "Faster, Better, Cheaper—Pick any Two." There are a great many successful, reliable, responsive, highly capable, low-cost space missions.
A2. Recognize that Low Cost is NOT Low Reliability	Eliminates major argument of wanting only "high reliability"	a) Low cost spacecraft are equally or more reliable than high cost spacecraft. b) Lower cost can dramatically improve reliability for the end user by: allowing multiple spacecraft on orbit, shortening the schedule, and reducing the probability the program will be cancelled.
A3. Recognize that Low Cost is Important	Assigns importance to low-cost and keeps it from being the last thing on the list of goals	Reducing cost has to be a high priority such that we are willing to give up something to achieve it. "We may want to explore Mars at a resolution of 10 m. But exploring Mars at a resolution of 20 m may be better than not exploring Mars at all."
A4. Recognize that Change is Critical to Reducing Cost	Reduces the opposition to "new ways of doing business"	We cannot buy or build the same spacecraft as last time with the same rules and processes and expect it to cost less.
A5. Recognize that Reducing Cost is Hard Work and Takes Real Engineering	Allows the allocation of resources and effort that are needed to achieve cost objectives	Like anything else of value, reducing cost is hard work and takes dedication, real engineering, and time, money, and attention devoted to it. There's a price to achieving low cost.
A6. Recognize that the Need to Change is Not a Criticism of Prior Programs or Practices	Reduces resistance to cost reduction programs	The Space Shuttle was a remarkable engineering achievement built by some of the best engineers in the world, but it didn't satisfy its end objective of greatly reducing launch cost. We have processes and rules in place for very good reasons, but collectively they, have created a space program that we cannot afford.
A7. Recognize that Virtually Any Technique Can Either Increase or Decrease Cost	Forces thinking about and evaluating changes within the context of individual programs—critical for successfully reducing cost	Cost reduction approaches must be implemented wisely with common sense. Virtually any of the techniques below can drive cost up if not implemented appropriately but this should not be used as an excuse for not doing them.

emphasized throughout Chaps. 1–5 the value of strong up-front mission engineering. This is a critical part of reducing both cost and schedule. Nonetheless, a blanket rule that requires a 2-year up-front mission engineering phase before anything can be implemented would be counter-productive by preventing new technologies from coming on board quickly, generally slowing down small, quick turn-around programs, and likely increasing their cost. Each of the recommendations must be implemented with some sense of balance to achieve the desired results.

13.5.1 Attitude

Large organizations would prefer to depend on processes and rules to reduce cost, but, as in most human activities, attitude and having the right people are critical to being successful. Table 13-8 summarizes the truly critical "attitude adjustments" that are needed to significantly reduce space mission cost and schedule. (People issues are discussed in Sec. 13.5.2.)

As discussed in Sec. 13.2, the largest single impediment to dramatically reducing cost and schedule is an attitude throughout the community most often expressed as "faster, better, cheaper—pick any two." This expresses the idea that we've done as well as we can do. In order to do better in one or two areas, we must do worse elsewhere, even though in virtually every other area of technology forward progress is real and moves at a much more rapid pace than in astronautics. The fact that technology advances and we have learned how to do things better over time should not be taken as a criticism of prior programs or prior management. Even though the space processes and technology that we have today have been created by some of the most capable and hard working engineers and managers in the world, this does not mean that we can't do better as time goes on.

Perhaps equally challenging is the idea that, so long as it works in the end, cost and schedule don't really matter. This is addressed quite well in the discussion on managing to a "reliability of zero" in Sec. 13.4. It doesn't matter to the soldier that was killed whether it was a remarkably capable system when it was finally launched, or would have been a great system when it was cancelled if in the end, the system wasn't there when it was supposed to be.

Table 13-9. Summary of Personnel-Related Approaches to Reducing Mission Cost and Schedule. Ultimately, it is people, not rules, that reduce cost and schedule. (For a more extended discussion, see Wertz and Larson [1996].)

PERSONNEL	Reducing Space Mission Cost—Personnel		© 2010, Microcosm, Inc.
	Technique or Action	Mechanism	Comment
	PI1. Improved Interpersonal Communications	Dramatically reduces errors and omissions. Conveys understanding as well as data.	Large programs use formal, structured communications through specified channels. Small programs use personal communications by creating close working relationships and personal responsibility.
	PI2. Small Team	Clear, nearly instantaneous communications; strong sense of personal responsibility	Can be a problem if a key person drops out, but in practice this rarely happens.
	PI3. Co-Located Team	Improves communications; reduces sense of "we vs. them"	Best communications are face-to-face, but AMSAT and some others don't seem to need it.
	PI4. Empowered Project Team	Rapid decision-making; strong sense of personal responsibility; can make "sensible" decisions	Eliminates a major function of the large, formal management structure. Encourages personal "ownership" of problems and issues, often expressed as "I own that problem."
	PI5. Reward Low Cost (both people and organizations)	Provides positive incentive to both people and organizations	Traditionally, if organizations spend less, they are rewarded by a reduced budget next year. Instead, they should get a larger budget next year or have half the savings put into an end-of-year bonus or party fund. (Rewards don't need to be monetary.)

In order to reduce cost and schedule, these have to be important to the system engineers, the program management, and the procuring organization. Finally, if you believe that something can't be done, or shouldn't be done, or isn't worth giving up something else to achieve, then it is likely that you won't be able to achieve it. Reducing cost and schedule is hard work and takes real effort and real engineering. **Reducing cost has a price.**

13.5.2 Personnel

Table 13-9 summarizes the second major pillar of reinventing space—personnel. Both the government and large organizations would much prefer to depend on rules, regulations, and processes, but the reality is that wars are won, inventions made, new businesses created, and creative ways to change how we do business in space are developed by motivated individuals who are out to get the job done. This means that individuals and small teams must be empowered to get things done and must be motivated to do them.

Personnel and groups must be rewarded for major reductions in cost and schedule. For example, the reward for not spending all of your department's computer budget by the end of the year is typically having a smaller computer budget next year. Amazingly, in most departments, the computer budget is always spent by the end of the year. A much better approach would be to split the savings between the organization, to reduce the overall budget, and fund the department end-of-year party.

Rewards don't need to be money. Give whoever came up with the idea for significantly reducing cost or schedule the parking space next to the door and let the department manager park in the back lot. (Actually, it would probably be best to give the department manager a bit of a reward as well. In the end, you want both the individuals and their managers to be pleased with the result.) Many small satellite builders take great pride in building high quality, low-cost satellites in a very short time. It is something they have learned how to do that much bigger, better-funded organizations do not know how to do. Recognizing this excellence can be a reward in itself. **The real secret to reducing space mission cost is to empower individuals and small teams, motivate them to reduce cost, reward them for achieving it, and get out of their way.**

13.5.3 Programmatic

Table 13-10 shows the primary programmatic approaches for driving down space mission cost. Traditionally, operational systems are very large with limited potential for changes to reduce cost or schedule, but these systems are both expensive and "fragile."

As discussed in Sec. 13.1.2, smallsats can help overcome this problem in multiple ways in the near term, and a mix of small satellites and traditional satellites can provide a robust capability with better performance, more flexibility, and greater mission assurance, at much lower cost. Another key programmatic approach is reducing the cost of failure. If we build a test engine that is so expensive, it can't be allowed to fail, we will never really learn what the failure mechanisms are. A major advantage of smallsats is that failures in test or early in the program are less of a problem than with traditional large satellites. The very reliable Soviet launch systems had multiple failures in early launches. These failures were used to improve the design and essentially eliminate nearly all failure modes. The demerit of this approach is that, unfortunately, failures early in a program tend to jeopardize funding for later phases or later missions.

Table 13-10. Summary of Programmatic Approaches to Reducing Mission Cost and Schedule. There are a large number of programmatic approaches that can help the problem. (See also Wertz and Larson [1996].)

PROGRAMMATIC

Reducing Space Mission Cost—Programmatic ©2010, Microcosm, Inc.		
Technique or Action	**Mechanism**	**Comment**
Pg1. Operational SmallSats	Allows low-cost operational satellites	Smaller, much lower cost spacecraft are becoming more competent and can do far more in the future. Can dramatically reduce mission risk by allowing satellite replacement and multiple satellites on orbit.
Pg2. Low Cost, Rapid Test Missions	Greater potential for breakthroughs and more rapid technology insertion	Allows genuine tests (i.e., possibility of failure), instead of just on-orbit demonstrations. Creates potential for major, near-term breakthroughs.
Pg3. Use Small Business Effectively	Small businesses are a major source of cost reduction technology and processes	Typically, small businesses are not used for creative processes, both because of the contracting difficulties and the perception that small businesses don't understand the "real world" of space missions.
Pg4. Buy Multiple Spacecraft	Uses both learning curve and continuity of production line	Much like buys of aircraft or ground vehicles.
Pg5. Provide Continuous, Stable Funding (i.e., avoid programs stops and starts)	Does not reduce cost, but avoids cost and schedule overruns	Stopping and restarting a program dramatically drives up cost and increases schedule well beyond the length of the schedule break. Typically not recognized by the program office. 3 key steps: a) Make major decisions away from funding boundaries b) If possible, provide multi-year funding c) Key programs funded while decisions are being made
Pg6. Make Cost Data Known	Drives more competitive cost proposals	It is very hard to reduce cost when cost data is known, and virtually impossible when it's not.
Pg7. Reduce the Cost of Failure	Allows both ambitious goals and calculated risk in order to make major progress	Fear of failure feeds cost growth spiral. Major breakthroughs require accepting the possibility of failure—particularly in test or in early mission trials. Example: development of Soviet launch vehicles.
Pg8. Build to Inventory	Reduces mission risk and permits higher system risk	Can dramatically reduce mission risk by allowing satellite replacement. Less of a target for enemy attack. Mission becomes less susceptible to system failures or orbital debris.
Pg9. Minimize Formal Documentation	Reduces programmatic overhead for creating, reviewing, and maintaining documents	Critical to document **reasons** for key decisions and as-built design. Minimizing documentation also allows documents to be given importance and maintained. AMSAT uses redlined schematics to document as-built design.
Pg10. Compress the Schedule	Less overhead costs and less time to spend money	Must be done with care—requires (a) reducing the amount of work required and (b) providing expedited decision making.

This is an example of the culture that we have all created preventing us from doing things that we all understand are important to reducing cost.

Making cost data known is also key to reducing long term cost, as discussed in Sec. 13.2.* Often it is the government which most strongly resists this step. For example, although we had the strong support of senior Aerospace Corp. personnel, we were not allowed to publish the current version of their Small Satellite Cost Model, SSCM, in this volume. (See Sec. 11.2.)

Buying multiple spacecraft simultaneously and launching from inventory is one of the major mechanisms to drive down cost, schedule, **and** mission risk. Here, it is important that we understand the meaning of "schedule" in terms of the end user. When buying a car, we don't really care how long it takes to build that car. We only care about how long it takes to get one after we have made the decision to purchase it. While buying from inventory is certainly not the same as building something quickly, this distinction is somewhat lost on the end user that simply needs to get their mission data as soon as they can. Similarly, if we want our new car to

* Note that here we use cost and price interchangeably. Making the profit (= price – cost) known may be counterproductive by keeping some suppliers out of the market and focusing attention on a relatively small budget element that has little potential for adjustment and won't bring about the large cost changes that are needed. Basically, beating on companies to minimize profit is not a good long-term substitute for finding ways to actually getting things done more economically.

have a new class of radio that just became available, we aren't concerned whether that radio was built into the car at the original assembly line, or was added by the dealer at some later stage in the process. What the end user wants is rapid functionality.

Buying multiple spacecraft, or at least one off of an assembly line allows us to take advantage of the learning curve (Sec. 1.3) to reduce cost per unit, often by a great deal. This involves not only learning, but tooling, spares, and production equipment. It also allows us to maintain a more-or-less steady manufacturing flow in an environment where use rates can change greatly. Inventory is effectively a buffer to draw from when launches or systems fail, are attacked by an enemy, or simply are in greater demand.

When using an assembly line, we aren't forced into a mode where everything has to work perfectly the first time. In addition, it means that we have people available (engineers, technicians, and supply chain managers) that understand the system, how it's built, and what the major problem areas are. We can fix problems that arise in early production (i.e., the Soviet approach) and can insert new technology as it becomes available. The use of an assembly line is a standard approach for aircraft, ships, and ground vehicles of all kinds. A key issue in nearly all other areas of government procurement is that once a manufacturing line is closed down, it is very expensive to start it up again. In traditional space systems, we effectively restart the assembly line with nearly every procurement.

Finally, the lack of funding continuity is one of the most damaging elements for a program and a major source of cost overruns. The causes of this are discussed in more detail in Sec. 13.6. Here we want to point out that, in large part, this is a management problem and not a funding problem. Basically, it requires rapid decisions in conjunction with planning ahead. Apparently, from the perspective of the program manager, it's worse to try and explain why you spent money on the early phases of a program that was later canceled than to stop the program and then explain cost and schedule overruns that can be blamed on the contractor or other outside factors. There is no doubt that some vacillation will be unavoidable, often because of funding profits, but much of it can be controlled.

13.5.4 Government/Customer

Table 13-11 shows the government (or other large customer) approaches to reducing cost and schedule. Of course one of the principal actions the government can take is to foster and encourage the other approaches described throughout this section. By making cost and schedule reduction a priority, being willing to allow some new ways of doing business to accomplish it, and devoting time and resources to creating lower-cost, much more rapid systems, a great deal can be accomplished.

Perhaps one of the more counter-intuitive approaches that the government can use to reduce cost and schedule is to decentralize space system procurement (item G2). On a fairly regular basis, calls for government reform point out "waste and inefficiency" presumably created by having multiple organizations working on a problem. Thus, there are regular calls for a "launch czar" or a "spacecraft czar" to reduce inefficiency by consolidating all of one activity into a single person or organization. In fact, this is likely to drive up cost, increase schedule, and be counterproductive relative to what we would like to achieve. We all recognize the value of competition in industry. With competition, multiple approaches are tested and we pick the ones that are best for our particular circumstances. But this same idea works within the government. If one person is in charge of launch, we will quickly eliminate all secondary programs in the name of efficiency, concentrate all of the work in a few large contracts (probably with the major primes), and, of course, make reliability the most important feature since if we have fewer systems it is more important that they work every time. This is a prescription for feeding the Space Spiral that we discussed at the beginning of the section. If, instead of a single launch czar, there are programs for small, responsive launch systems from the Army, Navy, Air Force, MDA, and NASA (and perhaps even separate ones from Marshall Space Flight Center, Goddard, and Ames), then we have the roots of a competitive environment in which low cost and fast response become what it takes to make your program proceed. If we need a launch vehicle that can put 10,000 kg into LEO, we could start designing that from the outset or we could start with multiple agencies working on ideas for putting 500 kg into LEO, select the most promising 3 or 4 of those to work on 2,000 kg to LEO vehicles, and so on. This approach gets us multiple small launchers that provide competition to hold down cost and develops and tests in flight alternate technologies that can be used to drive down cost on larger vehicles. Rather like airplanes or ships or computers, it doesn't necessarily make sense to have one supplier working with one government agency to solve the diverse needs of the space community.

The government controls most of the R&D spending in space technology. Unfortunately, there is a strong bias within the R&D community toward challenging new technology and away from practical systems capable of being implemented and able to reduce cost or schedule or both. For example, the *Small Business Innovative Research* (SBIR) program would be an excellent vehicle for small companies, for which a major strength is finding innovative approaches to reducing cost or doing things more quickly. Microcosm undertook a survey of both DoD and NASA space-related SBIR topics in 1996 for the book *Reducing Space Mission Cost* [Wertz and Larson, 1996] and again in 2010 for this volume. The rather discouraging results are shown in Table 13-12. It is likely that much more rapid advances in reducing cost and schedule would be possible if the government chose to sponsor more R&D oriented toward reducing cost, without demanding that it advance technology at the same time.

Another SBIR approach is to make more extensive use of SBIR Phase IIIs (G6). By law, the SBIR Phase III

Table 13-11. Summary of Government or Customer Approaches to Reducing Mission Cost and Schedule. Many of these involve implementing approaches described elsewhere in this section. (Notes in parentheses in the rightmost box of the first row refer to methods from other tables, i.e., Pg5 refers to Programmatic method 5.)

GOVERNMENT

Reducing Space Mission Cost—Government / Customer		© 2010, Microcosm, Inc.
Technique or Action	**Mechanism**	**Comment**
G1. Implement Actions Specifically to Reduce Cost	Demonstrates a real interest in reducing cost and willingness to take action to achieve it	Among those listed elsewhere that can be done directly by the government or major customer: a) Foster an attitude of wanting and rewarding cost reduction (A1 to A7) b) Force trading on requirements (SE1) c) Reward low cost (PI5) d) Provide funding continuity (Pg5) e) Make cost data available (Pg6) f) Reduce the cost of failure (Pg7) g) Develop a low-cost small launch vehicle (L5) h) Create low-cost smallsat programs for both test and operations (Pg 1 and Pg2)
G2. Decentralize Space System Procurement	Allows innovation and options that would not be allowed under a centralized procurement approach	Innovation often comes from small businesses and "secondary" organizations within the government. These should be encouraged as a way of providing positive and valuable "competition of ideas" within the government.
G3. Sponsor R&D to Reduce Cost	Main mechanism for finding lower cost solutions	Need to make reducing cost an alternative and acceptable objective for R&D, without demanding that it simultaneously "advance technology."
G4. Sponsor Knowledge Preservation and Dissemination	Actually using lessons learned rarely occurs, but is important to reducing cost	Space technology has dramatically fewer books, commercial software, or university programs than any other comparable major discipline. Knowledge is being lost at a rapid rate.
G5. Revise SBIR Objectives	An excellent source of innovative ideas for reducing cost	Currently, less than 4% of SBIR topics are directed specifically toward reducing cost. Can provide both innovative solutions and simpler, faster contracting mechanism via Phase IIIs.
G6. Use SBIR Phase III	Sole-source contract	Sole-source contract mandated by law. Can shorten schedule by a year or more. Contracting process short and simple.
G7. Assign the Task of Reducing Cost to an Individual or Organization	Allows cost reduction to be a part of the official hierarchy of organizational objectives	Typically, reducing cost is not in anyone's "job jar," which means that it is a secondary priority in a system where not all first priority tasks get done.
G8. Create a Program Intended Specifically to Reduce Cost	Allows cost reduction to be a part of the official hierarchy of organizational objectives	Similar to item above, but allows contractor participation. Can create specific objectives with a near-term schedule that can impact both near-term and longer-term, larger missions.

Table 13-12. Survey of the Principal Objectives of Space-Related Small Business Innovative Research (SBIR) Topics in 1996 and Again in 2010. (Study done by Microcosm for RSMC and again for this book.)

Principal Objective	1996		2010	
	No. of Topics	Percentage	No. of Topics	Percentage
New Software	33	15%	38	22%
Improvements to Existing Software or Technology	67	30%	20	12%
New Technology	113	51%	108	63%
Reduce Cost	**8**	**4%**	**6**	**3%**
Total	221	100%	172	100%

meets all of the competition in contracting requirements, is strongly encouraged by Congress, and has been endorsed within DoD [Finley, 2008]. This means that ideas developed under the SBIR program can go directly to being funded and built via a sole-source contract without another round of studies and competition, which can save more than a year and a large amount of funding and time in the competition and contracting process. Unfortunately, this isn't a popular law within the government bureaucracy because it eliminates another round of competition and contracting, and is often ignored.

Perhaps the most important thing that the government can do to reduce cost is either assign the task of reducing cost to an individual or organization (G7) or create a program intended specifically to reduce cost and schedule and fund some small programs to do that (G8). In both

Table 13-12, Fig. 13-5◀, Eq. 13-1◀

Table 13-13. Summary of Systems Engineering Approaches to Reducing Mission Cost and Schedule. These approaches have proven to be extremely successful on many past and current low-cost, high-reliability missions.

Reducing Space Mission Cost—Systems Engineering		© 2010, Microcosm, Inc.
Technique or Action	Mechanism	Comment
SYSTEMS ENG		
SE1. Trading on Requirements	Allows a balance between cost and benefit	Makes are traditional competition difficult. Allows the government to become an "intelligent consumer."
SE2. Concurrent Engineering	Allows schedule compression; increases feedback between groups	High non-recurring cost. Can achieve "local optimization," but reduces willingness to consider truly different approaches.
SE3. Design-to-Cost	Adjusts requirements and approach until cost goal has been achieved	Has been used, but not often (Discovery, New Frontiers). Arbitrary cost goals are unlikely to be successful.
SE4. Large Margins	Reduces testing; better flexibility; reduces cost of engineering, manufacturing, and operations	Margins traditionally kept small to maximize performance. Requires balanced implementation—forcing large margins in all components may drive up cost.
SE5. Fly New Computer Plus Same Computer Flown on Last Mission	Allows use of newer technology with higher capability without the associated risk.	Allows use of newest computer technology—both lower cost and more capability with very low risk. Can be used for other hardware as well. Approach used by SSTL.
SE6. Avoid "Design for a Reliability of 0"	Places much greater emphasis on cost and schedule as critical	Recognize that from a user perspective, every day that the system is late, it has a reliability of 0. A program that is cancelled due to cost or schedule overruns has a reliability of 0 to the end user.
SE7. Devalue Optimization	Allows multiple cost reduction methods	"Optimized solutions" prevent standardization and use of non-space equipment or processes and require that everything be uniquely designed for each specific application.
SE8. Use the Existing Knowledge Base	Reduces cost, schedule, and risk	Reinventing the wheel is rarely economical. According to John Mather, "6 months in the laboratory can save you a week in the library." Specific approaches to building on existing knowledge: a. Books and literature b. Courses and conferences c. Commercial software tools d. Become a part of the low-cost community e. Take advantage of the knowledge of others

cases this allows reducing cost and schedule to become a part of the official hierarchy of organizational objectives, to be reported on at meetings, to get some assigned budget, and to flow regular status reviews up the management chain. All of this makes it clear that this is something the organization genuinely wants to accomplish and will be judged on how well it is being achieved.

13.5.5 Systems Engineering

Systems engineering approaches to reducing cost and schedule are summarized in Table 13-13. Perhaps the most fundamental of these is trading on requirements (SE1), an approach that has proven exceptionally successful in low-cost programs. The process for doing this is shown in Table 13-14 and is discussed in detail in terms of mapping and pointing accuracy in Sec. 8.6. (See Table 8-15 and Sec. 8.6.3.) The critical issue is to find areas where there is increased performance available at increased cost and then balance need vs. cost. In many respects, this is equivalent to the process of an individual buying a car. We don't write out a set of specifications and send them around to car dealers to find out who can give us this car at the best price. Instead, we look in our wallet and typically find that there is a bit less budget available than we would prefer. We then go around to various dealers and find out what is available at what price, knowing that there will be a certain amount of negotiation later on. Ultimately, we try to find a reasonable compromise between what we want and what we can afford. Our goal in space is the same—we would like to make the government into an intelligent consumer that can balance cost, schedule, performance, and risk.

We have talked in Sec. 13.2 about the problem of culture and the difficulties it raises in trying to do business economically. This is an excellent example. The government wants to give an impression of being fair to all of the contractors. Assume they evaluate multiple proposals and

Table 13-14. Process for Trading on Requirements. The example shown is for geolocation accuracy as discussed in Sec. 8.6. (See Table 8-15 and Sec. 8.6.3.)

Step	Example	How Traditionally Done
1. *Document Reason for Requirements*	Geolocation accuracy for science data reduction	Not done
2. *State Functional Rather than Technical Requirements*	Provide geolocation to within 500 m at nadir	Not done (i.e., attitude determination required = 0.05 deg, 3 σ)
3. *Identify System Drivers for the Requirement*	Attitude sensor accuracy	Not done
4. *Identify Principal Options*	(a) $500K Earth sensor @ 0.10 deg (b) $2,000K star sensor @ 0.01 deg	(a) $500K Earth sensor @ 0.10 deg (b) $2,000K star sensor @ 0.01 deg
5. *Determine Consequences in Terms of Mission Performance*	(a) geolocation to 1 km (b) geolocation to 100 m	(a) N/A (b) Meets requirement with margin
6. *Select Preferred Option*	Discuss cost with customers and users. Is it worth $1.5 mil more per spacecraft to achieve geolocation of 100 m vs. 1 km?	Option (b) is only acceptable choice

select contractor A to build the next Mars lander. They work through the detailed design, trading on requirements as they go to get the most effective product at the best price. But when they're done the lander doesn't look very much like their original requirements and now contractor B comes back and says "if we had known that was what you wanted, we could have done it much cheaper," and, of course, that may be true. While we want to find ways around cultural (or legal) impediments, we also want to be sure that we understand why they exist so that we can find the right balance.

Trading on requirements typically demands other changes in the way we do business in space:

- Setting functional, rather than technical requirements (i.e., stating **what** we want done, not **how**)
- Documenting the source of requirements, so we know **why** they are needed
- Trading explicitly on the principal driving requirements
- Being willing to give up some level of performance in order to reduce cost or schedule

These are all things that we would do in normal buying decisions, but are much harder to do in a formal, rule-based system.

Another approach often used in low-cost systems is to use larger design margins (SE4). Traditional space systems are typically optimized for performance and, therefore, minimize margins. Everything is made as light and as thin and as small as possible, largely because space systems cost $10,000 kg to get to orbit and every fraction of a kg matters if we want to squeeze out as much performance as possible. But the net effect can be a system that is nearly impossible to build or change. Returning to our car analogy, assume we find the car we want, but we would like to put in a better radio because we will be making many long trips and the better radio gets more stations in remote areas. No problem. The radio is an extra $300. Of course, the new radio is heavier, so we will have to redesign the dashboard to be able to hold it. It also uses more power, so we'll need a larger generator. And then there's the heavier wiring for the rear speakers. And because everything has gotten heavier we can't meet our acceleration performance spec, so we will need a new engine. Fortunately, in the real world, cars are designed with lots of margin. For our spacecraft, large margins reduce cost, and the potential for cost and schedule overruns in quite a few ways:

- Less testing required
- Normal manufacturing tolerances acceptable
- Fewer rejects and reworks
- Fewer failures in both test and operations
- More robust design means less redesign
- Potential for standardized components
- Higher level of component and design reuse
- Can use more commercial grade components
- Can accept less certainty about the environment
- Reduces operations cost for planning and analysis

Most high quality products use large design margins to overcome the vicissitudes of the environment. In space, optimizing the design by driving margins as small as possible is one of the principal reasons that standardization has been dramatically unsuccessful. (In most cases, standardization means flying more capability and more mass than you actually need. Therefore, in space systems we tend to want the standard product, but with 3 of the 5 features removed to save mass and optimize performance.) Of course, this problem is made worse by the logic that says we must optimize performance so as to get the most bits (or whatever else it is we're buying) per spacecraft dollar. This leads not only to building Ferraris for every mission, but to customizing each individual Ferrari for the particular mission at hand. It results in a remarkably expensive way to observe, communicate, or predict the weather.

Table 13-14, Fig. 13-5◀, Eq. 13-1◀

Finally, a key element of reducing cost is to make use of the knowledge base of what people have learned before. Low-cost, high-reliability missions have been around since the beginning of the space program and are getting more competent very rapidly with modern advances in microelectronics, processors, and composite materials. There is very little value in trying to reinvent what has been done before. It is important to learn what has worked and what hasn't worked in terms of reducing cost and schedule in prior programs and why. This means taking advantage of the knowledge of others, becoming a part of the low-cost, responsive space community, and making use of conferences, courses, and the existing literature base. The major sources of information are listed in the tables in Sec. 1.6. The references and bibliographies on the book website, along with web references to where to buy or download individual references, provide background information.



In addition a specific "Bibliography of Reducing Space Mission Cost" is included on the website section for this chapter. This does not imply that every mission will want to follow every process developed for other missions. In many respects, each mission and organization is unique and will tailor what has been learned to their particular circumstances. But reinventing those rules is not the right approach.

13.5.6 Mission

Table 13-15 provides a summary of mission-related approaches to reducing mission cost and schedule. One of the most important elements of reducing mission cost is considering the possibility of alternative orbits. (See Sec. 13.1.1.) Traditional Earth surveillance missions want to last for a decade or more and, therefore, need to blanket the entire Earth with every sensor that will be needed in the future. (Because the spacecraft themselves are individually very expensive and effectively irreplaceable, the system as a whole tends to cost many billions.) If we are able, instead, to respond directly to world events, this cost can be dramatically reduced by both reducing the amount of coverage that is needed, but also by doing things, such as flying low, that can provide excellent performance at reduced cost with systems that are not intended to last for decades. This also lets us introduce new technology as it becomes available and employ that technology in those specific circumstances where it's needed. (See Sec. 13.1.1)

Similar considerations can apply to other missions as well. The orbit for science missions is often chosen as the best orbit for that mission irrespective of cost, in part because the "cost" of an orbit tends to be intangible. This is the reason for introducing the *orbit cost function* in Sec. 10.3.2 which is the ratio of the mass required in low Earth orbit (LEO), due east from the launch site to the total spacecraft mass needed in any given operational orbit. For example, going to GEO requires putting into LEO about 5 times the mass required in GEO. Going to the surface of the Moon requires about 8 times the mass in LEO that will ultimately end up on the surface of the Moon. This implies an orbit cost function of about 5 for GEO and 8 for the surface of the Moon. It may be that GEO or one of the Lagrange points is the ideal place for a scientific instrument due to excessive interference from the Earth. However, if we could get the same effect in LEO by tripling the mass of the spacecraft by adding shielding of baffling equal to the twice the spacecraft mass, we could potentially be much better off. We would still be launching only a bit more than half the mass of

Table 13-15. Summary of Mission-Related Approaches to Reducing Cost and Schedule. Cost reduction starts with the preliminary mission definition and design that largely determines many of the system costs.

MISSION

Reducing Space Mission Cost—Mission		© 2010, Microcosm, Inc.
Technique or Action	**Mechanism**	**Comment**
M1. Change the CONOPS to Use Existing Data Communications	Much lower cost. Systems are maintained at little or no cost to end user	Make the satellite simply another node on the Internet. Use commercial providers to get the data to and from the spacecraft.
M2. Fly Low	Low altitude is a dramatically lower cost substitute for large aperture for observations or high power for active payloads	a) Resolution is proportional to distance/aperture (small numbers are better). b) For SAR or lidar, power required is proportional to R^4 c) Also avoids the orbit debris problem. Requires propellant for drag make-up.
M3. Use Repeat Coverage Orbit	Much better coverage for a predefined location than SSO	Gives up global coverage for much better coverage of a specific latitude range. Coverage is tuned to provide 4 to 6 successive orbits of coverage at the latitude of interest.
M4. Use Orbit Cost Function as a Measure of Orbit Cost	Allows cost vs. benefit trade on the orbit selection	Orbit Cost Function is the ratio of the payload available at 185 km due east from the launch site to the payload available in a given operational orbit.
M5. Short Mission Design Life	Reduces redundancy and complexity	Most missions live much longer than their design life. Also short design life allows introduction of newer technology on a regular basis.

Table 13-15, Fig. 13-5◀, Eq. 13-1◀